

NORTHERN KENYA GRASSLAND CARBON PROJECT

Document Prepared By Soils for the Future, LLC, The Nature Conservancy, Native, A Public Benefit Corporation and The Northern Rangelands Trust

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Monitoring Period of this Report	01 January 2017 – 31 December 2020
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Gold Level Criteria	Climate Change Adaptation Benefits - increased productivity and water storage in soils; Exceptional Biodiversity Benefits - increased populations of trigger and endemic wildlife species and conservation of sky islands; Community Benefits – better livestock health, improving household livelihoods

Table of Contents

1	Summary of Project Benefits	3
1.1	Unique Project Benefits	4
1.2	Standardized Benefit Metrics	5
2	General	8
2.1	Project Description	8
2.2	Project Implementation Status	17
2.3	Stakeholder Engagement	25
2.4	Management Capacity	35
2.5	Legal Status and Property Rights	37
3	Climate	43
3.1	Monitoring GHG Emission Reductions and Removals	43
3.2	Quantification of GHG Emission Reductions and Removals	65
3.3	Optional Criterion: Climate Change Adaptation Benefits	80
4	Community	85
4.1	Net Positive Community Impacts	85
4.2	Other Stakeholder Impacts	95
4.3	Community Impact Monitoring	97
4.4	Optional Criterion: Exceptional Community Benefits	101
5	Biodiversity	102
5.1	Net Positive Biodiversity Impacts	102
5.2	Offsite Biodiversity Impacts	120
5.3	Biodiversity Impact Monitoring	120
5.4	Optional Criterion: Exceptional Biodiversity Benefits	122
6	Additional Project Implementation Information	124
7	Additional Project Impact Information	125
8	References	126
9	Appendix A	129
10	Appendix B	132
11	Appendix C	133
12	Appendix D	134

1 SUMMARY OF PROJECT BENEFITS

The Northern Kenya Grassland Carbon Project aims to remove greenhouse gases from the atmosphere by implementing sustainable grazing management over approximately 2 million hectares of northern Kenya savannas and grasslands. Active planned grazing and livestock movements in each of 13 different Conservancies, replaced baseline unrestricted grazing. Rapid movement allows the recovery of perennial grasses, which foster greater carbon inputs, higher retention of carbon as lignin and cellulose and greater belowground production, all of which lead to removal of greenhouse gases from the atmosphere to soil carbon.

Climate Benefits

The planned rotational grazing project activity will remove 0.6 to 1.1 tonnes CO₂e per hectare each year for at least thirty years. The project will be implemented across nearly 2 million hectares of rangeland.

Community Benefits

Past overgrazing by pastoralists from more than six indigenous ethnic groups in the region has depleted soils of organic matter, greatly reduced perennial vegetation cover and the potential production of forage for livestock. The sustainable planned grazing implemented in the project is restoring grass and other livestock forage, which will provide buffers against variable rainfall and reduce the need for long and risky migrations during droughts. Restoration of grass is increasing the climate adaptability of local communities because soils with more carbon store more water and sustain greater forage production through dry periods. Water retained in soils can improve surface water availability. Improved forage may also allow livestock owners to more easily sell livestock into food markets. Revenues from sale of carbon credits shared with communities and salaries/wages will benefit project employees and their families.

Biodiversity Benefits

Restoration of perennial grasses and soil water lead to increased productivity and cover, which will increase populations of four endemic, endangered wildlife species: reticulated giraffe, Grevy's zebra, beisa oryx and black rhino. Associated with the improvements in productivity will be increases in the diversity of grassland and savanna biodiversity indicator species from groups including plants, trees, dung beetles, grasshoppers, and birds.

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1) Local communities, oriented around 13 wildlife conservancies organized under an NGO called the Northern Rangelands Trust, engage in new planned rotational grazing practices, as opposed to repeated, permanent grazing simultaneously on all grazing lands.	Project activities have resulted in a halt in the decline of forage productivity that occurred prior to the start of the project. After accounting for the effect of annual variation in rainfall, forage production has nearly doubled in the project area during the period when the project activities were fully implemented across all 13 Conservancies (2015-2020).	3.2.2	Same as current monitoring period
2) Multiple communities and ethnic groups participate in an annual regional grazing plan that allows livestock to move over long distances and efficiently use wet and dry season range.	Planning through a hierarchy of grazing decision-makers was implemented in October 2015 and led to the explicit inclusion of grazing plans in overall conservancy management plans for 8 conservancies. Since 2015, conservancy managers have utilized these annual conservancy management plans in order to effectively carry out grazing plans for their respective conservancy. It also extended the decision-making process to all ethnic groups regardless of the conservancy in which they live.	3.2.2	Same as current monitoring period
3) Long-term benefits to both financial, social, and natural capital for both communities and individuals.	The project has demonstrated improved access to milk and water and greater indicators of security under the project scenario.	4.1.3	Same as current monitoring period
4) Enhancement of the region's role as a High Conservation Area for key endangered species.	Black rhinos (<i>Dicornis bicerors</i>) were successfully introduced into the Sera Conservancy in 2016. The number of illegally killed elephants has sharply declined	5.1.1	Same as current monitoring period

	since the start of the project, and three umbrella wildlife species (reticulated giraffe, Grevy's zebra, and beisa oryx) have expanded their range and increased in density since the start of the project.		
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1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emissions reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	4,179,523 tCO ₂ e	3.2	8,001,641 tCO ₂ e
	Net estimated emission reductions in the project area, measured against the without-project scenario	Not applicable	N/A	Not applicable
Forest ¹ cover	For REDD ² projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	Not applicable	N/A	Not applicable
	For ARR ³ projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	Not applicable	N/A	Not applicable

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

Improve d land manage ment	Number of hectares of existing production forest land in which IFM ⁴ practices have occurred as a result of the project's activities, measured against the without-project scenario	Not applicable	N/A	Not applicable
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	1,993,075 ha	2.1.7	1,993,075 ha (previously 1,993,078 ha- see PD deviation 3)
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	No data	N/A	N/A
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	No data	N/A	N/A
Employ ment	Total number of people employed in of project activities, ⁵ expressed as number of full time employees ⁶	No data	N/A	N/A
	Number of women employed in project activities, expressed as number of full time employees	No data	N/A	N/A
Liveliho ods	Total number of people with improved livelihoods ⁷ or income generated as a result of project activities	Over 86% of people surveyed said they experienced improved livelihoods (assets) from the	4.1.3	For livelihood: Increase in 1,527 people since baseline surveys

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (VCS Program Definitions)

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, subcontracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

		project activity. Thus 12,706 people benefitted.		
	Number of women with improved livelihoods or income generated as a result of project activities	Over 86% of women surveyed said they experienced improved livelihoods (assets) from the project activity. Thus 3,414 people benefitted.	4.1.3	For livelihood: Increase in 1,086 people since baseline surveys
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	No data	N/A	N/A
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	No data	N/A	N/A
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	No data	N/A	N/A
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	No data	N/A	N/A
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Mean time to gather water has declined by 20% under the project scenario	4.1.3	Same as current monitoring period
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities,	Over 68% of women surveyed said they experienced improved access to drinking water from	N/A	Increase in 2,149 people since baseline surveys

	measured against the without-project scenario	the project activity. Thus 2,894 people benefitted		
Well-being	Total number of community members whose well-being ⁸ was improved as a result of project activities	Over 82% of people surveyed said they benefited from the project activity through increased cattle ownership. Thus 12,677 people benefitted	4.1.3	Increase in 1,513 people since baseline surveys
	Number of women whose well-being was improved as a result of project activities	90% claimed the grazing plan was "Helpful" or "Very Helpful" (Figure 16).	4.1.3	
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ⁹ measured against the without-project scenario	80% of the project area experienced successfully implemented grazing plans, therefore 1,594,460 ha were better managed	3.2.2	Same as current monitoring period
	Number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities, ¹¹ measured against the without-project scenario	6 under project scenario	5.1.3	Same as current monitoring period

1.

2 GENERAL

2.1 Project Description

This project aims to remove greenhouse gases from the atmosphere by implementing sustainable grazing management over a large area of northern Kenya savannas and grasslands. Past

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

⁹ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

overgrazing by pastoralists from more than six indigenous ethnic groups in the region, including Maasai, Samburu, Burana, and Turkana, has depleted soils of organic matter, greatly reduced perennial vegetation cover and the potential production of forage for livestock. Marginal livestock production resulting from this degradation has put these pastoralists at risk from further climate change and from inter-ethnic tensions over grazing lands and water. The project has local communities, oriented around 13 wildlife Conservancies organized under a NGO called the Northern Rangelands Trust, engage in new planned rotational grazing practices, as opposed to repeated, permanent grazing simultaneously on all grazing lands. These new practices, which would not occur without significant carbon revenues to motivate change in herder behavior, will allow recovery of perennial grasses and the restoration of soil organic carbon.

Soil carbon storage that results from this improved grassland management is expected to result in significant removal of CO₂ from the atmosphere, as will be documented using the VCS VM0032 methodology ALM – Sustainable Grassland Management through Adjustment of Fire and Grazing, hereafter VCS VM0032 methodology. This methodology allows VCUs to be earned for the activity of improved grazing management, and thus verified as a consequence of demonstrating such improved management by measuring altered grazing intensities, records of the time, timing and number of livestock grazers using the project area, and vegetation.

The project was implemented in stages from portions of 3 Conservancies that adopted planned rotational grazing in 2012 to all 13 Conservancies in the project area (1.99 million ha project zone) participating in a region-wide drought response grazing plan by February 2014. With expected annual removals of approximately 0.88 metric ton CO₂e/ha on average, the project should remove more than 1.7 million tonnes CO₂e annually and 50 million tonnes CO₂e over its lifetime.

Improved livestock grazing and cash payments for carbon credits are expected to result in a number of social benefits and very few detriments. Clear long-term benefits will result from restoring perennial grasses. Rotational grazing, which will provide grass banks and a lower likelihood of catastrophic livestock loss during drought, sustained meat and milk production through the dry season, and heavier livestock that are saleable in the burgeoning Kenya beef market. Improved soil organic matter should help increase rainfall infiltration and lengthen the time surface water is available for people and livestock. Payments to communities through NRT will also be used to fund health care and bursaries that provide benefits to the poorest community members, such as paying school fees. In the short term, grazing plans have led to a stronger perception of community connection to land, reduced conflicts over grazing, and some re-claiming of Conservancy areas that were previously thought to be unsafe, as revealed by the results of focus group interviews and household surveys. These benefits to financial, social, and natural capital at the level of both communities and individuals have reduced tensions between ethnic groups and fostered better collaboration among different conservancies. Revenues to communities should help develop infrastructure such as schools, health clinics, water collection and storage facilities, thus reducing food and water insecurity.

The ecological vegetation and soil changes from the implementation of the project should also enhance the region's role as a High Conservation Area for key endangered species. Such species include Grevy's zebra (*Equus grevyi*), black rhino (*Dicornis bicerors*), and elephant (*Loxodonta africana*). Other endemic bird and ungulate species occur in the project area's biogeographical position connecting species from the Rift Valley to the south to the Sahel to the west. Also, the project activity of implementing planned rotational grazing should improve species diversity across a number of groups of key bio-indicator taxa, including plants, birds, and key insect groups, such

as grasshoppers (Orthoptera) and dung beetles (Coleoptera: Scarabeidae). The potential for the project to enhance biodiversity is evident from effects of improved grazing on these groups detected in biodiversity camps. Our baseline comparisons of the diversity of these groups among year-round grazed areas vs protected areas and those with limited grazing use supports the expectations from prior studies and justifies the project monitoring plan for biodiversity in savannas and grasslands. As the project develops isolated “sky island” forests on mountain ranges adjacent to the project area, such as the Mathews and Ngare Ndare Forests, should also have their biodiversity protected because improved grazing management should greatly reduce the need for herders to degrade the forests to provide forage for livestock during drought.

2.1.1 Implementation Description

Implementation of the project proceeds by:

1. Determination of grazing plans through initial interactive meetings between NRT personnel, Conservancy grazing coordinators, and Conservancy managers, followed by meetings between community members and Conservancy grazing coordinators and other leaders. Community meetings make full attempts to encourage attendance by women and ethnic minorities within Conservancies.
2. Individual herders, under the guidance and supervision of Conservancy grazing coordinators, move their herds among different grazing blocks within Conservancies, and during dry seasons among different areas of other participating Conservancies.
3. The number of livestock from particular conservancies present in different grazing blocks and in different areas of other participating conservancies is monitored by project-employed grazing coordinators. These coordinators supply monthly reports of locations and numbers to NRT and other professionals involved in the current project.
4. Conservancy grazing coordinators also monitor when herders from a Conservancy move their livestock off project area and record the number of days a number of livestock over which this occurs. These data are used to determine leakage.
5. Impacts of livestock are monitored using a project – developed protocol that employs satellite imagery to assess how frequently and with what intensity livestock grazing occurs throughout the project area. The project has an algorithm that converts the satellite imagery data into an assessment of whether project implementation goals were met across the entire project area. Essentially satellite data allows project administration staff to assess whether livestock graze an area once or not at all during a year. If livestock are determined to graze an area more than once, the satellite imagery “pixel” is classified as “not successfully implemented,” and removals are not claimed from the area represented by the pixel.
6. The area of successful implementation is determined for each Conservancy and multiplied by the expected mean carbon removals for that Conservancy to

yield the total annual removals for that Conservancy. This is then repeated for each of the 13 participating Conservancies.

7. Leakage is managed by project employed game scouts, which provide input from grazing coordinators to herders in remote sections of the project area and act as reminders that herders are expected to remain on the project area if forage is available. Herders are made aware of the negative consequences of going off the project area during annual grazing planning meetings. Finally, when herders do go off the project area, game scouts also act as first responders to help avoid conflicts and to signify when NRT personnel are needed to broker reciprocal agreements that allow project herders to legally and respectfully use pastures outside the project area.
8. Further feedback to control leakage is provided by annual distribution of revenues which is in part influenced for individual conservancies by whether their herders followed project grazing plans and/or left the project area. This mechanism issues economic penalties to whole communities as a consequence of individual herders' decisions and creates social pressure to follow Conservancy grazing plans.
9. Non-permanence threats from encroachment by herders from non-participating communities are also reduced by project border activities of game scouts and conduct of peace meetings by NRT personnel and Kenya County government agents.
10. The total net greenhouse gas removals claimed by this project in the current monitoring period are 4,179,523 tonnes CO₂e.

2.1.2 Project Category and Activity Type

This is an ALM project in the AFOLU sector under the VCS VM0032 methodology. It is not a grouped project.

2.1.3 Project Proponent(s)

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2.1.4 Other Entities Involved in the Project

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2.1.5 Project Start Date (G1.9)

The project Start Date is 10 December 2012 (Date of TNC/NRT charter). This date reflects the timing of initiation of yearly rotational grazing plans by the first three Conservancies (Il Ngwesi,

Kalama, and Westgate), and the signing of a joint MOU between NRT and the Nature Conservancy to administer and finance the implementation of the project.

2.1.6 Project Crediting Period (G1.9)

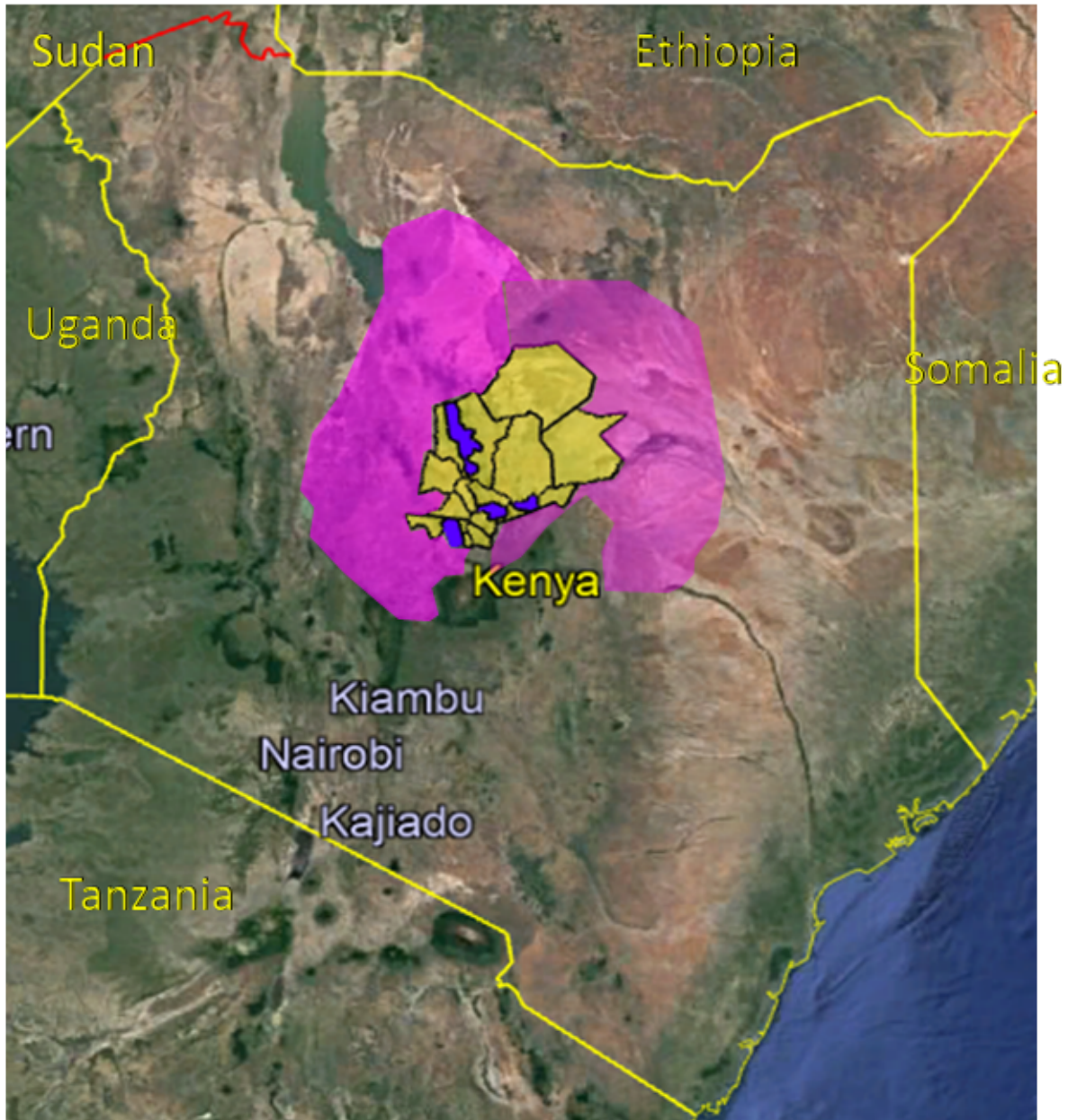
The project crediting period is 30 years from the implementation of yearly rotational grazing plans: from 1 January 2013 to 31 December 2042. The crediting period start date was moved from 10 December 2012 to 01 January 2013 as project monitoring period activities were not fully implemented until 01 January 2013.

2.1.7 Project Location

The project zone is 1,993,075 ha of savanna grassland in Kenya extending northward from the northern slopes of Mt. Kenya toward the Ethiopia border (VVB reviewed see attached KML file). The zone consists of a series of former group ranches and community trusts organized under the Northern Rangelands Trust to form wildlife conservancies of various sizes (Fig. 1). The project zone also includes the forested mountain ranges, which are protected forest reserves, found in the center of the project area. It also includes three National Game Reserves that protect wildlife access to permanent water, the Ewaso Nyiro River, which arises from the Aberdare Mountains 600 km to the southwest and flows from west to east, bisecting the project zone. Outside the project zone lie additional pastoral lands to the west, north and east, many of which are organized as Conservancies. Private ranches and other community-based Conservancies on the Laikipia plateau lie to the southwest, and a belt of commercial farms lies to the southeast between the project area and the forests of Mt. Kenya National Park^[1, 2].

It is important to note the process in which an area becomes a conservancy, as defined in this project. If a community, or collection of communities, wishes to join NRT membership, a formal written application is made by the community / communities in question to the NRT Council of Elders (C of E). The application details their status as a community / communities (land tenure / registration / governance structure), their collective decision to apply for NRT membership (internal FPIC with evidence), and the assets / entities they wish to protect / strengthen, with support from NRT (natural resources / wildlife / peace building / livelihoods / etc.). The NRT C of E then assesses the application, and if they are confident it is a well deserved application, the C of E will request NRT to undertake a visit to the proposed area and to draft a report with both recommendations and cautions. NRT cannot support infinite number of conservancies, as it requires secured funding (for a minimum number of years), so the assessment and feasibility study provides guidance on whether the application can or cannot be supported. If an application is successful, the formal process of membership starts - led by the C of E. If the application is declined - the reasons for this are clearly outlined, and feedback is given to the community and/or communities in question.

NRT member conservancies are statutory governance entities and as outlined above, conservancies are not formed by NRT but rather requested to NRT from communities. NRT solely facilitates the process of conservancy application and membership. All NRT conservancies involved in this project were formed prior to the project start date and without the initial intent of participating in the project.



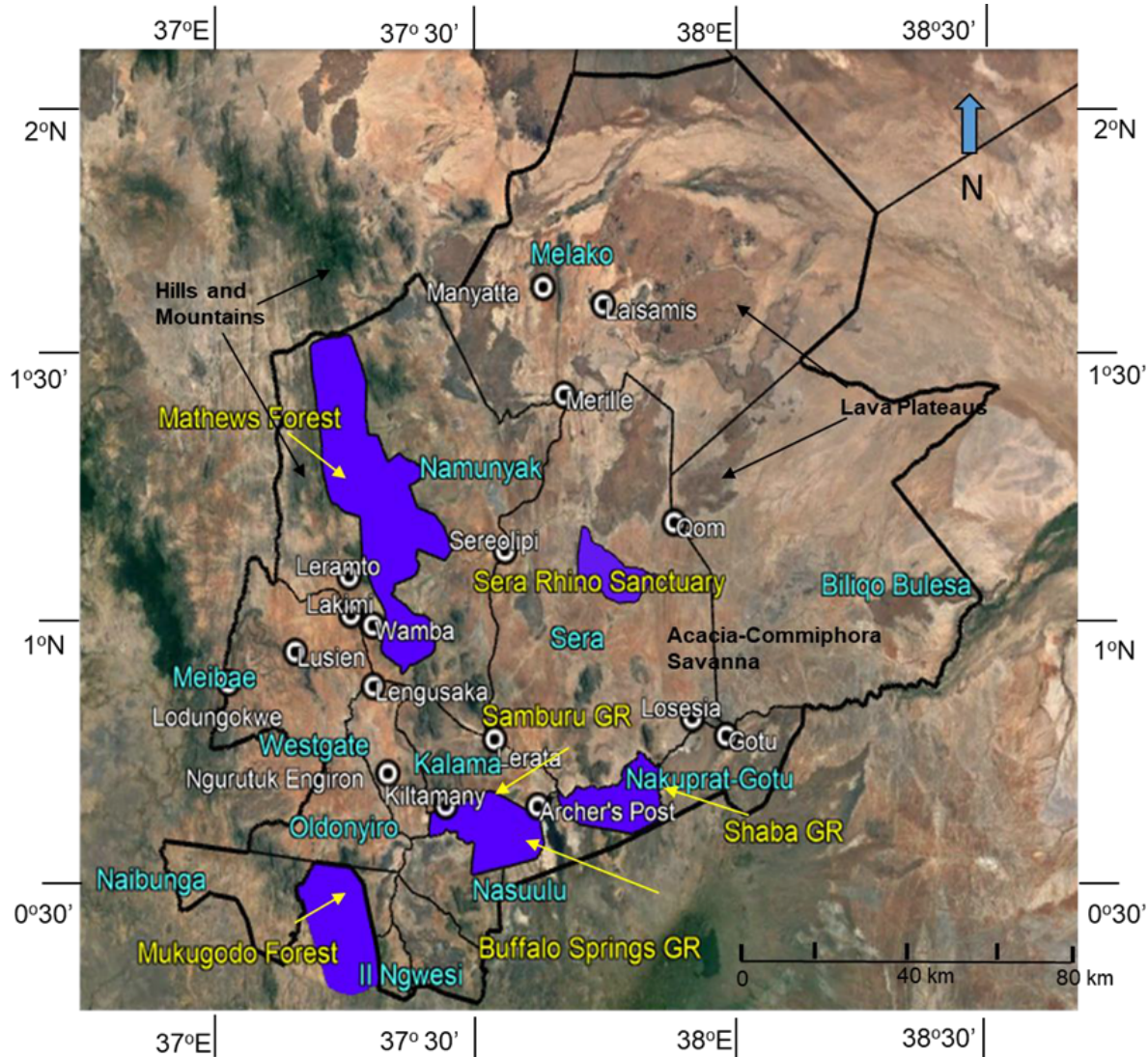


Figure 1 (Above). Larger scale view of the project zone (yellow Conservancies and purple biodiversity HCV areas). Regions surrounding the project zone (pink) include peoples that are likely to receive excursions of livestock from the project area or to be the source of encroaching livestock into the project area. However, livestock migrations can come from as far away as Somalia and Ethiopia, so the border area of influence on project activities and implementation is difficult to define.

(Below) Map of project zone (heavy black boundary) including the 13 participating Conservancies (black boundaries, blue text) in which project activities will occur (see Table 1 below). Principal communities are indicated by targets (white text) and Biodiversity HCV areas are indicated in purple (yellow text, arrows). Community HCV areas encompass the entire project zone since all these lands are used by pastoral community members to conduct their traditional lifestyle and

maintain their culture. Physiographic features evident in the background satellite (Landsat 8) image include rocky hills and mountains (dark patches), lava plateaus with “black cotton” clay soils (dark-gray brownish patches), black clay soils created by fluvial processes (swamps and deltas) and Acacia-Commiphora savannah (light reddish areas that comprise the majority of the landscape).

2.1.8 Title and Reference of Methodology

The methodology used in this project is VCS VM0032: Methodology for the Adoption of Sustainable Grasslands through Adjustment of Fire and Grazing v1.0¹²

The tools used in this project include those listed under VCS VM0032 (v1.0):

- CDM A/R methodological tool: *Calculation of the number of sample plots for measurements within A/R CDM project activities*
- VMD0016: *Methods for stratification of the project area (X-STR)*
- VT0001: *Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities*
- VMD0040: *Leakage from Displacement of Grazing Activities*

The tool used in the current monitoring period include the *VCS AFOLU Non-Permanence Risk Tool (v4.0)*.

2.1.9 Other Programs (G5.9)

The project is involved grassland management and therefore the credits are ineligible under existing Emissions Trading programs, such as the California Air Resources Board or the European Trading System.

The VCS is the only standard that allows carbon credits under agricultural land management to adjust grazing. The project will not pursue other forms of environmental credit.

The project has not applied for or been rejected by any other greenhouse gas accrediting program.

Double –counting on the voluntary market is avoided by having credits assigned on the VCS Registry, which assigns each credit a unique serial number which is held in retirement when the credit is used to offset greenhouse gas emissions.

2.1.10 Sustainable Development

From Kenya Millennium Development Goals¹³ the project specifically addresses five of eight stated goals

¹²

<http://www.v-c-s.org/methodologies/alm-adoption-sustainable-grassland-management-through-adjustment-fire-and-grazing>

¹³ <https://www.indexmundi.com/kenya/millennium-development-goals.html>

1. Eradicate extreme poverty and hunger. The project will route carbon revenues through Conservancies to address needs of the poorest 25% through bursaries. In addition, improvement of forage production should increase the wealth of even small livestock holders.
2. Promote gender equality and empower women. The project actively involves women in grazing management decisions and planning, has and will actively hire women in project employment, and improves sustainable livelihoods and access to water from rangeland improvement.
3. Reduce child mortality. Community revenues will at least be partially allocated to improving healthcare facilities. In addition, improved forage production will allow production of milk and meat even in the dry season and will thus improve child nutrition.
4. Improve maternal health. For reasons similar to items 1 and 3, the project will improve nutrition and health of women, particularly childbearing and lactating mothers.
5. Ensure environmental sustainability. Improvement of soils, forage production and water availability from project grazing management activities provides adaptation to climate change, allows production of meat and milk through dry periods and drought, and bolsters alternative community revenues through increases in wildlife populations and ecotourism revenues.

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G1.9)

The implementation of the project features three main components: Planning and monitoring grazing activities and impacts, monitoring biodiversity, and monitoring community benefits and/or costs. Together with periodic re-sampling of soil carbon to verify the SNAP soil carbon model on which removal calculations are based, the following schedule has been implemented:

Table 1. Key dates and milestones in the project's development and implementation.

Date	Milestone(s) in the project's development and implementation
1 Feb 2011	Project baseline soil data collected
10 Dec 2012	Project start date, MOU signed between NRT and TNC
1 Jan 2013	Implementation of planned grazing for first three conservancies (Westgate, Il Ngwesi, Kalama) and crediting period start date
1 Nov 2013	Implementation of planned grazing for six additional conservancies (Sera, Meibae, Namunyak, Oldonyiro, Biliqo Bulesa)
1 March 2014	Formation of Nasuulu, Leparua, Nakuprat-Gotu conservancies
1 June 2014	Implementation of planned grazing for Melako conservancy
1 July 2014	Baseline biodiversity monitoring in Kalama, Westgate, and Sera conservancies

1 Feb 2015	Baseline community monitoring in Il Ngwesi, Kalama, Nakuprat-Gotu, and Melako conservancies
1 March 2015	Validation and verification begin
15 May 2015	Initial conservancy solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff in the form of pamphlets, and an announcement on the NRT website (http://nrt-kenya.org). All responses to the proposed carbon project during this round were made by cell phone text to a standing repository, which was then downloaded for review and archiving.
1 June 2015	Implementation of planned grazing in Nasuulu, Leparua, Nakpurat-Gotu, and Naibunga conservancies
15 Sep 2015	VVB conducts first site review
1 Oct 2015	Project-wide grazing planning process and grazing implementation report
1 May 2016	SocialCoMMs monitoring in Westgate, Sera, Meibae, Namunyak, Oldonyiro, Biliqo Bulesa, Melako conservancies
15 Aug 2016	Second round of conservancy solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff. All responses to the proposed carbon project during this round were made by cell phone text to a standing repository, which was then downloaded for review and archiving.
Sept-Oct 2016	2015 and 2016 announcements rounds followed by community meetings, open to all members in each Conservancy, in each of the 13 participating Conservancies. Meeting times and dates were posted at all Conservancy headquarters and were communicated verbally in villages in each Conservancy by NRT SocialCoMMs supervisors.
1 Oct 2016	Expanded baseline soil carbon sampling in Melako, Biliqo Bulesa, Sera, Namunyak, and Meibae conservancies
1 Oct 2016	Project-wide grazing planning process and grazing implementation report
7 Nov 2016	VVB conducts second site review
1 Oct 2017	Project-wide grazing planning process and grazing implementation report
15 April 2018	VVB conducts third site review
1 Oct 2019	Project-wide grazing planning process and grazing implementation report
29 April 2020	Validation finalized by VVB and Verra
1 May 2020	Biodiversity Monitoring in Kalama, Westgate, Sera, Melako, Meibae
1 May 2020	SocialCOMMs Monitoring - 6 Conservancies, randomly selected

1 Oct 2020	Project-wide Grazing Planning Process and Grazing Implementation Report - All 13 Conservancies
2017-2020	At varying times, Annual Governance Meetings (AGM) were held with representatives from each project conservancy. Carbon project updates were discussed as part of each AGM's agenda.
28-30 March 2022	Soil carbon sampling 2 for 2021 Verification, VCS and CCB site visit by VVB
December 2022	2017-2020 VCS Verification Complete, finalized by VVB and Verra
July-August 2023	Anticipated 2017-2020 CCB Verification Completion, finalized by VVB and Verra
August 2023	Anticipated initiation of 2021 Verification by VVB

2.2.2 Methodology Deviations

There are two methodology deviations during this monitoring period.

1. The parameter $ED_{c,t}$ (number of days spent by encroaching [off-project] livestock of category c on the project area during year t) was monitored during this monitoring period. Monitoring of this parameter is not required by VM0032, but it was monitored during this monitoring period (per the request of Verra) to show that Applicability Condition 3 of VM0032 was met by showing that encroaching animal days is de minimis relative to total animal-days for project animals. See the second PD Deviation listed below (Section 2.2.4) for further details.
2. The methods used to measure parameter $PN_{c,t}$ (number of project animals of category c in the project area during year t) have changed from those presented in Section 3.3.2 of the PD. This parameter continues to be calculated using trends extrapolated from 2014 conservancy census data. These means for parameter $PN_{c,t}$ were assumed to be similar to 2014 in 2017-2020. This method was used because many Conservancies were highly inconsistent in reporting actual livestock numbers, even in months where all livestock were reported to be present on the Conservancy, or simply did not report actual livestock numbers. This method is more reliable than the previous estimation method presented in the PD and does not affect the applicability of the methodology. See Section 3.1.2 of this Monitoring Report. Additionally, the VVB has reviewed the 'Households Over Time' tab of the spreadsheet 'NKCP Methane Calculations 2017-2020 v1 Sept2021.xlsx' for further details.
3. The methods used to measure parameter $DN_{c,x}$ (number of project animals of type c off the project area during month x) have been updated from those presented in Section 3.3.2 of the PD. The update (Section 3.1.2 of this MR) clarifies how monthly grazing reports from each conservancy are used to determine the parameter value for the quantification of leakage emissions. Three reports for the former monitoring period were either missing or incomplete (Biliqo Bulesa, April 2014, Namunyak, Nov 2014 and Dec 2015). In each instance, leakage estimates for neighboring months were evaluated and the month with the highest reported leakage was used to substitute for the missing report.

This deviation will continue to be applied until additional monitoring measures can be implemented to record leakage without the use of the current “penalty” approach.

2.2.3 Minor Changes to Project Description (Rules 3.5.6)

There is one minor change to the Project Description. In the Project Description, it is stated that for biodiversity monitoring, “Surveys of key indicator species, sampling in savanna/grassland habitats will take place every 3 years in five permanent biodiversity monitoring platforms, or ‘camps.’” Baseline surveys were conducted in 2014 and the first project surveys were conducted in 2021. A survey did not take place in 2017 as initially stated in the Project Description. The justification for this change in surveying cadence is that the project did not generate VERs (from 2013-2016) to be sold until 2021, and therefore the project did not have revenue to pay for the biodiversity sampling.

2.2.4 Project Description Deviations (Rules 3.5.7 – 3.5.10)

There are six on-going project description deviations during this monitoring period:

1. Section 2.1.15 of the PD lists the project crediting period as beginning on 10 December 2012. For the first verification, the monitoring period was moved from to 01 January 2013. Project monitoring period activities were not fully implemented until 01 January 2013.
2. While Section 2.1.9 of the PD identifies project stakeholders based on occupation, socioeconomic status, and role in the community, Section 4.2.1 of the PD proposes to monitor impacts on the basis of ethnic group. During the first monitoring period, we identified stakeholders based on occupation, socioeconomic status, and role in the community and monitored impacts only on those stakeholder groups (Section 4.1.1 of this report).
3. The parameter $ED_{c,t}$ (number of days spent by encroaching [off-project] livestock of category c on the project area during year t) was not mentioned in the PD but has been added to Section 3.1.2 of this Monitoring Report. This parameter is used along with an approach modified from the “Tool for testing significance of GHG emissions in A/R CDM project activities” to show that project emissions from encroaching livestock for the current monitoring period are de-minimis (<5% of project methane emissions) in each year of the monitoring period and can therefore be excluded from the final calculation of net emissions and removals. This demonstration was reviewed by the VVB in the ‘NRT Methane Encroachment’ tab of the spreadsheet ‘NKCP Methane Calculations 2017-2020 v1 Sept2021.xlsx’.
4. The project area has decreased by 3 ha due to technological improvements in analyzing conservancy boundaries. The change from 1,993,078 ha to 1,993,075 ha is reflected in all calculation files and was reviewed by the VVB during the 2017-2020 verification event.

For deviations 5.) & 6.), please reference Appendix D for summary and full methodological details for the assessment of successful pastoralist grazing plans.

For the 2017–2020 monitoring period, there were two deviations (5 & 6) to the remote sensing method as compared to the 2013–2016 Monitoring Report. First (deviation 5), an adjusted set of

ungrazed reference polygons is used for calculating annual relative change in biomass. This is justified due to conservancies with lower mean annual rainfall (relative to wetter reference polygons) exhibiting high relative changes in NDVI and thus being judged in error as unsuccessful. Secondly (deviation 6), the threshold value of annual relative change in NDVI is adjusted from 0.05 to 0.1. This is justified because a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry season grazing areas.

5. Justification for Adjusted sets of reference polygons for calculating successful implementation of grazing plans during 2017-2020.

The method of assessing annual grazing plan implementation depends on comparing annual relative changes in NDVI, $SUM\Delta NDVI_{rel}$, determined by summing monthly $\Delta NDVI_m$ across the twenty-four 16-day intervals for each year, $SUM\Delta NDVI_m$ (see Eq. 2 in Appendix D) at each MODIS pixel, as compared with mean annual sums of $\Delta NDVI_m$ across the same time intervals ($meanSUM\Delta NDVI_{REF}$) from reference polygons that are not grazed or only slightly grazed by livestock (see Eq. 2 in Appendix D). These reference locations are embodied in multiple polygons that span the project area, including “Core” areas of Il Ngwesi and Kalama Conservancies, the tops of plateaus in

Kalama Conservancy that are relatively inaccessible to livestock, and a large ~1,000 ha fenced rhino sanctuary in Sera Conservancy. In most years, the mean annual relative change in NDVI ($meanSUM\Delta NDVI_{REF}$) in these reference areas is relatively similar despite differences in mean annual rainfall. Consequently, the Google Earth Engine (GEE) calculations of annual relative change in NDVI ($SUM\Delta NDVI_{rel}$) for individual pixels represent a good estimate of livestock impacts in drought years (2017, 2018). However, in wet years following droughts, such as occurred in 2019 relative to 2018, the $meanSUM\Delta NDVI_{REF}$ in Il Ngwesi Core area and Kalama Plateaus are much bigger than the values for Kalama Core and Sera Rhino Sanctuary reference polygons. Averaging in these higher $meanSUM\Delta NDVI_{REF}$ from those reference polygons virtually guarantees that conservancies with lower mean annual rainfall will exhibit frequent pixels with high $SUM\Delta NDVI_{rel}$ and thus be judged in error as unsuccessful. Therefore, reference polygons appropriate to the mean annual rainfall of the different Conservancies should be used for each year (2017-2020). Reference polygon sets are broken into higher rainfall (wetter) and lower rainfall (drier) conservancies:

Reference set 1 is to be used with the higher rainfall conservancies – Il Ngwesi, Naibunga, Meibae, Nanapishu (Oldonyiro), Leparua, Nasuulu, Namunyak

Reference set 2 is to be used with the lower rainfall conservancies – Kalama, Westgate, Sera, Biliqo Bulesa, Melako, Nakuprat-Gotu

Thus, we calculated relative change in NDVI, $SUM\Delta NDVI_{rel}$ (Eq. 1- see Appendix D) for the drier Conservancies Melako, Sera, Namunyak, Biliqo Bulesa, Nakuprat-Gotu, Nasuulu, Kalama, Westgate, and Leparua Conservancies using only the reference polygons from Kalama Core and the Sera Rhino Conservancy. Calculations for wetter Conservancies Il Ngwesi, Naibunga, Meibae, and Oldonyiro include all the reference polygons, as is coded in the current GEE model.

6. Justification for a Shift in the Threshold Value of Annual Relative Δ NDVI

The threshold for unsuccessful implementation was obtained from two logistic regressions, which indicate the estimated Δ NDVI values that correspond to whether substantial livestock were present in a grazing block (Fig. 3- see Appendix D), and whether substantial livestock were present for more than one two-week period during the year (unsuccessful implementation).

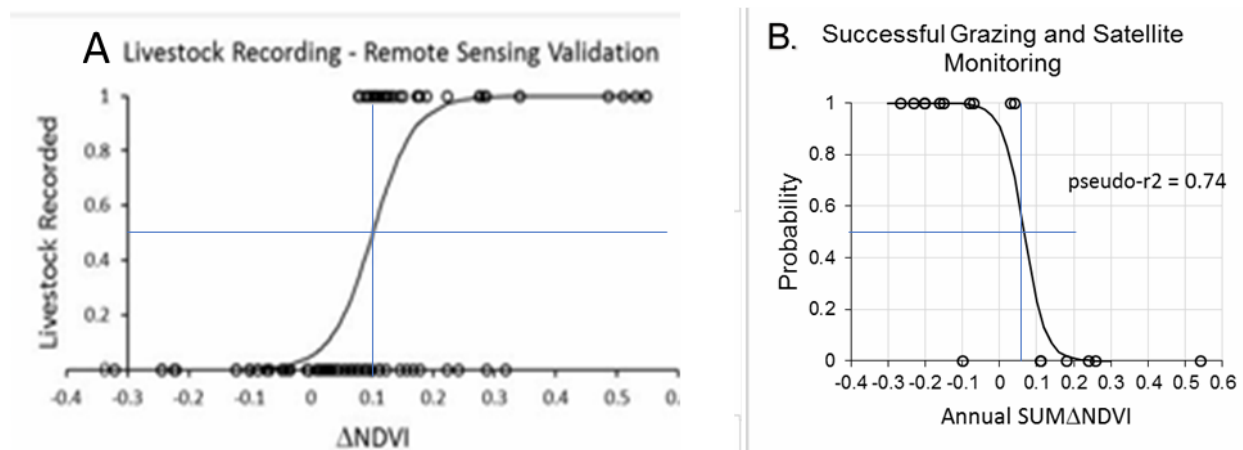


Fig. 1. Logistic regressions to assess successful implementation of grazing plans in the Northern Kenya Grasslands Carbon Project. Fig. 1.A. The logistic regression of livestock presence (livestock recorded = 1) and absence (livestock recorded = 0) versus calculated **monthly relative Δ NDVI** (Δ NDVI_{m,rel}, Eq. 1- see Appendix D), the value of monthly relative Δ NDVI at which there was greater than a 50% probability (livestock recorded > 0.5) of recording substantial livestock present is 0.1 (intersection of two perpendicular lines). This logistic regression was previously shown in Fig. 3C (see Appendix D). Also, livestock were only rarely encountered in map locations where Δ NDVI_{m,rel} < 0.1. Fig. 1.B. At 17 grazing blocks where presence/absence of livestock was monitored over an entire year, a logistic regression of the occurrence of only a single grazing event (probability = 1) or more than one grazing event (probability = 0) versus the relative annual sum of Δ NDVI for the 2014 calendar year (**annual relative Δ NDVI**, SUM Δ NDVI_{rel} from Eq. 2 - see Appendix D). This logistic regression was previously shown in Fig. 4B- see Appendix D.

The threshold of 50% probability of unsuccessful implementation occurs at approximately annual relative Δ NDVI = 0.08, but, due to the small sample size of grazing blocks, there is a large confidence interval around that estimate. Since monthly relative Δ NDVI (Δ NDVI_{m,rel}) must be > 0.1 for livestock to be detected at a location then it is reasonable that the annual sum, or annual relative Δ NDVI (SUM Δ NDVI_{rel}) should be > 0.1. This is especially true for dry season grazing areas because they do not regrow immediately after grazing. Since it requires a relative change in NDVI of 0.1 to detect a grazing event (Fig. 1.A.), it is difficult for dry season grazing areas to show a relative delta NDVI < 0.1 even if they are grazed only once. Thus using a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry season grazing areas.

This justifies a threshold of annual relative Δ NDVI (SUM Δ NDVI_{rel} = 0.1) for assessing whether grazing plans were successfully implemented at a location. Thus, where pixels have SUM Δ NDVI_{rel} ≤ 0.1, these areas are deemed to be successfully grazed in each year

2017-2020. Pixels with SUMΔNDVIrel > 0.1, are deemed to be unsuccessfully grazed in each year 2017-2020.

2.2.5 Grouped Projects

This section is not applicable, as the project is not a grouped project.

2.2.6 Risks to the Project (G1.10)

Short-term risk for this project is the failure to implement planned rotational grazing. Such failure is most likely if drought is severe, which would cause large-scale movement of livestock outside the project area and other grazing management decisions that might lead to over-use of rangelands and an incremental reversal of accumulated soil carbon stocks.

The principal mitigation strategy for such risk is to:

1. Rapidly build grass banks, particularly in extensive unsettled areas typically used only during the dry season, that can be used as forage during droughts.
2. The accumulation of soil carbon will lead to greater infiltration of rainfall, which will make communities progressively less vulnerable to climate variability and drought as the project proceeds.
3. Further build the livestock-to-market program instituted by NRT to provide cash compensation for de-stocking in advance of intense droughts, where droughts can be anticipated by using early warning systems such as the USDA FEWS network <http://www.fews.net/>.
4. Through education associated with monitoring social benefits of the project, continue to convey the community and personal revenue advantages linked to continued participation in the project.
5. Convey the risks of encroachment from herders from off-project communities, and train and implement strategies to handle encroachment and thus reduce risk. For example, game scouts are trained to deal with and communicate about poachers and encroaching livestock herders.¹⁴
6. Progressively entrain sustainable livestock management processes as a new “tradition” of planned rotational grazing.
7. Continue to demonstrate the indirect positive influence of rangeland vegetation and soil recovery on biodiversity and the concomitant benefits of biodiversity to people.

Longer-term risks include persistence of community willingness to participate in the project, and risks related to the ability to adapt to climate change and climate variability, etc. Community willingness to participate will depend on several factors. Lack of land tenure can discourage participation, but the Kenya Community Lands Act of 2016 creates drivers to encourage people to remain and care for land and to invest in management and governance strategies that yield sustainable productivity from it. Rotational grazing management can be challenging to govern, but with carbon revenues to support governance, it should be able to continue long enough for

¹⁴

https://static1.squarespace.com/static/5af1629f12b13f5ce97ca0b5/t/5b6330d9575d1f66316bf5f0/1533227321252/Wildlife-CoMMS+guide_complete_low_res.pdf

community members to see the returns of implementation in the form of improved forage, greater surface water distribution, development of livestock value chains, etc.

To that end, the objective of the project to meet CCB climate, biodiversity and community standards, and the need for auditors to gather social data, was communicated to community members prior to both the 2015 and 2017 site visits during public comment periods, as well as a February-March 2019 public comment period. In addition, the specific objectives of the standards and the need for the social community managed monitoring program (Social CoMMs) was communicated to Conservancy chairpersons and grazing coordinators from all 13 Conservancies during the September 2017 workshop.

Climate change represents an external risk that has the potential to cause irreversible degradation of the soil and grass resources if current management strategies of unrestricted grazing are pursued. However, again, the farther into the project the communities get the better buffered and more management options people will have (see list above) to deal with increased frequencies of droughts and more frequent incursions of peoples seeking livestock forage^[31].

Long-term benefits beyond the project timeline are the ultimate goal of the project. Restoration of perennial grasses and re-building of soil carbon should allow community members the opportunity for sustainable livelihoods from pastoralism and to dramatically improve livelihoods and well-being from livestock husbandry. These benefits would not depend on carbon revenues, as the capacity to sustain livestock should at least double from its current value, and wildlife and tourism opportunities should be higher than current. As highlighted in NRT's Strategic Plan 2018-2022, Conservancy governance structures put in place with carbon revenues to support and manage grazing within Conservancies will be funded beyond the project lifetime by interest from conservation trust funds built from carbon revenues. Newly established regional grazing plans and drought mitigation strategies will be coordinated by county governments and be funded by devolved national tax revenues rather than by carbon. Risks of encroachment from herders from outside the project area may continue. However, regional governance, which includes local elders and women trained in peace-making, sharing of forage agreements among Conservancies and mobile security teams drawn from across ethnic groups and managed at the County level will be well established and capable of resolving conflicts quickly (see Mid-Year Technical Report on NRT Operational Plan 2018, as supporting document).

2.2.7 Benefit Permanence (G1.11)

The project is part of a larger effort by NRT, TNC, and numerous donors, such as USAID and DANIDA (US and Danish international aid agencies) to restore rangelands and improve human well-being in the northern Kenya landscape. The project activity, improved grazing management, is at the core of educational efforts in the 13 Conservancies aimed at making long-term livelihood and land management changes that improve well-being. These changes have occurred at the Conservancy administrative and public decision-making level and are designed to help communities conduct sustainable grazing practices over a 30-50 year time horizon beyond the project lifetime. Incentives to continue these grazing practices include healthier and more valuable livestock, improved forage production, surplus production that can be used as grass banks against drought, improved small stock and milk production during dry seasons, and lower physical risks when migrating to dry season ranges with livestock. These incentives act as barriers against the return to continuous grazing near settlements that generated past degradation.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

Meetings were held to review the project's performance and status with stakeholders in April 2014, March 2016, and September 2017. During these meetings, the stakeholders were notified where further documentation could be found and provided the documentation during the meeting. The documentation can be found at NRT headquarters, each participating Conservancy headquarters, and posted at public kiosks in the major towns of Isiolo, Archer's Post, Sereolipi Wamba, and Merille. This process was also repeated in 2021 for the dissemination of the translated monitoring report summary for the 2013-2016 monitoring period. This will be the case for the 2017-2020 monitoring period once the monitoring report is approved and finalized.

2.3.2 Dissemination of Summary Project Documents (G3.1)

Given the expanse of the project, a summary of project documents was provided at multiple locations throughout the project area, including NRT and participating conservancy headquarters during three different public comment periods in September 2015, April – May 2017, and April – May 2018. Stakeholders were informed of where the documents could be found as hard copies and online through stakeholder meetings and postings on the NRT website, conservancy headquarters, and at kiosks in the major towns of Isiolo, Archer's Post, Sereolipi Wamba, and Merille. As was the case for 2013-2016, project documentation will be posted at conservancy HQ. This will include a 2017-2020 monitoring report summary translated in Kiswahili. Additionally, links will be provided to project documentation posted on the NRT website in both English and Kiswahili.

2.3.3 Informational Meetings with Stakeholders (G3.1)

The potential impacts of the project on the communities were evaluated in a number of ways. First, our focus has been on educating people about the project activity, which is planned rotational grazing. The education process began with workshops conducted by Grevy's Zebra Trust and Craig Leggett, then of Holistic Management International, first in Westgate and then in Kalama and Il Ngwesi Conservancies in 2009 – 2011. Workshops and advertised public meetings to discuss the need for rotational grazing and restoration of grazing lands were conducted at numerous venues throughout the project area in 2012, including Wamba (in Namunyak) and Isiolo. The average number of attendees at meetings ranged from 15 in 2009 to over 100 by 2012. In 2013 and 2014, as new conservancies (Melako, Leparua, Nasuulu, and Nakuprat-Gotu) were added to the NRT group, administrative training by NRT staff included training on planned rotational grazing. Finally, the advent of the livestock to market program in 2010 led to links between implementing planned rotational grazing and conservancy cattle being chosen for sale in the program.

In a given year, numerous opportunities are provided for stakeholders to engage in informational meetings and events. Each conservancy hosts an Annual Governance Meetings (AGMs) with the purpose of conservancy leaders providing conservancy and community updates, receiving questions and feedback and align on goals for the upcoming year. Special AGMs, held at village or zonal level across the conservancies, were utilized in 2021 to brief the wider community and ensure that there was general consensus within the community to continue participating in the project. During a given year between 2017-2020, each of the following meetings were held, at

least once, within or for each project conservancy: Annual General Meetings (AGMs), Grazing Awareness Meetings (educational visits to other project conservancies), Conservancy Board Meetings, Council of Elders Meetings, Grazing Committee Trainings, Grazing Committee Meetings (monthly), Regional Grazing Meetings (multiple times per year), Grazing Planning (Wet Season) Meetings (quarterly or bi-annually).

2.3.4 Community Costs, Risks, and Benefits (G3.2)

The project involves community members in planning and management operations in multiple ways, and includes efforts to involve **vulnerable and marginalized groups**, notably women and the poorest 25% of households, most of which lack livestock. Several different indigenous groups occupy the project zone (Figure 1), and the risks and values to “the community” are fully applicable to each group. The success of the project depends on the joint cooperation of the participating communities and the financial and livestock management benefits of the project apply equally to all the Conservancies.

Communities within the conservancies that are a part of the Northern Kenya Rangelands Carbon Project have been engaged since the inception of the project. Methods of engagement have ranged from education and support for changing grazing behavior to preserve grasslands, to participation in informative carbon discussions, and the signing of the agreements which have provided the permission for the sale of the credits.

At the outset, the engagement was primarily based around the improvement of soils and grasslands through sharing rotational grazing techniques. As new grazing techniques were successfully implemented across the conservancies, NRT, TNC, Soils for the Future and Grevy's Zebra Trust introduced to communities the potential for enhanced carbon sequestration through improved grazing management. The engagement with the communities on the benefits of these new grazing methods shifted to a conversation about the potential for the conservancies to generate revenue from soil carbon. In 2015, NRT held a series of public meetings and announcements in the 14 conservancies (in local languages) describing the carbon project, its potential to create revenue, how it linked to the grazing programs, and what the costs and benefits of embarking on this project would be. After these meetings each of the participating conservancies, through their democratically elected Boards, signed Letters of Intent to participate in the carbon project. In September of that year, the first VVB site visit commenced where just under 200 community members were interviewed either in focus groups or in individual households, and public meetings were held in six different towns across the project area to assess and promote project awareness.

In September 2017, NRT hosted a workshop whereby all the details were laid out of how the project would run, its key activities, and how its success and benefits depended on good governance and adhering to the grazing plans. Issues of leakage, encroachment, and revenue sharing were discussed with the members of conservancy leadership and rangeland coordinators. This was well received and NRT, TNC and Soils for the Future, acting on behalf of the conservancies, continued to engage with VCS as they pursued further auditing work to support the validation of the proposed project methodology. A second VVB site visit at this time and a further 50-100 community members were interviewed by the visiting team and public meetings were held at a further (and different to the previous) three locations in the project area.

Between 2017 and 2018 participating conservancies formally agreed for NRT to represent their carbon rights for the project.

In 2019, public notices were made at every conservancy HQ and in every major settlement area in the project area to publicly announce the project and provide a summary of the project monitoring report to that point.

As leadership in the conservancy changes every three years, through the conservancy board elections, NRT felt it was critical to host another briefing to ensure that current leadership was in a position to fully understand the project and for communities to decide whether they were still interested in pursuing the sale of carbon credits through their conservancies. Thus, in May and June 2021, NRT undertook a wide spread briefing and socialization of the project for all participating conservancies and their member communities. NRT presented a revised and renewed 'Project Implementation Agreement'* to set out a clear understanding between NRT and participating conservancies as the project launches the first issue of carbon credits. As part of the community consultation process NRT held briefings on the project for each conservancy's board chair, manager, grazing committee chair, and rangelands coordinator. These members then briefed their Boards and Grazing committees, and following that, 'Special Annual Governance Meetings' were held at village or zonal level across the conservancies, to brief the wider community and ensure that there was general consensus within the community to continue participating in the project. These meetings were extensively documented both through written minutes and visually recorded through photographs. Based on a strong consensus, all communities approved a resolution for the conservancy chairperson to sign the Project Implementation Agreement on behalf of their community. The 14 chairpersons met on 24 June 2021 and formally signed the Project Implementation Agreement. Since this time, the Carbon Project Oversight Committee (CPOC) formed by representatives of each of the participating conservancies, has met at least every three months to oversee the management of the project and full minutes of these meetings are shared with each participating conservancy.

It is important to note that NRT as the project proponent fully recognises that free, prior and informed consent for communities is not something that is delivered on a one-off basis and is something that must be continually invested in by the project on behalf of the participating communities throughout the project lifecycle. NRT seeks to continually make improvements to this process and seeks to encourage conservancies to support best practices for free, prior and informed consent in all communications with community members, including giving appropriate notice for public meetings and endeavouring to seek representation / endorsement from those members of their pastoralist communities who may be far from home at the time when such public meetings are held.

**The Project Implementation Agreement outlines both the project activities involved and provides detail outlining the compliance and reporting requirements of the project. It provides a schedule of the reporting requirements a conservancy agrees to partake in and it includes detail on access for project monitoring, the validation process, verification, registration, records and other reporting relating to CCB Gold Standard, and sets out the benefit sharing mechanism for revenue.*

2.3.5 Information to Stakeholder on Verification Process (G3.3)

Within-project stakeholders were identified by monthly community meetings held by the Conservancy managers, bi-annual meetings of grazing chairmen and the NRT grazing coordinators with herders, workshops held in villages and at Conservancy headquarters.

Off-project stakeholders were identified from annual County public meetings in the towns of Baragoi (Samburu County), Isiolo (Isiolo County), Laisamis (Marsabit County), and Nanyuki (Laikipia County), from individual interviews with government representatives, and from oral reports of Conservancy chairman who negotiate grazing rights for herders from off-project communities

Meetings were advertised by email, calls and texts on networks of personal cell phones, and posting on the NRT website. Some meetings were announced with public postings of flyers and in public meetings within communities prior to project stakeholder meetings.

In February 2022, prior to the 2017-2020 verification auditor site visit, documentation was sent and translated to inform stakeholders of the upcoming March 2022 site visit.

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

To that end, the objective of the project to meet CCB climate, biodiversity and community standards, and the need for auditors to gather social data, was communicated to community members prior to both the 2015 and 2017 site visits during public comment periods, as well as a February-March 2019 public comment period. In addition, the specific objectives of the standards and the need for the social community managed monitoring program (Social CoMMs) was communicated to Conservancy chairpersons and grazing coordinators from all 13 Conservancies during the September 2017 workshop. As mentioned in the section above, in February 2022, prior to the 2017-2020 verification auditor site visit, documentation was sent and translated to inform stakeholders of the upcoming March 2022 site visit.

2.3.7 Stakeholder Consultation (G3.4)

Stakeholder consultation within the project area is multi-layered; it happens at multiple levels to ensure points raised from the management team, or points raised from a herder and/or community member, are known and shared with all entities involved.

Changes since the 2013-2016 monitoring period revolve around the fact that the landscape has become much more coordinated - regional meetings and dialogue between conservancies and communities on the move has improved significantly in the past few years. The project has also focused heavily on building capacity at the village level to ensure understanding and ownership at that level. There is also the development of county government rangelands policies which ensure local ownership but with support from the authorities when it comes to imposing by-laws.

Grazing planning continues to be undertaken quarterly - it is based around seasonal plans (wet vs dry). The planning starts at the village / zonal level where herders sit together to develop their local plans - these are then communicated at a conservancy level through the conservancy grazing committee (representatives from each zone) to ensure the zonal plans are coordinated within the greater conservancy plan. The coordinated conservancy plan is then relayed back to the zonal level (through the grazing committee) to confirm the plans with the herders who will be

the ultimate enforcers- as in, herders are the ones who either comply or go against their own plans and it is the herders who decide when penalties / by-laws should be imposed.

The conservancy grazing plan is then relayed at a regional level between neighboring conservancies to coordinate the larger landscape - and also outside of the project area where applicable. Herders and communities continue to influence the project implementation as they are the "users" of the grazing plans and all plans evolve from provided input at the village / zonal level. As mentioned above, these grazing plans are self generated and supported plans which originate from the herders and communities and continue to be critical for project implementation.

Every meeting held will continue to be minuted and filed for record keeping and future VVB review.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

During the first monitoring period, dedicated personnel positions were created within NRT and the Conservancies to continuously monitor herder activity, detect encroachment from off the project area, and increase communication between Conservancy leaders and grazing coordinators and herders. Efforts have continued to build capacity to ensure greater effectiveness in these areas through ongoing training and structured mentoring from the rangelands department within NRT as well as better coordinate regional level planning ensuring project activity success.

During the first monitoring period, dedicated personnel positions were created within NRT and the Conservancies to continuously monitor herder activity, detect encroachment from off the project area, and increase communication between Conservancy leaders and grazing coordinators and herders. In addition, the project plans to devote some project revenues to establishing livestock finishing infrastructure (protected pastures and fencing, transportation hubs, etc.) to help shift community values from the number of livestock to the quality of livestock. Over time, this adaptation should reduce grazing pressure and increase the ability to reserve forage as grass banks and increase revenues and improve livelihoods of community members, including women and the poorest 25%.

Information about project activities continues to be disseminated from conservancy leaders and grazing coordinators throughout the local communities through established communication channels. Conservancies are owned and run by the communities that they represent; as such the governance structure within a conservancy focuses on the representation of the communities that live within the conservancy boundaries. Conservancies are led by an elected Board typically 12-14 members strong. Each of these Board members represent a village or zone (cluster of small villages) within a conservancy. A Chairperson is elected from within the Board and he / she represents the conservancy within the broader NRT Council of Elders. Once appointed, Board members undergo Leadership and Management Planning (LAMP) training and are tasked to not just represent the voices of the members of community they were elected by, but also to convey information concerning the management of the conservancy back to their communities. This is a critical step in the "communication loop".

A similar structure is applied to each conservancy's Grazing Committee, where each village or zone elects a representative, but instead of LAMP training, the committee undertakes more specific training on grazing management and the implementation of rotational grazing (project activity focussed) for the project which leads to improved rangeland management and subsequent increased sequestration of carbon.

Outside of both the Board and Grazing Committees, the Conservancy Manager and the Rangeland Coordinator (who are appointed by the Board and are formal conservancy employees) also provide channels for information flow to communities, but also back to conservancy management and leadership. Examples of this would be around carbon project updates, grazing coordination with neighboring conservancies, in advance of livestock movement, results from routine surveys (e.g: Veg-CoMMS).

2.3.9 Stakeholder Consultation Channels (G3.5)

As part of baseline social data gathering, community elders, women and young herders (morani) were asked a series of questions in focus groups in three Conservancies, and over 1000 households were interviewed in a household survey of four Conservancies in October 2013. Baseline data were gathered from the most populous ethnic groups.

These social data illustrated community concerns about encroachment by herders from outside the project area and the problem of security in effectively implementing planned grazing. These comments increased project costs devoted to paying grazing assistants, who patrol in designated grazing blocks to inform herders from outside the project area about expectations and rules for grazing within the Conservancy and to otherwise diffuse any conflicts.

Both baseline and project scenario data were collected through NRT Social CoMMs surveys and focus groups. It is important to note Social CoMMs surveys were designed for use throughout all NRT conservancies and are not exclusive to the conservancies participating in the project. Baseline data were collected in 2014 from conservancies that had just joined the project and had not yet implemented grazing plans. Project scenario data were collected in 2015 and 2016 from conservancies that had implemented planned grazing for at least two years. SocialCoMMs is an outgrowth of training of community members to conduct social surveys by professional social scientist Dr. Jessica Musengezi of The Nature Conservancy, supported by permanent NRT staff. Through the guidance of TNC and NRT personnel, focus groups are chosen by community chairmen to represent the public, rather than by the grazing committee. Household surveys are conducted at randomly selected households, and every effort is made to interview at least the matriarch of each household. The results of groups and surveys were (are and will continue to be) incorporated in the management plan for the project as follows

- a. Responses to household surveys collected, with emphasis to include vulnerable and marginalized groups, notably women and the poorest 25%.
- b. Data are analyzed with regards to perceived risks and detriments of project activities.
- c. Actions to mitigate risks or detriments, as discovered from Social CoMMs data are originated and discussed in quarterly carbon project management meetings.
- d. Operating procedures are defined and updated on the basis of suggested actions from management meetings.

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

Extra efforts are made by community leaders to invite women, the poorest 25%, ethnic minorities and other vulnerable groups to within-Conservancy public and grazing planning meetings. These invitations are both general, in the form of public notices or verbal public announcements as well

as personal. Community leaders identify vulnerable households and communicate directly with the heads of these households to emphasize the importance of their participation.

This planning process is now established to operate throughout the project lifetime.

Planned livestock movements include some movements outside the project area. Mainly these movements are transitions through HCV areas such as movements through the Mathews and Mukugodo Forests during wet to dry season transitions. These are kept as brief as possible but are nevertheless counted as leakage.

Cultural division of labor places grazing issues in the purview of men, so including women and households without livestock in the decision process can be challenging. However, focus groups that included women and morani revealed that women influence decisions about livestock sales and the distance and timing of livestock movements within their household, and thus receive information in an additional within-household communication channel beyond step 4. Women heads of household, many of which have significant livestock herds, are also contacted by village elders, Conservancy grazing committee members, NRT grazing coordinators, and NRT Social CoMMs supervisors.

Planning through a hierarchy of grazing decision-makers was implemented in October 2015 and led to the explicit inclusion of grazing plans in overall conservancy management plans for 8 conservancies. Since 2015, conservancy managers have utilized these annual conservancy management plans in order to effectively carry out grazing plans for their respective conservancy. It also extended the decision-making process to all ethnic groups regardless of the conservancy in which they live.

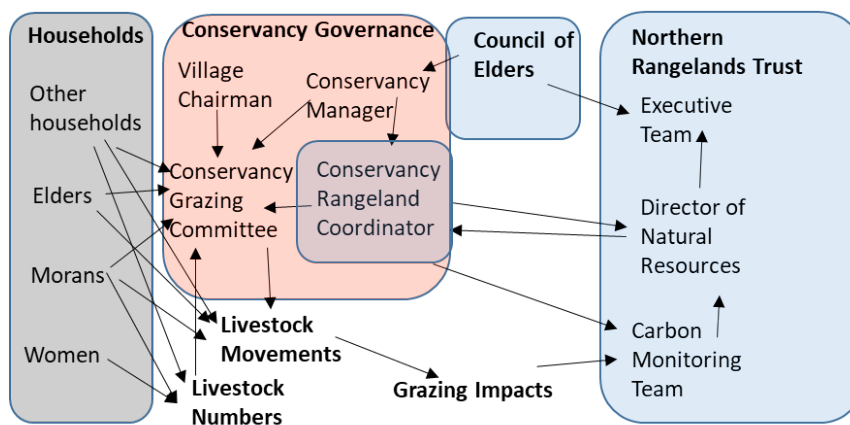


Figure 2. Schematic of how different stakeholders influence grazing management decisions. These stakeholders operate in a hierarchy of organizations from NRT down to the household level and are centered around grazing committees and the Conservancy grazing coordinator. The carbon monitoring team and NRT operate as a key information feedback loop.

2.3.11 Anti-Discrimination Assurance (G3.7)

All NRT and Conservancy endeavors are required by their charter to follow the 2007 Kenya Occupational Safety Act ¹⁵. Compliance is monitored by NRT human resources staff. Field employees are provided with functioning 4WD vehicles with licensed, trained drivers and/or motorcycles for navigating on and off road with helmets and other protective gear. Staff work in pairs for safety and the ability of a partner to transport another to health care providers distributed among different Conservancies. Game scouts are highly trained and provided uniforms, rifles, operating radios, and other appropriate field gear, and also work in teams transported, if necessary by trained and licensed drivers.

Particular attention is placed on avoiding discrimination and sexual harassment. Procedures and policies are already established to avoid sexual harassment are in place for conservation game scouts, which regularly mix genders in isolated field settings, and the same policies apply to other workplaces, as specified in sections 11.1-11.5 of the NRT Human Resources policy¹⁶. This policy is accompanied by a grievance procedure in sections 11.6-11.9 that include provisions for reporting and resolving incidents, representation by other employees, and sanctions against offenders.

2.3.12 Grievances (G3.8)

We have no records of grievances by community members related to project activities during the monitoring period.

The grievance mechanism in place, as stated in the PD, allows for any issue arising to be communicated - this can come from anyone who is a member of the conservancies, and therefore the project.

There is also an "informal / undocumented" grievance mechanism where grazing committees, who represent every herder within the project through their zonal distributions, can be approached by their respective communities with any issue they have. Grazing committee members can then raise the issue to the conservancy level, who can then raise it to the carbon project leadership team for further review at that level.

In regard to reporting disputes and/or grievances related to cattle-raiding and unplanned grazing within the project zone, it is important to note that cattle-raiding is theft of someone else's property - and it is therefore dealt with by the national government of Kenya. At a project level, when there is a case of cattle-raiding the incident is reported through the formal conservancy chain of command; from the reporter (most likely a community member, or conservancy ranger) to the conservancy warden, to the conservancy manager and then to the JOCC (Joint Operations Command Center) based at Lewa. The JOCC has Kenya national police embedded within, and it is their responsibility to address every incident of cattle-raiding reported, using lawful means.

When it comes to unplanned grazing, this is dealt with at a project level; conservancy grazing committees and relevant elders representing the respective communities involved in the incident are mobilised to engage one another in advance of any conflict arising. Dialogue is the mitigating

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factor - where the herders are then addressed to ensure they comply with the grazing plan in question. If the incident escalates and conflict arises, the incident is reported as per the example of cattle-aiding above, and the relevant national authorities intervene where they feel necessary.

2.3.13 Worker Training (G3.9)

Worker training occurs for monitoring employment in Social, Veg and Wildlife CoMMs. Training occurs as guided by the NRT Human Resource Policy Manual 2015 (see below). Training is implemented by Conservancy members to others, as these are Conservancy managed programs. All procedures have evolved from experienced practice by local community members and are specifically designed to build local capacity that transcends personnel turnover. More details are available in the guide “Wildlife CoMMs: Ranger-Based Monitoring of Wildlife and Illegal Activities” Grazing coordinator training has involved university courses and off-site training, such as the US Forest Service rangeland management training or programs at the Savory Institute. The grazing coordinator team has survived almost complete staff twice since 2013.

Formal training is encouraged by NRT, leave is allowed, and explicit policy within its HR manual codifies expectations (pp. 38-41).

Training in grazing management continues at the level of grazing committee members occurs on an annual basis. In addition, training in grazing management is now occurring at the village level. Three-hour workshops given to primarily women and elders in 10 villages in each of Kalama and Namunyak Conservancies occurred in June of 2016. This village-level training was also expanded to other Conservancies in 2017.

Project employment opportunities feature the following jobs, with associated risks and mitigation:

Job	Risks	Mitigation
Rangeland coordinator	Attack by wildlife, herders from rival ethnic group or outside the project area, Traffic accident	Accompanied by armed askari, Enclosed 4-wheel drive vehicle maintained by NRT
Monitoring field assistant	Attack by wildlife, herders from rival ethnic group or outside The project area, Traffic accident	Accompanied by armed askari, Enclosed 4-wheel drive vehicle maintained by NRT
Social survey technician	Attack by belligerent subject Traffic accident	Surveys made by appointment, Approval of village chairmen Travel in enclosed 4-wheel drive

2.3.14 Community Employment Opportunities (G3.10)

These laws and policies have been and will continue to be implemented in the project through implementation of NRT's hiring policies, which include:

1. Open announcements of jobs associated with NRT and carbon project positions¹⁷.
2. Coverage of all employees under Workmen's Compensation
3. The National Hospital Insurance Act of 1988, which ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights.
4. Occupational Safety and Health Act of 2007 further expanded the range of types of accidents covered and is used as the basis for education and training and employees. (NRT HR Policy Manual p. 49).

The project will provide numerous employment opportunities for Conservancy members. 45% of NRT staff come from the local Conservancies, and searches for grazing coordinator, monitoring specialist, field assistant, driver, interviewer, data entry, accountant and other positions will actively seek Conservancy members. Some project revenues will be used by village or Conservancy administration to hire other Conservancy members as game scouts or other non-project jobs. These jobs will be announced and posted at Conservancy Headquarters, on the NRT website, and in standard job posting sites, and special attention will be given to applicants from vulnerable or marginalized groups, such as women and members of households without cattle. NRT has a strong track record of hiring women, as 33% of their workforce is composed of women comparable to the national average¹⁸ but well above employment levels in the Conservancies and in most pastoralist communities in Kenya.

2.3.15 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

The National Hospital Insurance Act of 1988 and Occupational Safety Act of 2007 ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights. How the project addresses compliance of this Act is covered in Section 2.3.14.

Per section 12 of the NRT HR Policy Manual (p. 49), employees are reminded of their rights to medical care in the event of job-related accidents at hiring. Employees have access to basic understanding of policy; responsibilities for Health and Safety (H & S) at work are all documented. There are up-to-date Health and Safety risk assessments and fire risk assessments conducted annually. The provision of necessary health and safety information, instruction and training is systematically undertaken along with these assessments and there is provision and correct use of necessary personal protective equipment (PPE) at all times.

Per the National Hospital Insurance Act, NRT has a Health and Safety Committee (NRT HR Policy Manual p. 50). NRT arranges for emergency response/evacuation, first aid and

¹⁷ <http://www.nrt-kenya.org/wp-content/uploads/2012/11/NRT-North-JOB-ADVERTS.pdf>;
<https://static1.squarespace.com/static/5653e896e4b0a689b3fafd97/t/56bdc8c7859fd0e5ed8c2a57/1455278301174/HR-Policy-and-Procedures-Manual-November-2015-+%281%29.pdf>

¹⁸

occupational health and employees are trained in documented procedures/safe systems of work, where necessary to establish and maintain a safe and healthy working environment.

Health and Safety protocols are extended to clients and visitors. These protocols are documented and known and communication and consultation with employees on H & S issues is consistent throughout the organization. There is consistent monitoring, investigation and reporting of any incidents, accidents or occupational ill-health; and corrective and preventative actions where any incidents, accidents or occupational ill-health occur are a priority. Accidents are required to be reported on a standard form.

Third party contractors perform regular audits of all safety measures, equipment, fire exits, among others, to ascertain the extent to which NRCL's Health and safety measures meet desired standards and legislation. Staff may be required to participate in safety-related activities, whenever called upon to do so.

2.3.16 Occupational Safety Assessment (G3.12)

The summary of worker risks and mitigation is covered in Section 2.3.13. These are indigenous communities living and working where they always have done, so the project does not impose any "new" threats as it supports and promotes their traditional practices. In terms of risks the communities might face; these include:

- Each-other through resource based conflict (this is not a new thing and the project activities are aimed at increasing peace between the communities through the sharing of a common incentive)
- Wildlife - as wildlife numbers increase due to project activity and improved grasslands, conflict is also likely to arise
- Politics - Kenya continues to face risks around land tenure and use, infrastructure development and taxation
- Pressure from communities outside the project area - this could be called the "green magnet effect" due to improved grasslands This is to be mitigated by development of new projects in the region.

2.4 Management Capacity

2.4.1 Required Technical Skills (G4.2)

The project team has experienced personnel covering all the major areas of expertise, including monitoring, modelling and carbon accounting (SftF, NativeEnergy, TNC), marketing (NativeEnergy), financing (NativeEnergy, TNC, NRT), social surveys (NRT, TNC), community relations and communications (NRT, SftF) and education and training of community members (NRT). Additional expertise is available from other stakeholders for monitoring (ICRAF education and training (Grevy's Zebra Trust).

2.4.2 Management Team Experience (G4.2)

The project proponent is **Northern Rangelands Trust (NRT)**, **NRT** facilitates and coordinates a governance relationship with multiple indigenous communities across the 2 million ha project zone. NRT helps different communities establish wildlife and biodiversity conservation activities in the form of Conservancies that cooperate with private vendors to benefit from tourism in return for

setting aside land from year-round cattle grazing. The communities engaged in these Conservancies are prepared to collectively participate in planned rotational grazing of livestock in both protected areas of their lands as well as open lands used communally by their members. Consequently, the communities of NRT are full participants in the carbon project. NRT also brokers peace agreements between different ethnic groups within and outside the project area, and is instrumental in conflict resolution, all of which clearly contribute to community well-being.

Skills Provided: community engagement, livestock management advice, peace-making and conflict resolution, administrative support, and social, vegetation, and wildlife monitoring.

Soils for the Future, LLC (SftF) is a carbon project developer that is responsible for the design and monitoring of project activities. SftF is working with its associated commercial enterprise **NRT Trading, Ltd, The Nature Conservancy (TNC)** and the 13 Conservancies. **Skills Provided:** biodiversity and soil and vegetation monitoring, carbon accounting, carbon modelling, advice on grazing management.

NativeEnergy is a project marketer and developer that builds customized projects and purchase **arrangements** with private companies seeking to offset emissions or invest in community development projects. **Skills Provided:** marketing, sales transactions.

TNC facilitates community-based conservation projects throughout eastern and southern Africa, including two REDD+ projects involving local community protection of forests in Tanzania. They implement a strategy of building local capacity to manage lands and resolve conflicts between competing land uses and conservation. **Skills Provided:** Fund-raising, administrative support, and social monitoring.

NRT Trading is a for-profit commercial enterprise associated with NRT and is responsible for enhancing, financing, and operational oversight of commercial business enterprises facilitated by NRT. **Skills Provided:** sales transactions, financial risk abatement.

The Conservancies are each a set of communities, typically dominated by a particular ethnic group that have joined together to set aside lands for wildlife and other types of conservation. Their goal is to provide security for livestock and wildlife and make better use of lands to maintain cultural traditions, diversify livelihoods, and reduce poverty. **Skills provided:** livestock management, wildlife and social monitoring.

2.4.3 Project Management Partnerships/Team Development (G4.2)

Grevy's Zebra Trust (GZT) has been and will continue to be instrumental in education of pastoralist communities regarding the rotational grazing of bunched herds as a desired management objective. Their training led to adoption of rotational grazing in the first two Conservancies in NRT, and subsequent workshops helped expand rotational grazing concepts to additional Conservancies.

Contact: Belinda Low Mackey, Executive Director, Grevy's Zebra Trust, Nairobi,
belinda@grevyszebratrust.org

International Centre for Tropical Agriculture (CIAT). This organization has and will provide the analyses of soil carbon and plant tissue lignin and cellulose for the project. CIAT also conducted some of the soil sampling used to determine the baseline carbon stocks in the project area.

Contact: Dr. Leigh Winowiecki, Soil Scientist I, International Centre for Tropical Agriculture, Nairobi, leigh.winowiecki@gmail.com

International Centre for Agroforestry (ICRAF) conducted some of the soil sampling used to determine the baseline carbon stocks in the project area and has developed a remote-sensing based model for estimating soil carbon stocks.

Contact: Tor-Gunnar Vagen, Senior Scientist, International Centre for Agroforestry, Nairobi, tvagen@cgiar.org

2.4.4 Financial Health of Implementing Organization(s) (G4.3)

The principal implementing partner in this project is NRT, whose annual budget is approximately \$10 million from both donor and commercial sources (for more details see NRT 2017 Annual Report¹⁹). NRT has grown substantially since its inception in 2004 and has a sustainable network of donors for livestock management and rangeland restoration featuring USAID, DANIDA, and The Nature Conservancy (TNC). TNC provides more than US\$1 million annually in support of NRT management, and recently arranged \$7 million in low interest loans to support the livestock-to-market program operated by NRT Trading, a commercial adjunct of NRT <http://www.nrt-kenya.org/nrt-trading/>. Project implementation costs not covered by these donor sources have been re-paid by revenues from the first issuance after the first monitoring period (2013-2016).

2.4.5 Avoidance of Corruption and Other Unethical Behavior (G4.3)

The Project Proponent and any of the other entities involved in project design and implementation are not involved in or are not complicit in any form of corruption such as bribery, embezzlement, fraud, favoritism, cronyism, nepotism, extortion, and collusion. Proceeds of sales will be distributed according to pre-established contracts signed by each project proponent and the managers of each Conservancy.

2.4.6 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

No commercially sensitive information has been excluded from the public version of this Monitoring Report.

2.5 Legal Status and Property Rights

2.5.1 Recognition of Property Rights (G5.1)

Grazing and grazing resource (water, soil) rights are conferred by land ownership associated with Group Ranch status via Kenya Land Act of 1968 (revised in 1970) or by Africa customary law for communities on Trust lands via the Kenya Trust Land Act of 2010. These rights are further conferred and superseded by the Kenya Community Lands Act of 2016. Rights to benefit from soil conservation are conferred by the right to register lands as Conservancies by the Kenya Wildlife Conservation and Management Act of 2013 and the Kenya Community Lands Act of 2016 (see also section 1.3.2).

¹⁹

https://static1.squarespace.com/static/5af1629f12b13f5ce97ca0b5/t/5b727401575d1f3a6fb55fec/1534227491888/LOW_Res_2017_State_Of_Conservancies_Report.+FINAL.pdf

Rights to benefit from soil conservation as a component of wildlife conservation are conferred by the right to register lands as Conservancies by the Kenya **Wildlife Conservation and Management Act of 2013**²⁰ and by the **Community Lands Bill of 2016**²¹. The former law confers the rights of communities to form Conservancies and benefit from conservation activities such as wildlife management and tourism and land restoration, and the latter confers this right on any public land occupied by communities, including trust lands and designated community lands. Conservancies are defined as a “community-owned and community-run institution which aims to improve biodiversity conservation, land management and the livelihoods of its constituents over a defined area of land traditionally owned, or used, by that constituent community.” As Conservancies and members of the Kenya Wildlife Conservancy Association (KWCA), communities have the unrestricted, and unencumbered right to benefit financially from conservation activities, which would include restricted, planned and managed grazing and building soils through carbon storage²². The Kenya Wildlife Service administers the formation of Conservancies and all 13 Conservancies have submitted applications for the formal formation of Conservancies.

2.5.2 Free, Prior and Informed Consent (G5.2)

Participation in this project is completely voluntary. Participation by community members is presumably motivated by the prospect of benefits to forage availability, livestock health and marketability in addition to benefits to community infrastructure and retention of land rights. The decision about where and when to move livestock will be made by a community decision-making process through existing mechanisms of livestock management, and not by NRT, TNC, or other stakeholders. The voluntary nature of the decision has been made clear by the announcement of the project during the public comment period during consultations with key informants, opinion leaders, and community administrative leaders.

Encroachment onto private lands is discouraged by expected payment of lease fees by project participating community members to the landowners. Excursions on lands outside the project area traditionally is not compensated as per cultural norms, but expectations of paying leases in those cases are increasing^[6]

The need for geographical integrity of the project for carbon accounting and coordinated, planned grazing at a regional scale creates a project of core participating Conservancies surrounded by Non-participating communities (NPC's). As the project develops and rangelands and soils are restored, the project area may become a magnet for encroachment from NPCs during droughts. Likewise, members of participating Conservancies members may also choose to move livestock onto range used by NPCs (this is treated as leakage in project carbon accounting); project protocols request such excursions to be legal and justified through compensation (leases or rents) to NPCs. Such negotiations thus far have been mostly informal (no legal documents or paper trail).

Individual conservancies may on their own volition choose to relocate settlements within their Conservancy, but this will follow established mechanisms of decision-making within the

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http://www.kws.org/export/sites/kws/info/publications/acts_policies/The_wildlife_conservation_and_management_bill_2013.pdf

²¹ http://info.mzalendo.com/media_root/file_archive/Community_LandBill_2015.pdf

²² <https://kwcakenya.com/conservancies/conservancies-policies/>

community. At no time will NRT, SFF, or TNC request community members to relocate. This will be made clear by the announcement of the project during the public comment period through consultations with key informants, opinion leaders, and community administrative leaders.

2.5.3 Property Right Protection (G5.3)

Land tenure in northern Kenya is community-based, in which government-owned land allows occupying communities ownership rights to resources. Some communities (Il Ngwesi, Kalama, Leparua, Meibae, Naibunga, Namunyak, Sera, Westgate) exist as or have applied for registration as Group Ranches under the Kenya Land (Group Representatives) Act of 1968²³. This law conferred rights through group representatives who are “authorized to hold land and other **assets** on behalf of the group, and to act on behalf of and for the collective benefit of all group members.” (Section II.8.). Soil carbon would be considered an asset for the group members, since it has value in the voluntary carbon market.

Other communities, such as those near Buliqo Bulesa, Melako, Oldonyiro, Nasuulu, and Nakuprat Gotu Conservancies are located on Trust lands, which confer rights to grazing and by inference resources that promote grazing (such as water, soil carbon) under customary law conferred by Section VIII item 59 of the Kenya Trust Land Act of 2010²⁴. These rights are now further conferred by passage of the Community Lands Act of 2016, which confers rights to resident communities to graze and conduct conservation activities even on Trust lands²⁵.

Unfortunately, all communities have suffered historically from an inability to defend ownership or customary rights from either government initiatives or land use by non-member individuals but NRT efforts and support have helped reduce conflicts and organize governance and management activities in these communities²⁶. As part of this process, all the participating NRT communities are now registering their lands, including settlements, as Conservancies. These are defined by the Kenya Wildlife Conservation and Management Act of 2013²⁷ as a “community-owned and community-run institution which aims to improve biodiversity conservation, land management and the livelihoods of its constituents over a defined area of land traditionally owned, or used, by that constituent community.” As Conservancies, communities have the right to benefit financially from conservation activities, which would include restricted, planned and managed grazing and building soils through carbon storage.

Most Conservancies are dominated by a single ethnic group, but the dominant group varies among Conservancies. Generally, interactions among ethnic groups in the project area are congenial, as evidenced by perennial dry season movements of livestock from Conservancies dominated by one ethnic group (e.g. Samburu from Kalama, Westgate and Namunyak) to lands on Conservancies dominated by others (Rendille in Melako and Borana in Biliqo Bulesa). Within-community distribution of resources generally proceeds without bias towards ethnic

²³ <http://faolex.fao.org/docs/pdf/ken62430.pdf>

²⁴ <http://faolex.fao.org/docs/pdf/ken62437.pdf>

²⁵ http://kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/CommunityLandAct_27of2016.pdf

²⁶

http://usaidlandtenure.net/sites/default/files/USAID_Land_Tenure_Kenya_Rangeland_Costal_Conservation_Program_Assessment_0.pdf

²⁷

http://www.kws.org/export/sites/kws/info/publications/acts_policies/The_wildlife_conservation_and_management_bill_2013.pdf

groups, as evidenced by a lack of indicated bias about revenue distribution among different ethnic groups within a Conservancy in baseline household surveys.

As part of community property protection rights, community members are never required to relocate. Movement of habitation is voluntary, such as when families voluntarily moved to villages and along the borders of Westgate Conservancy to facilitate implementation of Conservancy grazing plans in 2014. Likewise, ethnic minorities within Conservancies are allowed to select their home location.

2.5.4 Identification of Illegal Activity (G5.4)

The project depends on voluntary actions by community members to herd livestock according to planned rotational grazing principles. By communal grazing rights afforded to each Conservancy, altered grazing management is a legal activity. Monitoring of biodiversity involves no capturing or handling of vertebrates, and sampling of plants and insects is not covered under Institutional Animal Care regulations. Human surveys by NRT staff are covered under research methods approved by an external Institution Review Board covering TNC and NRT-sponsored social science research.

Traditional pastoralist practices of stealing livestock, armed takeover of grazing areas or water points, or killing pastoralists of other ethnic groups will not be tolerated. Conflicts, when they do exist have been and will be addressed by combined NRT, Conservancy, and County government response teams. When response team intervention fails to resolve the conflict, NRT calls a peace meeting²⁸, such as occurred in July of 2015. Peace meetings have been highly successful in reducing conflict in the area since the advent of NRT in 2008.

No significant illegal activities were identified between 2017-2020 and all activities were tracked through “conflict incident reports.” These reports are generated monthly.

2.5.5 Ongoing Disputes (G5.5)

The decline in traditional governance and wider insecurity in the Horn of Africa have combined to make illicit firearms readily available. There is a general willingness throughout the region for people to use them to enforce resource access rights. The proliferation of firearms has combined with tribal traditions in which raiding livestock from a neighboring tribe is a male rite of passage and means by which a *moran* (warrior) may obtain social status and marry. However, in recent years, the availability of fire-arms has led to more frequent raids and inter-tribal killings. In Samburu District, 88% of respondents reported that they had used firearms in their possession for cattle raiding with comparable figures likely elsewhere in the region. Cattle-raiding is estimated to result in the loss of US \$1 million annually in Samburu District, and is a significant factor constraining economic development in the region^[7]. Insecurity from the threat of cattle-raiding has fed the cycle of increased environmental degradation, which in turn fuels the conflict^[7]. Because much of the rangeland has become unsafe for grazing livestock and nomadic herders, pastoralists have become further sedentary in secure areas^[7]. Furthermore, stealing cattle may be the only viable way for many impoverished people to rebuild herds lost from droughts. Drought further exacerbates conflict because herders may move to areas distant from their home

28

https://static1.squarespace.com/static/5653e896e4b0a689b3fafd97/t/57cff831ebbd1adbaf21edd6/1473247301208/SocReport_2015_Final_Spreads_LowRes.pdf

Conservancies to find water or grass, thus bringing historically antagonistic ethnic groups into proximity with little governance as to how limited resources are allocated.

Ethnic groups with the most frequent history of conflict are Samburu with Turkana and Maasai and with Borana. Two major sources of conflict highlight certain localities in the project area that have received more peace meetings and pre-emptive consultation during droughts when conflicts are more likely to occur. First Conflicts between Turkana, Maasai and Samburu date back 400 years^[8, 9], and both maintain cultural traditions of livestock theft from each other. These three ethnic groups are juxtaposed along the Ewaso Nyiro River where Turkana have settled among otherwise Maasai-dominated communities but are in close proximity to Samburu north of the river. This confluence of three antagonistic ethnic groups (Turkana, Maasai, and Samburu) around a critical water resource (the river) fosters persistent livestock theft, especially during drought conditions.

The other ethnic conflict “hotspot” is along the Sera- Biliqo Bulesa border, where Samburu herders bring livestock from the west simultaneously with Borana bringing livestock from the east during the dry season. Borana, together with Gabra are a subgroup of the larger group of Oromo people centered in north central Kenya and in Ethiopia and differ sharply in religion, livestock preferences (camels vs cattle) and language from the Samburu. Conflicts over forage and water intensified between these two groups in the late 19th Century, and animosity persists.

These geographic foci allows NRT staff, Conservancy grazing coordinators and game scouts, and County government staff to more efficiently deal with conflicts as they arise, with effort put into specific localities at different times. For example periods of persistent conflict occur in July, when Samburu herders typically cross Sera Conservancy and approach two locations with reasonably predictable availability of water, Kisima Hamsini (50 wells in Kiswahili) and Qom, placing them in proximity to Borana and Gabra herders approaching from the east^[1]. The other major conflict period is September- October near the end of the dry season, when livestock of Samburu in Kalama, Turkana in Nakuprat-Gotu and Nasuulu often have significant herds in juxtaposition^[1]. In 2017 and 2018, NRT, working with Samburu County government attempted to anticipate conflict in these localities with peremptory peace meetings, but severe drought conditions and erratic movements of herders from different Conservancies required repeated post-conflict resolution from NRT and Samburu County.

Project activities are highly unlikely to either cause or influence the outcome of such disputes. First, most conflicts occur because of livestock theft by members of different ethnic groups, and resolution of these disputes involves returning stolen livestock, peace meetings, intervention by County officials, etc. Project activities involve movement of livestock according to local Conservancy grazing plans or regional grazing plans in the event of extended dry seasons or drought. These plans allow anticipation of livestock movements 2-4 weeks ahead of schedule so that conflict areas can be cleared of unscheduled livestock or range-sharing verbal agreements can be arranged. Conflicts are more likely when herders are not following planned project activities, e.g., arrive at grazing areas or watering holes too early or too late, or if herders steal livestock, which is never tolerated and is prosecuted by Conservancy game scouts as part of the project activities.

Such meetings are largely at the site of conflict, impromptu, and conducted in local unwritten languages, so meeting records are not available. However, since May 2015 grazing coordinators meetings are attended by project proponents, Conservancy managers have increasingly

mentioned the outcome of meetings in their reports, and these are available as supporting documents. Conservancy manager reports from early 2018 describe multiple peace meetings with successful resolution in which stolen cattle were returned, agreements were reached with encroaching herders from communities outside the project zone, and disputes were settled between elders and morani (young adult males) from within the same Conservancies. In addition, programs in which morani are interviewed and involved in grazing committee decisions began to be implemented in early 2018, which further reduced conflicts during the failed short rains of November 2018.

There is a history of conflict between ethnic groups in the project area based on access and proximity to water leading to encroachment of herders from other conservancies and outside the project area and persistent livestock theft between groups. Conflicts are more likely to occur during droughts. The Project is unlikely to either cause or influence the outcome of these conflicts. Project activities involve movement of livestock according to conservancy or regional grazing plans and conflicts are more likely when herders are not following planned grazing activities. As a result of the Project, there are more game scouts within the conservancies who patrol conservancy borders and deal with conflicts as they arise.

2.5.6 National and Local Laws (G5.6)

Grazing and grazing resource (water, soil) rights are conferred by land ownership associated with Group Ranch status via Kenya Land Act of 1968 (revised in 1970) or by Africa customary law for communities on Trust lands via the Kenya Trust Land Act of 2010. These rights are further conferred and superseded by the Kenya Community Lands Act of 2016. Rights to benefit from soil conservation are conferred by the right to register lands as Conservancies by the Kenya Wildlife Conservation and Management Act of 2013 and the Kenya Community Lands Act of 2016 (see also section 2.5.1).

The relevant labor laws include the Employment Act (Cap.226); the Regulation of Wages and Conditions of Employment Act (Cap. 229); the Trade Unions Act (Cap. 233), the Trade Disputes Act (Cap. 234), the Workmen's Compensation Act (Cap. 236), These have been and will continue to be implemented in the project through implementation of NRT's hiring policies, which include:

2. Environmental Management and Co-ordination Act, 1999. The project has been registered with and reviewed by the National Environmental Management Authority. Additional laws protect the integrity of Conservancies and their ability to engage in contracts
3. Companies Act, (Law of Kenya Cap. 486). NRT is a registered non-profit organization and the Conservancies are registered as limited liability corporations and are in good standing to operate in Kenya.
4. The Trespass Act (Law of Kenya Cap. 294). This is a law that is seldom enforced but could be used by Conservancies to help enforce the project area borders and reduce project livestock encroachment into the project area.
5. The National Hospital Insurance Act of 1988, which ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights.

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

Per the VCS VM0032 methodology, the following data and parameters were presented at validation:

1. Maps of the project area, indicating all land parcels included in the project, as indicated in accompanying shape files with vector coordinates of project and stratum boundaries.
2. Map(s), with accompanying georeferenced shape files, of the locations of the permanent sampling stations overlaid on a map of project strata.
3. Results of analysis to determine the number of sampling units and their allocation among strata.
4. Results of cluster analysis to determine project strata.
5. Table of all project strata, their description, and area, PA_m
6. Legal statements of the usage rights of Conservancy members to graze livestock and benefit from carbon sales, and governmental permissions for conducting the project.
7. Justification of planned rotational grazing practices.
8. Justification of methane as the major emission source and methane and soil carbon as the major sink for carbon dioxide in the project.

Project Design

Data / Parameter	$PA_{m,g}$
Data unit	ha
Description	Project area across 13 participating conservancies (strata)
Source of data	NRT shapefiles
Value applied	1,993,075
Justification of choice of data or description of measurement methods and procedures applied	Using shapefiles in a GIS or from known coordinates of stratum boundaries
Purpose of data	Determination of baseline and project emissions Calculation of project soil carbon removals

Comments	

Baseline Methane Emissions

Data / Parameter	GWP _{CH₄}
Data unit	tCO _{2e} /tCH ₄
Description	Global-warming potential for CH ₄
Source of data	Obtained from the IPCC Fifth Assessment Report
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Most recent value published by IPCC Fifth Assessment Report
Purpose of data	Determination of baseline and project emissions used to calculate the final project soil carbon removals
Comments	

Data / Parameter	W _{c,t}
Data unit	kg
Description	Average body weight for animals of category c in year t
Source of data	Measured in project area
Value applied	Varies
Justification of choice of data or description of measurement methods and procedures applied	Necessary to estimate emission factor for grazing animals using allometric equations. Measurements must be taken in accordance with the procedures described in 9.1.2 of methodology (VM0032).
Purpose of data	Calculation of baseline emissions
Comments	W _{c,t}

Data / Parameter	N _{c,j}
Data unit	Number
Description	Baseline number of animals of category c in census i
Source of data	Measured in project area
Value applied	Varies

Justification of choice of data or description of measurement methods and procedures applied	The number of animals in each census i must be measured to calculate the harmonic mean of the multiple counts i of n censuses of animals in category c . At least four measurements within the baseline period, with at least two during the period 5- 10 years prior to the project start, must be available. Measurements must be taken in accordance with the procedures described in Section 9.1.2 of methodology (VM0032).
Purpose of data	Calculation of baseline emissions
Comments	This document will provide a table of historical estimates for number of grazing animals, BN_c , for each year in which counts or estimates are available, sorted by the c categories of animals in the project area: species, breed, sex, and age, plus the respective live body weights (W_c) of each category, with 95% CI and uncertainties). The project description also provides a data table showing calculations of methane emissions based on the equations in VM0032 for each animal category for each year the data are available. The table includes calculated total emissions for that year and a cell containing the harmonic mean of total annual calculated methane emissions. This is the baseline BEM. The harmonic mean appropriately and conservatively weights the average methane emissions towards the lower values of a time series of measurements. The table also shows the uncertainty in daily methane emissions and the harmonic mean and its uncertainty.

Parameters for Baseline Calculation of Emissions from Burning of Biomass

The project does not plan to increase fire intensity, therefore the data/parameters within VM0032 for monitoring burning of biomass are not applicable.

Parameters for Calculation of Baseline SOC

Data / Parameter	$DEPTH_{m,j,0}$
Data unit	cm
Description	Soil core depth at station j in stratum m at time $t = 0$ (i.e., at the start of the project or since the last verification)
Source of data	Measured in project area
Value applied	20
Justification of choice of data or description of measurement methods and procedures applied	At each sampling station j , according to standard methods, soil must be taken from at least 3 soil cores to a depth that accounts for the vast majority (> 80 percent) of SOC in the soil column, reflects depth to hardpans or bedrock, or matches calculations

	from soil carbon models. Multiple cores may be well-mixed into a single composite sample for analysis.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	SOC% _{j,m,0}
Data unit	Dimensionless proportion
Description	Proportion soil organic carbon at station j in stratum m at time t = 0 (i.e., at the start of the project or since the last verification)
Source of data	Measured in project area
Value applied	Varies
Justification of choice of data or description of measurement methods and procedures applied	The baseline for the measured offset approach is based on increasing SOC. Tracked at the level of j = 1 to individual sampling stations in each stratum. Offset will be based on demonstrating changes in SOC at individual stations and then summing increments. At each sampling station j, according to standard methods, soil was taken from at least 3 soil cores to a depth that accounts for the vast majority (> 80 percent) of SOC in the soil column, reflects depth to hardpans or bedrock, or matches calculations from soil carbon models. Multiple cores were mixed into a single composite sample for analysis. Multiple stations were sampled within each stratum to determine a 95 percent confidence interval.
Purpose of data	Organic carbon concentrations were measured at an academic laboratory using Mid-InfraRed (MIR) Spectrometry as calibrated by chemical combustion
Comments	Calculation of baseline emissions

Data / Parameter	BULK _{m,j,0}
Data unit	g/cm ³
Description	Bulk density at station j in stratum m at time t = 0 (i.e., at the start of the project or since the last verification)
Source of data	Measured in project area
Value applied	Varies across sampling stations
Justification of choice of data or description of measurement methods and procedures applied	At each sampling station j, according to standard methods soil was taken from at least 3 soil cores and known volumes of soil from the cores were sieved to remove rocks, pebbles, and coarse fragments. The remainder was dried (5 days at 45°C or equivalent) and weighed to determine bulk density.
Purpose of data	Calculation of baseline emissions
Comments	

Parameters for Soil Carbon Models

Data / Parameter	MAP _m
Data unit	mm/yr
Description	Mean annual precipitation in stratum m
Source of data	Read from CHIRPS database for applicable months and years http://chg.geog.ucsb.edu/data/chirps/
Value applied	Varies across sampling stations Appendix C for project area rainfall map
Justification of choice of data or description of measurement methods and procedures applied	A key variable that affects carbon inputs (plant growth) and carbon losses (respiration) from soil.
Purpose of data	Calculation of baseline and project emissions
Comments	

Data / Parameter	SAND% _{j,m}
Data unit	Dimensionless proportion expressed as percent
Description	Proportion of soil that is sand, at station j in stratum m
Source of data	Measured in project area
Value applied	Varies across sampling stations
Justification of choice of data or description of measurement methods and procedures applied	Soil collected to desired depth at each sampling station should be mixed, and a subsample analyzed for clay, silt, and sand fractions in a professional laboratory. SAND% for soils was estimated by ICRAF laboratories Nairobi using a MIR reflectance method calibrated for Kenya soils.
Purpose of data	Calculation of project removals
Comments	

Data / Parameter	GI _{j,m}
Data unit	Dimensionless proportion
Description	Mean annual grazing intensity at station j in stratum m
Source of data	Based on assumptions of McNaughton 1985 Ecological Monographs, Veldhuis et al 2019 Science
Value applied	0.8
Justification of choice of data or description of	The project removals were calculated using an estimated grazing intensity thought to be likely to occur under planned grazing based

measurement methods and procedures applied	on measured grazing intensities in protected areas with high densities of migratory wildlife species (e.g. Serengeti). ²⁹
Purpose of data	Calculation of project removals
Comments	

Data / Parameter	MAPLC _{j,m}
Data unit	Dimensionless proportion
Description	Mean aboveground plant cellulose plus lignin at sampling plot <i>j</i> in stratum <i>m</i>
Source of data	Measured in project area
Value applied	Varies
Justification of choice of data or description of measurement methods and procedures applied	SOC is often closely related to inputs of these forms of carbon because they resist microbial decomposition. Lignin and cellulose content for each sampling station was estimated on the basis of cover of different plant functional types: annual grass, forb (herbaceous dicot), or perennial grass. Five leaf and stem samples from each of the five most common species of each plant functional type were collected from 30 different sampling stations in Il Ngwesi, Kalama, Sera and Westgate Conservancies. To measure lignin plus cellulose fraction, dried plant material from the entire clipped quadrat was ground through a 0.9 mm (40 mesh) screen in a Wiley mill. Ground material was then subjected to a sulfuric acid hydrolysis method[10], i.e., sequential digestion in first neutral detergent manufactured by Ankom Technology Corp., Macedon, New York, USA (sodium lauryl sulfate, 50.02%, EDTA disodium, dehydrate, 31.03%, sodium borate, decahydrate 11.35%, and sodium phosphate, dibasic, anhydrous, 7.6%), followed by Ankom manufactured acid detergent (20 g cetyl trimethylammonium bromide (CTAB) diluted in 1 L 1.00 NH₂SO₄), and then in 72% sulfuric acid, followed finally by ashing in a muffle furnace at 500°C for 24 h. Lignin and cellulose fractions were inferred by the mass lost between the remainder following neutral detergent digestion and the remainder after ashing, expressed as a proportion of the original sample mass.
Purpose of data	Calculation of project removals
Comments	

²⁹ (Mean 60%, McNaughton 1985 Ecological Monographs, Veldhuis et al 2019 Science) discounted for the likelihood that herded livestock use areas for a longer time than wildlife. The expert assessment used in estimating GI for the baseline is better at assessing grazing intensity over previous decades rather than short term estimates, which are sensitive to a random factor: number of days since livestock last grazed at a sampling station.

Parameters for Removals from Woody Plant Biomass

Woody Plant Biomass Removals are conservatively assumed to be *de minimis* because removals are largely used for fuel wood (see compliance with applicability conditions and therefore, the woody plant biomass parameters are not applicable to this project. See section 3.2.4 for discussion on Woody Plant Biomass removals.

3.1.2 Data and Parameters Monitored

Data and parameters monitored fall into three major categories: (1) data and parameters needed to calculate project greenhouse gas removals and emissions, (2) data gathered to demonstrate impacts of the project on biodiversity, and (3) data gathered to demonstrate project impacts on communities.

Data / Parameter	$PA_{m,t}$
Data unit	ha
Description	Project area in stratum m in year t
Source of data	Measured in project area
Description of measurements methods and procedures to be applied	Using shapefiles in a GIS
Value applied	1,993,075 (the project area has been updated resulting in a decrease of 3 ha during the current monitoring period)
Monitoring equipment	None
QA/QC procedures to be applied	The project area has been determined from accurate GIS layers of conservancy boundaries (strata) and has not changed since the project's validation
Purpose of data	Calculation of project emissions
Calculation method	Measured directly
Comments	None

Parameters for Calculating Project Methane Emissions

Data / Parameter	$PN_{c,t}$
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Data unit	number
Description	Mean number of project animals of category <i>c</i> in the project area during year <i>t</i>
Source of data	Measured in project area
Description of measurement methods and procedures to be applied	Calculated as a product of estimated number of households in conservancy <i>y</i> during year <i>t</i> and the measured number of livestock of category <i>c</i> per household. Number of households in conservancy <i>y</i> during year <i>t</i> estimated from results of 2014 conservancy census and extrapolated trends over time. Number of livestock of category <i>c</i> per household measured during 2014 census.
Frequency of monitoring/recording	Livestock and household numbers measured directly during 2014 conservancy census. Values for remaining years of current monitoring period extrapolated from 2014 results.
Value applied	Depends on year: see 'Households Over Time' tab of 'NKCP Methane Calculations 2017-2020 v1 Sept2021.xlsx'
Monitoring equipment	Binoculars
QA/QC procedures to be applied	All data is backed up and stored in Google Drive
Purpose of data	Used to calculate project emissions and to determine available pools of livestock to be allocated to on-project or off-project categories for calculation of leakage emissions.
Calculation method	Calculated as a product of the estimated number of households in conservancy <i>y</i> during year <i>t</i> and the measured number of livestock of category <i>c</i> per household.
Comments	None

Data / Parameter	$ED_{c,t}$
Data unit	days
Description	Encroachment-days. Number of days spent by encroaching (off-project) livestock of category <i>c</i> on the project area during year <i>t</i> .
Source of data	Measured in project area
Description of measurement methods and procedures to be applied	Calculated based on livestock encroachments documented by conservancy grazing coordinators in monthly grazing reports. Number and class of encroaching livestock for a monthly report is determined using encroachment decision tree (Appendix A). $ED_{c,t}$ calculated as the product of number of encroaching livestock in a given month and number of days in that month.
Frequency of monitoring/recording	Data collected monthly to facilitate annual calculation.

Value applied	Depends on month and conservancy. See 'Encroachments Summary' tab of 'Leakage and Encroachment Verification Calculations 2017-2020 July2021.xlsx'.
Monitoring equipment	Binoculars
QA/QC procedures to be applied	Monthly grazing reports are reviewed by project administrators (NRT, SftF) and stored in duplicate online databases, Google Drive and Dropbox.
Purpose of data	Calculation of project encroachment methane emissions
Calculation method	Number of encroaching livestock in month m x number of days in month m .
Comments	None

Project Emissions from Burning Biomass

As discussed in section 3.2.1, emissions from burning biomass are expected to be *de minimis* and the parameters are therefore not applicable for monitoring.

Parameters for Calculating SOC Removals

After crediting periods long enough to detect changes in SOC at sampling stations, e.g. 5-7 years, the soil organic carbon $SOC_{m,j,z}$ will also be re-measured in order to re-validate and recalibrate the SNAP soil carbon model.

Data / Parameter	$DEPTH_{m,j,t}$
Data unit	cm
Description	Soil core depth at station j in stratum m at time $t = 0$ (ie, at the start of the project or since the last verification)
Source of data	Measured
Description of measurement methods and procedures to be applied	At each sampling station j , according to standard methods, soil must be taken from at least 3 soil cores to a depth that accounts for the vast majority (> 80 percent) of SOC in the soil column, reflects depth to hardpans or bedrock, or matches calculations from soil carbon models. Multiple cores may be well-mixed into a single composite sample for analysis.
Frequency of monitoring/recording	During each soil sampling at permanent plots
Value applied	20
Monitoring equipment	Meter stick

QA/QC procedures to be applied	Depth cored must be the same as for baseline soil carbon sampling. However, the depth used in calculating SOC after Y years of project activities must be adjusted to account for changes in bulk density such that $DEPTH_{m,j,Y} \times BULK_{m,j,Y} = DEPTH_{m,j,0} \times BULK_{m,j,0}$. This ensures that equal masses of soil are compared between year 0 and year Y
Purpose of data	Calculation of project emissions
Calculation method	Not calculated
Comments	None

Data / Parameter	$SOC\%_{j,m,Z}$
Data unit	Dimensionless proportion
Description	Proportion soil organic carbon at station j in stratum m at time (year) = Z, the number of years sufficient for changes in SOC at individual sampling stations
Source of data	Measured in project area
Description of measurement methods and procedures to be applied	Measured SOC necessary to detect changes in SOC in the project that will be used to re-validate and if necessary, recalibrate-the soil carbon model of choice. Tracked at the level of $j = 1$ to z_m individual sampling stations in each stratum because that provides the most rigorous test of a soil carbon model.
Frequency of monitoring/recording	Measured once at the beginning of the project and then at 5-10 year intervals thereafter
Value applied	Varies with location from 0.2 – 3%
Monitoring equipment	Equipment for taking soil cores (augers, metal pipes, etc) and lab equipment for doing loss on ignition (drying ovens, furnaces) or autoanalyzers for estimating carbon loss on combustion, or spectrophotometers for measuring infra-red light reflectance
QA/QC procedures to be applied	Precision weighing is required for soils with low carbon content, paper and digital archiving of data, cross-calibration of different methods, e.g., samples analyzed by loss on ignition and spectrometry.
Purpose of data	Calculation of baseline and project removals
Calculation method	Not calculated
Comments	None

Data / Parameter	$BULK_{j,m,Z}$
Data unit	g/cm^3
Description	Bulk density in stratum m, station j, year Z
Source of data	Measured in project area

Description of measurement methods and procedures to be applied	Necessary to convert proportion of SOC in soil to mass of SOC/volume following changes in SOC, after Z crediting years 5-7 crediting years
Frequency of monitoring/recording	Measured once at the beginning of the project and then at 5-10 year intervals thereafter
Value applied	Varies with location from 0.4 – 1.4 g/cm ³
Monitoring equipment	Metal pipe of known volume, sledgehammer, metal plate to prevent soil from leaking, drying oven, sieve (2mm mesh). Heavy duty plastic bags, graduated cylinder
QA/QC procedures to be applied	A consistent system of soil storage to prevent loss of mass prior to weighing, accurate estimation of rock volume, paper and digital archiving, corroboration with literature values
Purpose of data	Calculation of project and baseline removals.
Calculation method	Total dry mass of soil / (Volume of pipe – volume of rocks)
Comments	None

Parameters for Project Soil Carbon Models

Data / Parameter	MAP _{m,t,Y}
Data unit	mm/yr
Description	Mean annual precipitation in stratum m over the crediting period Y years
Source of data	Read from WorldClim, World Resources Institute, https://www.worldclim.org/
Description of measurement methods and procedures to be applied	Parameter in the SNAP soil carbon model.

Frequency of monitoring/recording	Not monitored – it represents a long-term average climate variable
Value applied	Varies by positions across the project area from 300 mm to 1000 mm/yr
Monitoring equipment	Acquired from digital databases
QA/QC procedures to be applied	Databases are maintained by government institutions for public use and are re-calibrated as new data are gathered
Purpose of data	Calculation of soil carbon removals
Calculation method	Not calculated
Comments	None

Data / Parameter	$GI_{j,m,Y}$
Data unit	Dimensionless proportion
Description	Mean annual grazing intensity at station j in stratum m over the crediting period Y years
Source of data	Measured in project area at each sampling station every three years prior to verification by comparing clipped biomass at least at the end of the growing season, or more frequently for some models, inside and outside small (1 m^2) fences. $GI_m = 1 - (\text{biomass outside/biomass inside})$. Biomass is clipped, dried at $25 - 50 \text{ }^\circ\text{C}$, and weighed
Description of measurement methods and procedures to be applied	Parameter in the SNAP soil carbon model.
Frequency of monitoring/recording	Measured once as an indicator of long-term impacts on plants
Value applied	Varies by project location from 50-99%
Monitoring equipment	Tools for clipping vegetation, drying oven, fencing to generate plant production in the absence of grazing animals, computer and software to download and analyze satellite data
QA/QC procedures to be applied	Visually estimated by trained experts with estimated guesses calibrated. Corroborated with calibrated satellite data and literature measurements
Purpose of data	Calculation of baseline and project soil carbon removals
Calculation method	Not calculated
Comments	None

Data / Parameter	MAPLC _{j,m,Y}
Data unit	Dimensionless proportion
Description	Mean aboveground plant cellulose plus lignin at sampling station j in stratum m in project crediting year Y
Source of data	Measured in project area by MIR spectral analysis of clipped plants (two 0.0625 m ² quadrats) at ICRAF laboratories in Nairobi
Description of measurement methods and procedures to be applied	LIGCELL% is a parameter in the SNAP soil carbon model.
Frequency of monitoring/recording	Measured for different dominant plant types at the beginning of the project
Value applied	22-36% depending on dominant plant type (A, F, N/A, P, M)
Monitoring equipment	Tools for clipping vegetation, drying oven, chemical digestion system, microbalance muffle furnace
QA/QC procedures to be applied	Compare measured values with other published values, written and digital data archived
Purpose of data	Calculation of baseline and project removals
Calculation method	(Mass lost in acid digestion – mineral content) / initial dry mass
Comments	None

Leakage

Data / Parameter	DN _{c,x}
Data unit	Number of project animals
Description	Number of project animals of type c off the project area during month x
Source of data	Measured in project area using reports from grazing coordinators documenting monthly movements of project livestock

Description of measurement methods and procedures to be applied	Grazing coordinators report livestock movements each month. Project livestock, as estimated by parameter $PN_{c,t}$ are allocated to one of three locations (on-conservancy, on-project, or off-project) based on monthly submitted grazing reports and the project's decision tree (Appendix A) for interpreting these reports.
Frequency of monitoring/recording	Monthly
Value applied	Depends on conservancy, month, and animal class. See 'Leakage and Encroachment Verification Calculations 2017 to 2020 v1 Sept 2021.xls'
Monitoring equipment	None
QA/QC procedures to be applied	Archival of digital versions of grazing reports in online databases, conversion of grazing report data to spreadsheets also stored in online databases, follow-ups from NRT personnel upon submission of reports by grazing coordinators
Purpose of data	Used to calculate leakage emissions under penalty approach of VM0032 (equation 17)
Calculation method	Off-project livestock days calculated as product of $DN_{c,x}$ and number of days in month x .
Comments	None

Data / Parameter	d
Data unit	Days
Description	Number of days any livestock were off the project area
Source of data	Measured in project area from reports of grazing coordinators as to the location of livestock from each Conservancy
Description of measurement methods and procedures to be applied	Grazing coordinators report livestock movements each month. Project livestock, as estimated by parameter $PN_{c,t}$ are allocated to one of three locations (on-conservancy, on-project, or off-project) based on monthly submitted grazing reports and the project's decision tree (Appendix A) for interpreting these reports.
Frequency of monitoring/recording	Monthly
Value applied	Depends on month
Monitoring equipment	None
QA/QC procedures to be applied	Archival of digital versions of grazing reports in online databases, conversion of grazing report data to spreadsheets also stored in

	online databases, follow-ups from NRT personnel upon submission of reports by grazing coordinators
Purpose of data	Used to calculate leakage emissions under penalty approach of VM0032 (equation 17)
Calculation method	Not calculated
Comments	None

Additional Monitored Parameters

In addition, we will measure an **ecosystem service variable**, gravimetric soil water content, to the depth soil is sampled (20 cm) at stations where biodiversity is measured or where soil carbon and vegetation is measured. Measures can be calibrated to recent rainfall MAP obtained from CHIRPS integrated satellite and weather station data and compared against NDVI-based measures of livestock impacts and movement.

Data / Parameter	SOILWATER% _{j,m,z}
Data unit	Dimensionless
Description	Soil water content as a% in stratum m, station j, year Z
Source of data	Measured in project area ³⁰
Description of measurement methods and procedures to be applied	Gravimetric method – save soil extracted for SOC determination in plastic bag, weigh wet. Air dry for 1-5 days (35-40°C), weigh dry. Water content in % = (1 – dry mass/wet mass)x100
Frequency of monitoring/recording	Once at the beginning of the project and then every 5-10 years as soils are re-sampled.
Value applied	Varies by location, 0-30%
Monitoring equipment	Plastic and paper bags, drying oven, shovel or soil auger or corer
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project benefit
Calculation method	Dry mass / wet mass of soil
Comments	None

³⁰ Note soil moisture was measured during the latest model assessment soil sampling, but was almost universally < 2% due to the extended drought prior to sampling. Thus there is no basis for comparing soil moisture between baseline and project scenarios.

In addition, data to verify activity were obtained from 72 MODIS satellite images and 468 monthly grazing reports for each conservancy during each month and year. These data include satellite images showing bi-monthly 15-day composite NDVI calculations and used to calculate mean NDVI scores for every 15-day period each year within designated grazing blocks within each Conservancy over the monitoring period. These data were used to calculate a change in NDVI ($\Delta\text{NDVI} = \text{earlier NDVI} - \text{later NDVI}$, with more positive values indicating stronger impacts, Fig. 3) between images, which serves as a measure of livestock impact on forage during the 15 days between composite images, as compared to the baseline and calibrated to ungrazed inaccessible areas to estimate grazing intensity.

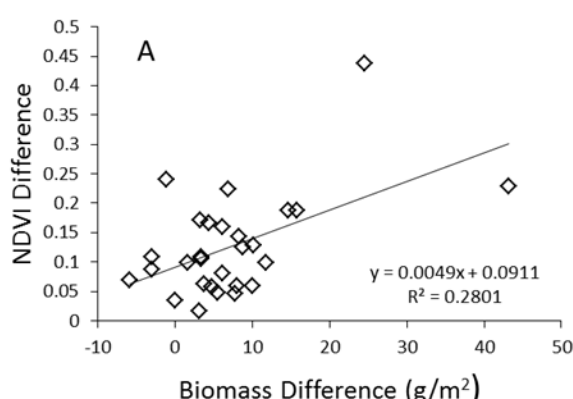


Figure 3. Relationship between measured forage biomass difference before and after grazing by 50 or more cattle and ΔNDVI between 28 July and 12 August, 2014 for thirty MODIS pixel areas within Kalama and Westgate Conservancies

Data / Parameter	NDVI _{jm,z}
Data unit	Dimensionless fraction
Description	Vegetation index calculated from MODIS spectral imagery (available at NOAA website)
Source of data	MODIS 15-day composite calculations of NDVI for tile 21,9 which provides NDVI index at 250 x 250 m resolution following atmospheric corrections and cloud masks.
Description of measurement methods and procedures to be applied	Metric associated with standing herbaceous biomass, ranging from negative 1 (water) to positive 1 (densest green rainforest canopy). Allows estimate of forage availability (Fig. 3) and grazing intensity (Fig. 6,7) from satellite imagery
Frequency of monitoring/recording	Every two weeks
Value applied	Varies by location 0.05 – 0.5 in grassland/savanna areas

Monitoring equipment	Data are free, by download from various sources, such as NASA in the US
QA/QC procedures to be applied	NDVI is projected in WGS89 , the same as the projection used for project boundary polygons. Images are produced on 16-day intervals and at multiple spatial resolutions, providing consistent spatial and temporal comparisons of vegetation canopy greenness, a composite property of leaf area, chlorophyll and canopy structure. The vegetation indices are retrieved from daily, atmosphere-corrected, bidirectional (both top and bottom of atmosphere) surface reflectance. The VI's use a MODIS-specific compositing method based on product quality assurance metrics to remove low quality pixels. From the remaining good quality VI values, a constrained view angle approach then selects a pixel to represent the compositing period (from the two highest NDVI values it selects the pixel that is closest-to-nadir).
Purpose of data	Calculation of project emissions/removals
Calculation method	Calculated as MODIS Data Product from reflected light of different spectral bands (frequencies) detected by satellite sensors Equation: $NDVI = (NIR - Red) / (NIR + Red)$
Comments	None

3.1.3 Monitoring Plan

The monitoring plan features four main components. First, monitoring of livestock numbers and movements using rangeland coordinator reports and satellite imagery is necessary to establish that Bunched Herd Planned Rotational grazing has been successfully implemented for at least a majority of livestock in the Conservancies. Secondly, vegetation cover and composition need to be monitored every 5-7 years to establish a new, reduced grazing intensity, so that the change in soil carbon (CO₂ removals) can be calculated with the SNAP model. Thirdly, project impacts on biodiversity are monitored through Conservancy-wide ranger surveys and by comparing biodiversity of indicator taxa across time (Fig. 24). Finally, the impacts of the project on communities and key groups (herders, elders, gender classes, and the poor) will be measured through focus groups and household surveys.

Organizational structure, responsibilities and competencies

The monitoring of project activities, consequences of these for vegetation and soils, project impacts on biodiversity, and changes in key social variables will be administered and coordinated by a joint carbon project task force, spearheaded by Soils for the Future and NRT personnel.

The Conservancies are groups of communities that occupy either designated Group Ranch lands or Trust lands to which they have acquired or are acquiring formal status as Conservancies with

rights to benefit from conservation activities, such as wildlife conservation that supports tourism. These communities are almost exclusively pastoral in culture, but are committed to education and to conservation-oriented grazing practices and land zoning. The Conservancies also serve as the source of most project employees involved as grazing assistants, biodiversity field technicians, data entry specialists, etc.

The Northern Rangelands Trust (NRT) (<http://nrt-kenya.com>) is a non-profit non-government organization devoted to helping the northern Kenya Conservancies develop governance, infrastructure and livelihoods that will bring peace, and prosperity to the region. NRT has in place professional personnel, such as grazing coordinators, and administrative competencies to help Conservancies develop and implement their own grazing plans, security teams, and monitoring capabilities. NRT has in place mechanisms to gather data from all Conservancies to build narratives that benefit all Conservancies through external funding, and in this case the data necessary to support claims of greenhouse gas removals. NRT also has established revenue sharing protocols to communities that allow substantial self-determined community-level benefits and projects. These protocols are a key mechanism of delivering carbon benefits to vulnerable and marginalized groups, namely women and the poorest 25% of households.

Soils for the Future, LLC (<http://www.soilsfuture.com>) (SftF) is a for-profit carbon development company, with established expertise in carbon accounting, field data collection, especially of soils and vegetation, and analysis of satellite images. Dr. Mark Ritchie, as President and author of more than 95 scientific publications, provides expertise in a variety of types of ecosystem and biodiversity modelling and satellite image interpretation. He also developed the methodology used in this project: VCS VM0032 methodology. Other SftF personnel have expertise with GIS, soil, ecosystem, and biodiversity measurements based on experience in Serengeti National Park and other field sites in Tanzania. Finally SftF has personnel with pastoralist backgrounds who speak many of the local languages and have experience with education about soil carbon projects at community and village levels.

The Nature Conservancy provides quality assurance in social data collection, with baseline household surveys and focus groups designed and conducted by Dr. Jessica Musengezi, a TNC social scientist with extensive experience working in Africa. Native Energy is the marketing partner with more than 18 years' experience marketing credits from nearly 800 projects.

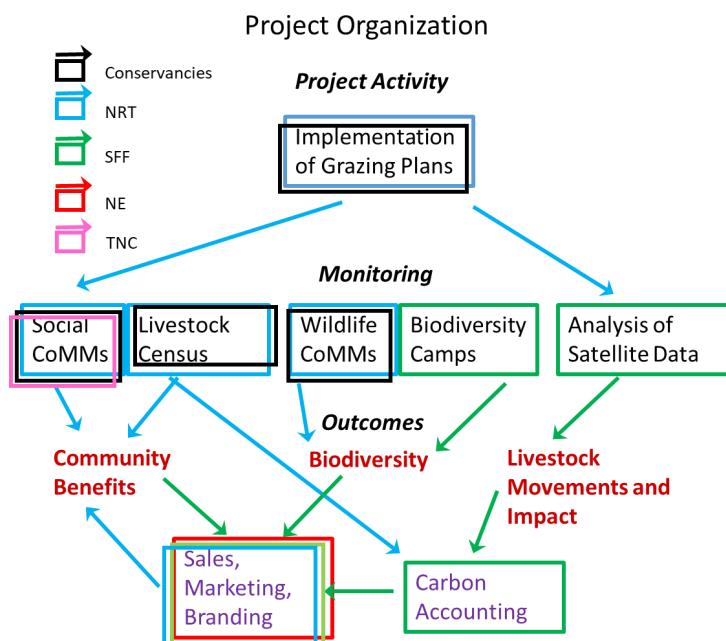


Figure 4. Diagram of project organization, including color-coded responsibilities (boxes) and information flows (arrows). Project actions are divided into project activity, monitoring, and accounting, with red text indicating project outcomes. Black boxes indicate Conservancy responsibilities, blue boxes and arrows indicate NRT responsibilities and sources of information, green boxes and arrows indicate Soils for the Future, LLC responsibilities and sources of information, and red boxes indicate TNC responsibilities. Boxes with multiple colors indicate joint responsibilities.

The overall project organization, task responsibilities and flow of information is given in Fig. 4. The project is truly a joint effort between the three organizations, as indicated by an available MOU between the three parties.

Risks to community participants are communicated upon hiring as part of standard operating procedures. For example, see the procedures for game scouts³¹ who also monitor borders and engage with encroaching off-project herders. Procedures address risks and mitigation previously discussed in section 2.3.

Within-Conservancy governance structures

Within-Conservancy governance links to the Conservancy partnership with NRT and administration of the carbon project through a Council of Elders made up of community members that oversees and approves NRT initiatives and conduct (Figure 2). Within Conservancies, individual herders are advised as to time and timing of livestock movements by Conservancy grazing committees staffed by elected chairs. The grazing committees coordinate recommendations and requirements with village chairs and with Conservancy managers. Rangeland coordinators are hired by the Conservancies to directly monitor, advise, and implement the planned rotational grazing program. Individual herder compliance is leveraged by

31

https://static1.squarespace.com/static/5af1629f12b13f5ce97ca0b5/t/5b6330d9575d1f66316bf5f0/1533227321252/Wildlife-CoMMS+guide_complete_low_res.pdf

social conformity pressures and eligibility for community benefits and for the “livestock to market” program conducted by TNC to facilitate sales of livestock to producers in Laikipia County who can “finish” livestock to make them market worthy.

A schematic of this complex and highly public decision-making is provided in Section 2.3.10 (Figure 2) and illustrates the central roles of public grazing committee meetings and the NRT grazing coordinators.

Methods for generating, recording, storing, aggregating, collating and reporting data on monitored parameters

All field data is recorded by hand into field notebooks. Data is scanned and stored electronically and then entered into Excel or other similar spreadsheets or digitized onto GIS layers analyzed in QGIS, a free GIS software program. The project database will be kept jointly by Soils for the Future and NRT. Meta-data will be built so that it will be transparent how all raw data can be explicitly linked into the equations that use it, per methodology VCS VM0032.

Soil carbon (SOC%) is measured to 20 cm depth with a standard steel pipe (5 cm diameter). Augured soil from two cores taken 3 m apart is combined and mixed in a plastic tub and sorted to remove rocks and clods are broken with a wooden rolling pin. Two 100 g samples are taken from the mixed soil and one is submitted for analysis using a Mid-InfraRed (MIR) at the International Centre for AgroForestry (ICRAF), Nairobi, and the other is archived under locked storage at NRT headquarters. Calibration of the MIR method is done by regressing MIR spectral frequencies against SOC percentage estimated by a combustion method at Crop Nutrition Specialists, Nairobi.

Soil bulk density is estimated to a depth of 20 cm with a steel cylinder (made of cut 5 cm diameter pipe) from one core at each sample location. A metal plate is welded to the top to allow the pipe to be pounded into soil with a sledgehammer. A separate 5 x 5 cm plate, welded to a 1 m x 1.25 cm diameter pipe handle, is inserted underneath the steel cylinder after it is pounded to 20 cm depth to prevent soil loss upon lifting the cylinder and its soil sample out of the ground. Soil from the cylinder is spread on a table or board and rocks > 2 mm in size are sifted from the sample. Soil clods are broken up with wooden rolling pins. Rocks will be added to a volumetric cylinder containing a known initial volume of water, and the volume of rocks will be measured from the increase in volume of the water column inside the cylinder. Soil, after rocks are removed, is weighed while moist M_w and then air-dried in paper bags (at approximately 40°C) for 1-5 days and then weighed again to obtain dry weight M_d . Soil moisture(%) is calculated as $100 \cdot (1 - M_d/M_w)$. Bulk density is calculated as the rock-free dry weight divided by the (volume of the cylinder - volume of rocks). All volumes are cm^3 .

Likewise, spatial data such as sampling stations, biodiversity sampling camps, observation locations of key species that confer High Conservation Value and satellite remotely sensed images will also be maintained in the project database. Files will be stored as both shape files (readable only in GIS software programs) and as bitmap or TIFF files with meta-data that would allow geo-referencing by future software programs.

Analyses of all social, livestock census, biodiversity, vegetation, and satellite data for Monitoring Reports are done primarily by SftF with assistance from NRT monitoring personnel. All these data are stored on shared Dropbox accounts.

To reduce the risk that sampling stations cannot be relocated, all sampling stations were photographed with a GPS digital camera (images available upon request) from both a northward and southward view.

All documents and records will be kept in a secure and retrievable manner for at least two years after the end of the project crediting period. Data will be stored as paper copies at NRT headquarters and electronically in off-site electronic data storage vehicles, such as Dropbox and Syracuse University backup servers. Over the project lifetime, these off-site storage vehicles may change as technology develops; the objective is to maintain the data independently of particular personnel or offices involved in the project.

Procedures for handling internal auditing and non-conformities

Data arrives from multiple field crews and sources (different satellite images), and such data faces some risk of error in reporting and converting into spreadsheet databases. Data checking protocols will be devised at three levels – supervisors of field crews, NRT and SftF scientists. Multiple personnel within the project will check data and calculations to ensure transparency and internal auditing. Evidence for project activities that reduce greenhouse gas removals are reviewed and cross-checked with supervisors of field crews, grazing coordinators, Conservancy managers per guidelines provided in VCS AFOLU Requirements 3.7^[11].

Specifically, all raw field data and satellite images are gathered and maintained by Soils for the Future. Copies of raw data will be stored online (currently in Microsoft DropBox) and shared with all partners. Data analyses will be submitted as reports to appropriate supervisory personnel, such as Kieran Avery, Rangeland Coordinator for NRT, or Tom Lalampaa, NRT CEO, who consult with specialists to confirm if monitoring results reconcile with on-the ground knowledge.

In the case of discrepancies, project leaders meet to discuss the scope of the problem, evaluate solutions, and decide a course of action. For example, when satellite imagery detected grazing impacts in the Mathews forest, and NRT personnel questioned whether this was true, SftF met with NRT personnel to discuss possible errors and use rangeland coordinator grazing reports as a second source of data. After evaluation, virtual meetings via skype discussed the ability to infer livestock impacts, and the different personnel agreed that it could, and satellite-based impacts of livestock on the Mathews Forest were included in annual reports of NRT for the first time. These same data are part of the evidence for exceptional biodiversity benefits.

The implementation of sampling approaches, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures. Where applicable, demonstrate whether the required confidence level or precision has been met.

Sampling points are based upon the stratified random method and defined by the pre-sampling study design. Individual “strata” comprise conservancy boundaries, and sample points are in some cases positioned to account for local variation in biophysical characteristics.

- Sampling personnel must be familiar with local conditions, soils, and flora and be able to extract intact soil samples, preserve all sample material, and label all samples clearly and correctly.

- Sampling locations are reachable by coordinates identifiable from a GPS device. Each location comprises one bulk density sample and two SOC samples (the two SOC samples are pooled at each location). The intact core method is used, and samples are to a maximum depth of 20 cm.
- Successive sampling at each location occurs at a predefined interval (e.g., every 5 years).
- Repeated surveys should measure soil bulk density and SOC concentration using identical methods.
- The average of the measured SOC% values is compared to the average SOC% predicted by the model. If measured values are outside of the range allowed by the methodology, the model must be recalibrated using the steps outlined.

3.1.4 Dissemination of Monitoring Plan and Results (CL4.2)

Given the expanse of the project area and the paucity of reliable telecommunications, the monitoring plan is still disseminated in a series of separate actions. 80% of monitoring is performed by communities and is subdivided into wildlife, vegetation, and social, with each having a separate community-based management system (CoMMs); the results of these surveys are shared annually with the conservancies who then communicate the results to communities. Livestock monitoring is performed by grazing coordinators who are members of the communities they monitor, and their activities are communicated to the public in grazing committee meetings within Conservancies; grazing committee meetings occur quarterly.

Larger and broader monitoring plans, such as those involving satellite imagery and cross-Conservancy issues such as seasonal migratory movements, have not been presented since May 2016 and September 2017 as their relevance to the communities is limited without the context of the verification and issuance of VERs which can then be subsequently sold. There are however plans to communicate these in greater detail at a forthcoming briefing meeting with Conservancy Leadership and Management teams in late January 2022 so that this information can be appropriately conveyed to communities through each respective conservancy AGM held during the course of the year (most will occur Feb-April). It is important that these results are communicated to communities in the context of the project to enhance the understanding and application of the project activities, and now visible results.

There were additional meetings during the 2017-2018 drought period focused at the regional level to bring all conservancies together across the region. Individual and regional plans were shared during these meetings and plans outlined for the movement of livestock during the drought. Such meetings were mostly held quarterly during this time and areas of high livestock concentrations (such as in Sera and Biliqo) were identified as high priority and afforded additional planning so that the surrounding conservancies could support them achieve their own grazing plans, as well as foster peaceful coexistence among different ethnic communities grazing in those areas.

Most of the conservancies in the project area were impacted by the locust invasion 2019-2020 in Northern Kenya. However, communities themselves cannot do much to control the movement of locust except to report the location of the swarms to authorities to facilitate a coordinated control effort between the government and Food and Agriculture Organization of the United Nations. This

included spraying swarms of locust in affected areas. However, communities did not carry out any additional meetings on the locust invasion itself.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Methodology VM0032 calculates baseline emissions as the sum of baseline methane emissions from livestock and baseline emissions from biomass burning (VM0032 equation 19). Biomass burning is not affected by this project and emissions from burning biomass are therefore counted as *de minimis*. Baseline methane emissions from grazing animals were calculated from equations 1 and 2 in methodology VM0032. Animal count data was collected for grazing animals of category c of each of three animal types: ruminants (sheep, goats, cattle, buffalo, and camelids (camels), donkeys, and pigs.

Number of people, NP, for each Conservancy at the end of the monitoring period, 2020 were obtained from numbers reported by each Conservancy in the NRT website interactive map³² and are reported in Table 1. The number of households NHH in the project area was calculated as:

$$NHH = NP / MHHN \quad (1)$$

Where mean household size MHHN was found to be 5.59 (+ 0.09 SE, uncertainty 0.06) in household surveys conducted in 1030 households in four different Conservancies in 2014.

Table 2. Reported Northern Kenya Grasslands Carbon Project Conservancy population sizes in 2020.

Conservancy	People	Households ¹
Biliqo Bulesa	5,300	948
IINgwesi	2,137	382
Kalama	9,958	1,781
Leparua	8,590	1,537
Meibae	12,235	2,189
Melako	20,974	3,752
Naapu-Nanapisho	6,833	1,222
Naibunga	19,606	3,507
Nakuprat-Gotu	6,734	1,205
Namunyak	32,504	5,815
Nasuulu	5,200	930
Sera	7,214	1,291
Westgate	4,494	804
Total	141,779	25,363

³² <https://www.nrt-kenya.org/interactive-map>

1 <https://www.nrt-kenya.org/interactive-map>

1 Based on estimates of number of people per household in 2014 Household Survey of 1030 households

Conservancy population sizes in the years 2017 and 2019 were estimated by regression of reported population size in 2009, 2011, 2014, 2018 and 2020 (see example in Fig.5, all relationships shown in Appendix B) versus year since 2009. and number of households was determined by multiplying each of these population estimates by mean household size (assumed to be constant across years)

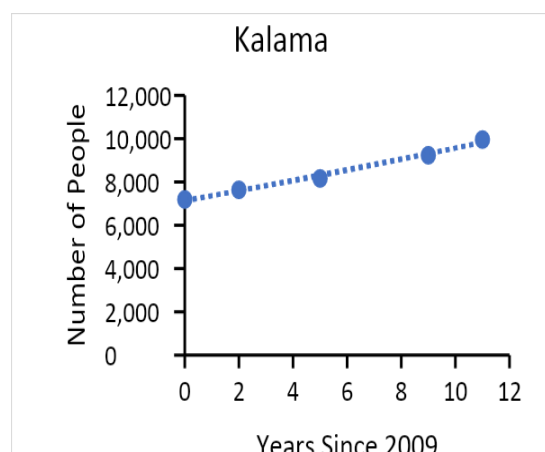


Figure 5. Example analysis of population trends versus years since the 2009 Kenya census in each Conservancy, shown here for Kalama. Regression yields an equation for exponential population growth, including the per capita growth rate in the exponent: $NP = N_0 \times e^{\text{coeff} \times (\text{Years Since 2009})}$

These population trend equations were multiplied by mean household size (Eq. 1) to estimate the number of households in each Conservancy for each year of the monitoring period (Table 3).

Conservancy	Coefficient, N_0	Per capita growth rate, coeff	Estimated Households			
			2017	2018	2019	2020
Biliqo Bulesa	4265	0.0204	898	917	936	955
IlNgwesi	1231.5	0.0443	314	328	343	359
Kalama	7163.1	0.029	1,616	1,664	1,713	1,763
Leparua	7810.3	0.009	1,501	1,515	1,529	1,543
Meibae	8773.7	0.0337	2,055	2,126	2,199	2,274
Melako	10939	0.0595	3,150	3,343	3,548	3,765
Naapu-Nanapisho	4953.3	0.0299	1,126	1,160	1,195	1,231
Naibunga	10181	0.0575	2,887	3,059	3,240	3,432
Nakuprat-Gotu	4864.7	0.0227	1,044	1,068	1,092	1,117
Namunyak	23358	0.0305	5,344	5,510	5,682	5,858
Nasuulu	3743.9	0.0275	835	859	883	907
Sera	6154.2	0.011	1,202	1,215	1,229	1,243
Westgate	3241	0.0253	710	728	747	766
			22,683	23,491	24,333	25,212

Table 3. Regression parameters for population trends and estimated number of households for each Conservancy in the Northern Kenya Grasslands Carbon Project.

Next the mean number of livestock of different classes c (cattle, shoats, donkeys, camels) per household was determined from a survey of 1030 households in 2014. These means were assumed to be similar to

2014 in 2017-2020. This method was used because many Conservancies were highly inconsistent in reporting actual livestock numbers, even in months where all livestock were reported to be present on the Conservancy, or simply did not report actual livestock numbers.

The total livestock of class c in year t , $N_{c,t}$ is given by

$$N_{c,t} = HSN_{c,t} \times NHH \quad (2)$$

where

$N_{c,t}$ = total number of livestock of class c in year t

HSN_t = the mean number of livestock of class c per household in the 2014 herder's survey for year t

NHH = number of households in the project area.

These calculations yielded estimates of livestock numbers for each Conservancy (Tables 4A-D) except for Namunyak. Livestock numbers reported on monthly grazing reports for each of the three Units in Namunyak were highly consistent among months when there was no report of encroachment and all livestock were reported to be on each Unit. However, the numbers reported were substantially (4-5x) higher than those estimated from Eq. 2. The likely reason for this is that herders in Namunyak routinely access relatively high rainfall savanna and forest areas in their seasonal movements, which allows herders to maintain higher than average herd sizes, especially of cattle (camel and donkey numbers reported in grazing reports were similar to those estimated from Eq. 2). Consequently, for Namunyak, we used the estimates for each Unit from the first month of each year in which livestock were reported to be fully present on the Unit and no encroachment was indicated.

Table 4. Estimated livestock numbers for cattle, sheep and goats (shoats), camels, and donkeys in the 13 Conservancies of the Northern Kenya Grasslands Carbon Project for the 2017-2020 monitoring period. Green shading indicates numbers taken directly from grazing reports.

A						B					
Conservancy	2017					Conservancy	2018				
	Estimated Households	Cattle	Shoats	Camels	Donkeys		Estimated Households	Cattle	Shoats	Camels	Donkeys
Biliqo Bulesa	898	4,974	22,778	682	705	Biliqo Bulesa	917	4,974	22,778	682	705
Il Ngwesi	314	1,739	7,963	238	246	Il Ngwesi	328	1,739	7,963	238	246
Kalama	1,616	8,949	40,981	1,227	1,268	Kalama	1,664	8,949	40,981	1,227	1,268
Leparua	1,501	8,315	38,077	1,140	1,178	Leparua	1,515	8,315	38,077	1,140	1,178
Meibae	2,055	11,381	52,119	1,560	1,612	Meibae	2,126	11,381	52,119	1,560	1,612
Melako	3,150	17,443	79,878	2,391	2,471	Melako	3,343	17,443	79,878	2,391	2,471
Naapu-Nanapisho	1,126	6,233	28,543	855	883	Naapu-Nanapisho	1,160	6,233	28,543	855	883
Naibunga	2,887	15,990	73,221	2,192	2,265	Naibunga	3,059	15,990	73,221	2,192	2,265
Nakuprat-Gotu	1,044	5,779	26,464	792	819	Nakuprat-Gotu	1,068	5,779	26,464	792	819
Namunyak	5,344	72,658	130,734	8,889	6,232	Namunyak	5,510	72,658	130,734	8,889	6,232
Nasuulu	835	4,625	21,181	634	655	Nasuulu	859	4,625	21,181	634	655
Sera	1,202	6,658	30,487	913	943	Sera	1,215	6,658	30,487	913	943
Westgate	710	3,931	18,001	539	557	Westgate	728	3,931	18,001	539	557
	22,683	168,677	570,426	22,053	19,834		23,491	168,677	570,426	22,053	19,834

C	2019					D	2020				
	Estimated Households	Cattle	Shoats	Camels	Donkeys		Estimated Households	Cattle	Shoats	Camels	Donkeys
Conservancy						Conservancy					
Biliqo Bulesa	936	5,181	23,727	710	734	Biliqo Bulesa	955	5,288	24,216	725	749
Il Ngwesi	343	1,900	8,701	260	269	Il Ngwesi	359	1,986	9,095	272	281
Kalama	1,713	9,484	43,428	1,300	1,343	Kalama	1,763	9,763	44,706	1,338	1,383
Leparua	1,529	8,466	38,768	1,161	1,199	Leparua	1,543	8,543	39,119	1,171	1,210
Meibae	2,199	12,175	55,752	1,669	1,725	Meibae	2,274	12,592	57,663	1,726	1,784
Melako	3,548	19,648	89,972	2,694	2,783	Melako	3,765	20,852	95,488	2,859	2,954
Naapu-Nanapisho	1,195	6,617	30,302	907	937	Naapu-Nanapisho	1,231	6,818	31,222	935	966
Naibunga	3,240	17,942	82,162	2,460	2,542	Naibunga	3,432	19,006	87,033	2,606	2,692
Nakuprat-Gotu	1,092	6,047	27,693	829	857	Nakuprat-Gotu	1,117	6,186	28,329	848	876
Namunyak	5,682	65,248	121,049	9,222	9,342	Namunyak	5,858	71,078	147,854	12,704	8,927
Nasuulu	883	4,888	22,383	670	692	Nasuulu	907	5,025	23,009	689	712
Sera	1,229	6,806	31,165	933	964	Sera	1,243	6,881	31,510	943	975
Westgate	747	4,135	18,936	567	586	Westgate	766	4,241	19,421	581	601
	24,333	168,538	594,037	23,383	23,974		25,212	178,259	638,664	27,398	24,110

The document reviewed by the VVB: “NKCP Methane Calculations 2017-2020 v1 Sept 2021” shows these calculations in detail. The total baseline emissions for each year of the monitoring period are 251,918 tCO₂e. Uncertainty in baseline emissions was calculated according to the equations in section 8.4.2.1.1 of VM0032, resulting in an annual uncertainty of 7.12%.

Baseline Soil Organic Carbon

Baseline soil organic carbon (SOC) was calculated by estimating the maximum SOC in the previous 10 years at each sampling station. To do this, we back-casted the SNAP model over the 10 years leading up to the project start date, as required by VM0032. Following the back-casting procedure, baseline SOC values were on average 30% higher than measured SOC, resulting in a conservative estimate of baseline SOC. See the Project Description for a detailed overview of the back-casting procedure. In addition, the VVB has reviewed the spreadsheet ‘NKCP Verification Calculations 2017-2020 v1 Sept2021 for the final calculations.’

Baseline soil organic carbon was calculated using the requirements outlined by the modeled approach of VM0032. The project uses the SNAP model of soil carbon dynamics^[10], a carbon mass balance model developed from the grassland savanna – grazing system in Serengeti National Park^[14]. This model was locally calibrated to the project area in northern Kenya, as is allowable by VM0032 (see Errata and Clarifications) to account for differences in dominant plant cover (more annual plants in Kenya than in Serengeti) and grazing intensity (higher in Kenya than in Serengeti). Under this calibration, the model meets the baseline performance requirements outlined by VM0032 (Figure 6). See the spreadsheet NKCP Verification Calculations 2013-2016 v2 June2020 for all calculations and the Project Description for a detailed review of the local calibration.

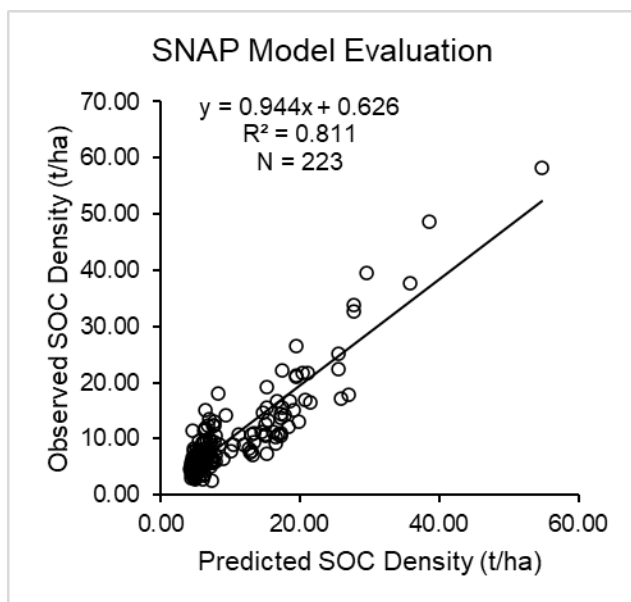


Figure 6: Evaluation of SNAP soil carbon model against requirements of methodology VM0032. VM0032 requires that the chosen model predict baseline SOC stocks with $R^2 > 0.80$ and that 95% confidence intervals for slope contain 1 (0.884 – 1.005 range for the project) and intercept contain 0 (–0.062 – 1.314 for project).

3.2.2 Project Emissions

Net annual GHG emissions and removals under the project scenario, NPR_t is determined, in the absence of any biomass loss to burning, by the annual net methane emissions, PER_t , (equation 19 in VM0032, page 38) and net removals from SOC sequestration, PRS_t

$$NPR_t = PER_t + PRS_t \quad (23)$$

Methane Emissions

A. Methane emissions from project livestock.

Livestock numbers were used to estimate project scenario emissions during the monitoring period 2017-2020. The attached document: "NKCP methane calculations 2017-2020 v1 Sept 2021.xls" shows these calculations in detail, as summarized in Table 5. The total baseline emissions for each year of the monitoring period are 251,918 tCO₂e, an amount 110,945 tCO₂e per year higher than baseline. Uncertainty in baseline emissions was calculated according to the equations in section 8.4.2.1.1 of VM0032, resulting in an annual uncertainty in methane emissions of 7.12%.

Grazing Animal Category					Animal Census (Number of Animals)						Methane Emissions		
Species	Sex/Age	Weight (kg)	Annual Methane Emissions/Animal, tCO ₂ e	Per Animal Uncertainty in Methane Emissions %	2017	2018	2019	2020	Mean	Uncertainty in Animal Counts	Methane Emissions for Category (tCO ₂ e)	Uncertainty in Methane Emissions %	Weighted Uncertainty
Cattle ¹	Adult	255	0.91	9.5	127,623	130,758	127,518	134,874	130,193	5.90	118,714	11.18	1.76.E+12
	Juvenile	175	0.63	9.5	41,053	42,062	41,019	43,385	41,880	5.90	26,505	11.18	8.78.E+10
Shoats ²	Adult	35	0.13	9.5	430,273	442,579	448,083	481,745	450,670	10.83	59,865	14.41	7.44.E+11
	Juvenile	18	0.07	9.5	140,153	144,161	145,954	156,919	146,797	10.83	10,231	14.41	2.17.E+10
Donkeys	Adult	225	0.22	28.2	17,088	17,697	18,331	18,994	18,027	10.09	3,971	29.95	1.41.E+10
	Juvenile												
Camels ³	Adult	452	1.59	9.5	15,775	15,070	16,726	19,599	16,793	26.27	26,679	27.94	5.56.E+11
	Juvenile	249	0.89	9.5	6,278	5,997	6,656	7,799	6,683	26.27	5,954	27.94	2.77.E+10
Total Livestock					778,244	798,324	804,289	863,315		Total Emissions	251,918	Total Uncertainty %	7.12

Table 5. Calculations of methane emissions for the Northern Kenya Grasslands Carbon Project during the monitoring period 2017-2020 from sums of estimates of livestock numbers for each Conservancy.

B. Methane Emissions from encroaching livestock.

Project methane emissions from off-project livestock that encroached onto the project area, $PEM_{t,encroach}$ were accounted for using encroachment-days (parameter $ED_{c,t}$) and Eq. 3, a modified version of Equation 8 in VM0032 that adjusts annual methane emissions by the number of days encroaching livestock were on the project area. Uncertainty in methane emissions from encroaching livestock was calculated by following Section 8.4.2.2.1 in VM0032.

$$PEM_{t,encroach} = \sum_{c=1}^K (ED_{c,t} \times DMEf(W_{c,t})) \times GWP_{CH_4} \times 6.26 \times 10^{-7} \quad (3)$$

Where:

$PEM_{t,encroach}$ = Project emissions of CH₄ from encroaching grazing animals during year t (tCO₂e)

$ED_{c,t}$ = Number of encroachment-days of each livestock category c in year t

$DMEf(W_{c,t})$ = Daily emission factor as a function of animal weight for category c (L CH₄ / day).

Functions used come from Table 4 of VM0032 (pg. 24).

$W_{c,t}$ = Average body weight during year t for animals of category c (kg)

GWP_{CH_4} = Global warming potential of methane (tCO₂e / tCH₄)

6.26×10^{-7} = Conversion factor for L CH₄ / day to tCH₄ / day

The key variable is $ED_{c,t}$, the number of animal days off-project livestock were on the project area, as estimated from narratives, map symbols and arrows indicating movements and sources of off-project livestock and/or reporting of additional livestock numbers well in excess of estimated livestock numbers for a Conservancy. Monthly incidents of encroachment are provided for VVB review in the attached file “Leakage and Encroachment Verification Calculations 2017 to 2020 v1 Sept 2021.xls”, Encroachment Records tab, and sums of animal days for each livestock class by year is provided in the Encroachment Summary tab of the same file.

Table 6. Animal-days of encroaching livestock from off the project area onto the project area.

	2017	2018	2019	2020	Total
Cattle Days	261,652	-	2,508,532	792,625	3,562,808
Shoat Days	599,084	52,700	3,790,460	2,230,112	6,672,355
Donkey Days	18,532	-	2,078	120,801	141,411
Camel Days	17,936	60,450	183,960	879,521	1,141,867
Total	897,203	113,150	6,485,030	4,023,058	11,518,441

Table 7. Calculations of methane emissions by encroaching livestock for the Northern Kenya Grasslands Carbon Project after converting animal days into an equivalent annual number of livestock (animal census columns)

Grazing Animal Category					Animal Census (Number of Animals)						Methane
Species	Sex/Age	Weight (kg)	Annual Methane Emissions/Animal, tCO ₂ e	Per Animal Uncertainty in Methane Emissions %	2017	2018	2019	2020	Mean	Uncertainty in Animal Counts	Methane Emissions for Category (tCO ₂ e)
Cattle ¹	Adult	255	0.91	9.5	542	-	5,200	1,643	1,846	280.73	1,684
	Juvenile	175	0.63	9.5	174	-	1,673	529	594	280.73	376
Shoats ²	Adult	35	0.13	9.5	1,242	109	7,857	4,623	3,458	224.67	459
	Juvenile	18	0.07	9.5	399	35	2,528	1,487	1,112	224.67	78
Donkeys	Adult	225	0.22	28.2	37	125	381	1,823	592	312.40	130
					-	-	-	-			
Camels ³	Adult	452	1.59	9.5	38	-	4	250	73	361.00	116
	Juvenile	249	0.89	9.5	12	-	1	81	24	361.00	21
Total Livestock					2,446	270	17,645	10,436		Total Emissions	2,864

Encroachment events occurred, but they accounted for only slightly more than 1% of methane emissions from project livestock and thus were considered negligible.

Encroaching livestock accounted for emissions of 2,864 tons of methane (Table 6), which is only 1.12% of total (project + encroaching livestock) methane (251,918 tons CO₂e (Table 5). Therefore methane emissions by encroaching livestock can be considered *de minimis* (negligible).

Project Changes in SOC Density

Project changes in SOC density were calculated using the SNAP model in accordance with section 8.2.3 of methodology VM0032. The proportion of the project area that experienced successful implementation of project activities was calculated using the previously calibrated and reviewed method that associates change in a satellite measure of green biomass (NDVI) with the number of grazing events each year.

Analysis of satellite imagery demonstrated substantial variation, 45 – 90%, in the successful implementation of planned rotational grazing and seasonal migrations over the years. This variation was driven by an extreme drought from 2017 into 2018 and possibly the influence of widespread locust swarms in 2020, both of which led to reduced success in implementing grazing management.

Assessing grazing events and successful implementation of project activities.

We used the criterion of $\Delta NDVI < 0.05$ (500 in the raw data) to identify MODIS pixels (250 x 250 m) where rotational grazing practices were successful each year during 2017-2020. For a given pixel i , NDVI measures are determined for two consecutive months, $m, m+1$, in **project** pixels where project success is being evaluated and

$$\Delta NDVI_{i,Y} = \sum_{m=1}^{12} (NDVI_{i,m,nonref} - NDVI_{i,m+1,nonref}) - \sum_{m=1}^{12} \sum_{n=1}^n (NDVI_{i,m,ref} - NDVI_{i,m+1,ref}) / n \quad (4)$$

where

n = the number of pixels in the reference area(s) where little or no grazing occurs
 $NDVI_{i,m,nonref}$ = NDVI measure from pixel i in month m in non-reference areas where livestock routinely graze.
 $NDVI_{i,m,ref}$ = NDVI measure from pixel i in month m in reference area
 $\Delta NDVI_{i,Y}$ = Net relative change in NDVI in pixel i for year Y .

Reference pixels are those in designated polygons that receive little to no grazing because of management (Core Areas in Il Ngwesi, Westgate, and Sera Conservancies) or topographic inaccessibility (e.g., steep-walled plateaus in Kalama). Monthly changes in NDVI in reference pixels help control for weather related changes in NDVI, such that changes in NDVI in project pixels. These calculations were performed in Google Earth Engine³³. The results for the monitoring period years 2017, 2018, 2019, and 2020 are presented in Table 8. Calibration and ground-truthing of delta NDVI with grazing coordinator observations of livestock numbers and movements is presented in Figures 7-9.

The outcomes reflect the broad narrative of climate variation that occurred during the monitoring period. The project area received little rain during the April-May and November peak rainfall months in 2017, which led to repeated grazing events in many portions of the project area during November 2017-April 2018 and, as we show in the next section, substantial leakage. However abundant April-May rains in 2018 and 2019 allowed livestock to remain on their home Conservancies for 8-16 months, and thus successful implementation of grazing plans. Consequently, 2017 featured a relatively low successful implementation, while 2019 featured relatively high success. In December 2019, April-May 2020 and again in November 2020, portions of the project area experienced widespread, dense locust swarms that may have depleted forage in some localities and may have disrupted planned livestock grazing rotations and migration (Fig. 10). Thus successful implementation in 2020 occurred on only approximately 75% of the project area.

Figure 7. Example correspondence between recorded livestock use (left hand map) and the $\Delta NDVI$ measured by MODIS satellite (right hand map) for grazing in the wet season (A. Namunyak Conservancy) and in the dry season (B. Biliqo Bulesa Conservancy). Black outlined polygons indicate major areas where livestock were reported by CGCs and $\Delta NDVI$ indicated significant livestock impacts. Red outlined polygons indicate areas where no livestock were reported but $\Delta NDVI$ indicates significant impacts. Areas

³³ <https://earthengine.google.com/>

without polygons showed no impacts as detected by ΔNDVI and no livestock were reported. Across the conservancies, livestock recordings by rangeland coordinators corresponded strongly with measurements of ΔNDVI as indicated by C., the logistic regression of probability of livestock use $PL = 1/(1 + \exp(2.95 - 29.25 \cdot \Delta\text{NDVI}))$, $R^2 = 0.48$, $N = 116$ random points, $P < 0.0001$.

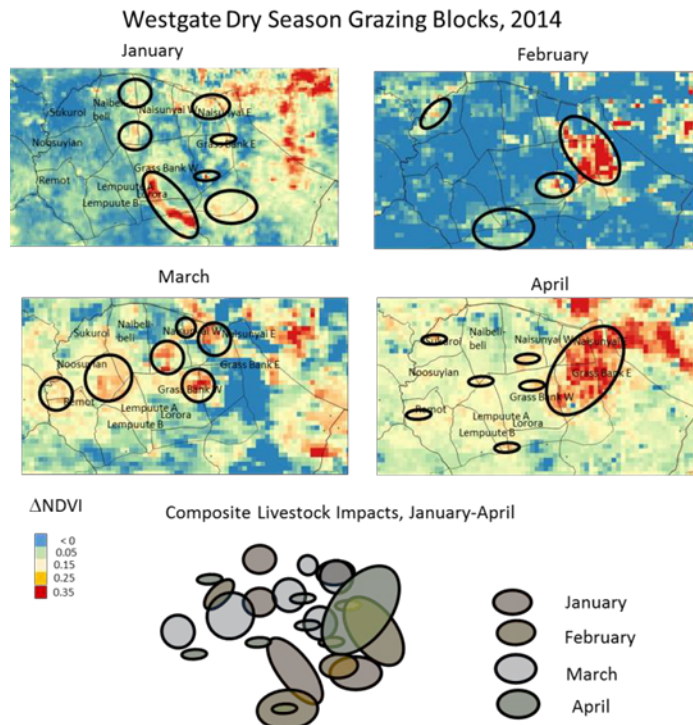


Figure 8. Example of shifting grazing impacts (ΔNDVI) during livestock movements in the first third of 2014 in Westgate Conservancy, northern Kenya. Ellipses indicate areas of grazing impact ($\Delta\text{NDVI} > 0.1$, conferring a probability of livestock presence $> 50\%$). These same ellipses are overlaid and color-coded for each grazing month and indicate significant overlap only in the far northeastern portion of Westgate.

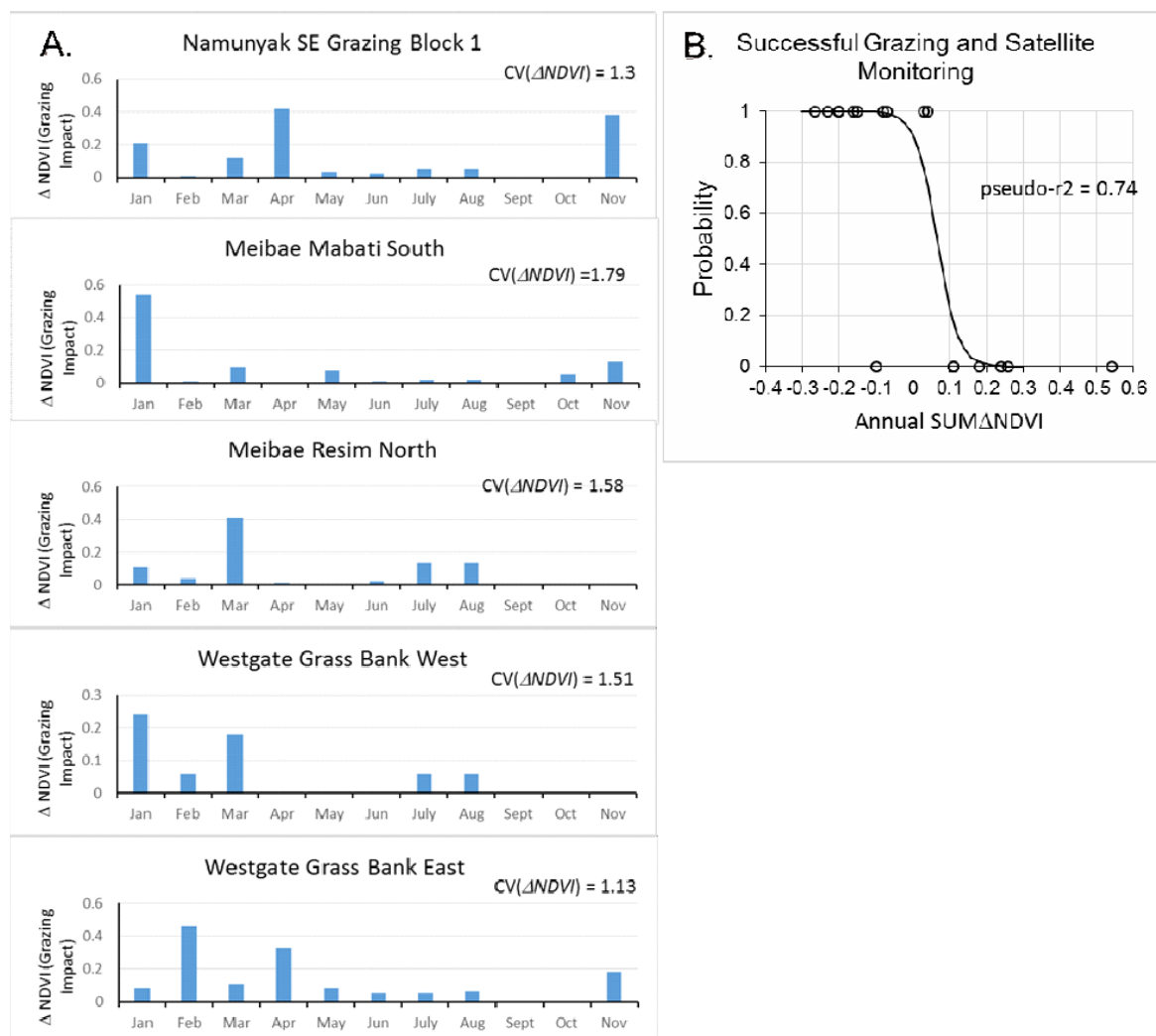


Figure 9. A. Example temporal patterns of monthly grazing impacts, measured as the mean decrease in NDVI from the previous month (Δ NDVI), on selected grazing blocks in southern Samburu County, Kenya from Meibae, Westgate, and Namunyak Conservancies for 2014. These examples clearly show the difference in temporal pattern between unsuccessful and successful rotational grazing practices. Unsuccessful patterns include Namunyak SE Grazing Block 1 and Westgate Grass Bank East, which were grazed (Δ NDVI > 0.1, see Figures 8 and 9) three times during the year, and especially twice during the wet season. Successful patterns included Meibae Mabati South and Meibae Resim North, which were grazed only once. B. Logistic regression to determine probability of successful grazing implementation, p , as a function of annual SUM Δ NDVI: $p/(1-p) = 2.3 - 35.2(\text{SUM}\Delta\text{NDVI})$, approximately 74% of the variation explained.

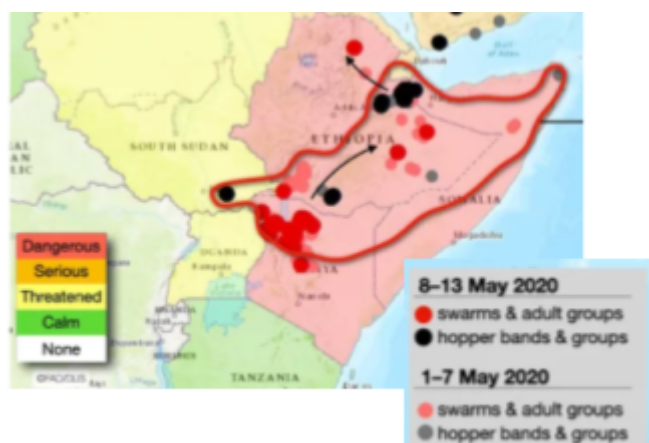


Figure 10. Locations of known locust swarms in the Horn of Africa during the April-May wet season 2020. Note the many adult swarms in north of central Kenya that occur on some parts of the project area.

Table 8. Proportion of each Conservancy that successfully achieved a single high intensity grazing event, as evaluated from monthly changes in NDVI.

Transcribed from report of Google Earth Engine model that calculated proportion of pixels with delta NDVI > 0.05 for each Conservancy relative to delta NDVI in reference lightly grazed polygons					
NDVI Threshold = 0.05		2017	2018	2019	2020
Name	Total area ha	Proportion Successful	Proportion successful	Proportion Successful	Proportion Successful
Biliqo	378,751	0.00	0.75	0.97	0.75
Il Ngwesi	9,432	0.39	0.54	0.93	0.77
Kalama	49,676	0.60	0.48	0.82	0.83
Leparua	32,833	0.22	0.89	0.89	0.76
Meibae	101,618	0.84	0.13	0.82	0.86
Melako	549,354	0.54	0.47	0.98	0.75
Naibunga	47,085	0.88	0.65	0.82	0.85
Nakuprat-Gotu	71,282	0.00	0.90	0.97	0.74
Namunyak	288,232	0.59	0.55	0.87	0.92
Nasuulu	34,601	0.34	0.95	0.88	0.76
Naapu-Nanapisho	54,442	0.70	0.25	0.93	0.78
Sera	339,520	0.27	0.63	0.97	0.75
Westgate	36,250	0.69	0.31	0.88	0.77

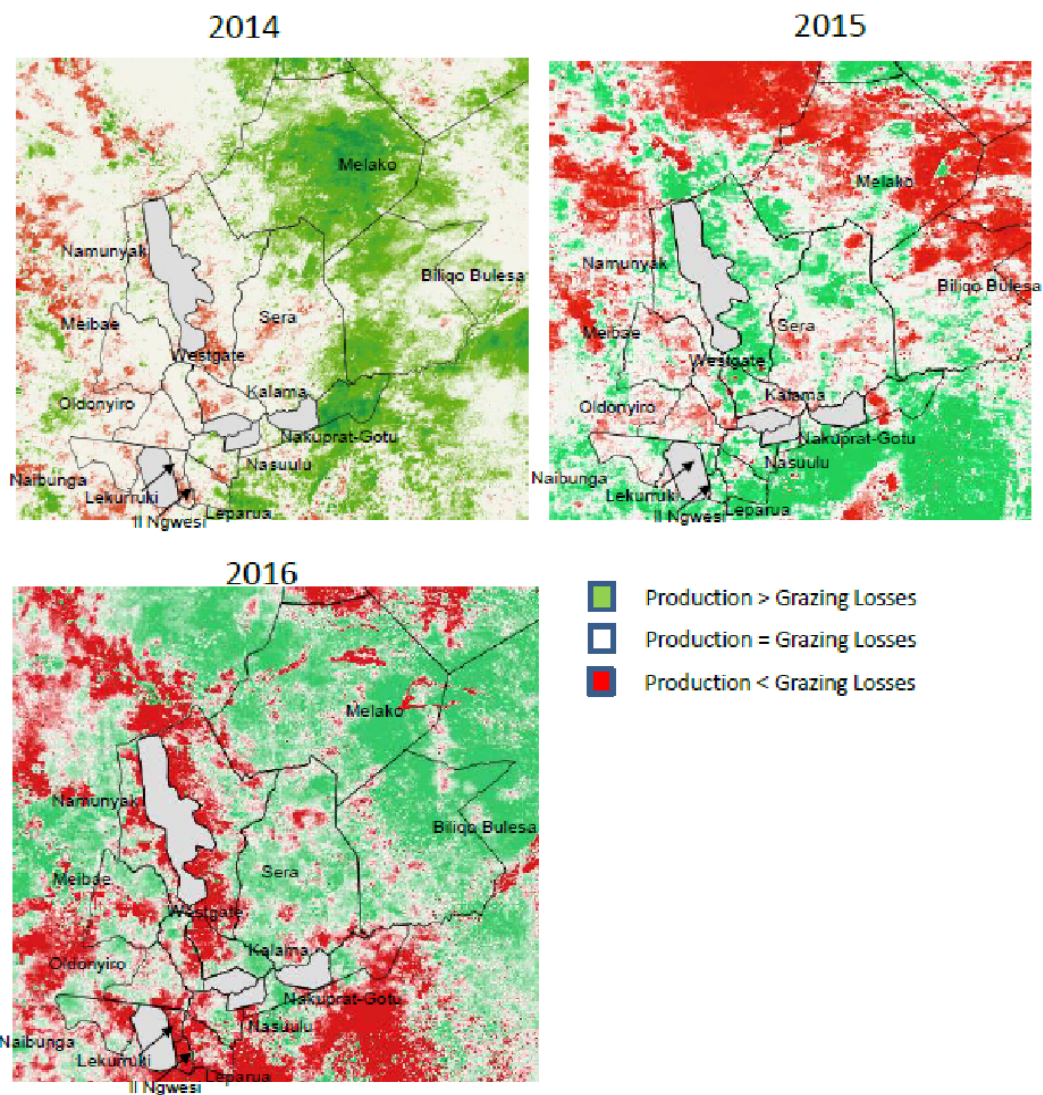


Figure 11. Evidence of successful project implementation (green and white areas) for 2014-2016 in the project zone

Table 9. Per hectare and total removals to soil carbon for each of the participating Conservancies in the Northern Kenya Grasslands Carbon Project during the second monitoring period, 2017-2020.

			2017	2018	2019	2020
Conservancy	Average Annual Removal, tCO ₂ e/ha	Project Area, ha	Conservancy Removals, tCO ₂ e	Conservancy Removals, tCO ₂ e	Conservancy Removals, tCO ₂ e	Conservancy Removals, tCO ₂ e
Biliqo	0.994098038	378,751	376,515	220,321	26,611	376,435
IL Ngwesi	0.638514018	9,432	6,022	4,326	1,056	6,018
Kalama	0.856418023	49,676	42,212	37,766	15,953	42,150
Leparua	0.687031125	32,833	22,557	8,773	4,470	22,526
Meibae	0.678926726	101,618	67,595	67,861	32,082	68,743
Melako	1.136236588	549,354	604,738	550,675	32,847	624,006
NaapuNanapisho	0.705840857	54,442	38,428	36,168	15,763	38,277
Naibunga	0.692916309	47,085	32,599	24,156	12,569	32,468
Nakuprat	0.828621374	71,282	58,940	16,718	4,311	59,050
Namunyak	0.863481680	288,232	242,729	182,179	67,700	244,807
Nasuulu	0.819186855	34,601	28,339	8,748	5,430	28,336
Sera	0.973407228	339,520	330,176	236,916	28,500	330,461
Westgate	0.712518944	36,250	25,612	24,938	10,371	25,732
Totals		1,993,075	1,876,464	1,419,546	257,663	1,899,008

3.2.3 Leakage

Leakage occurs when project livestock leave the project area but not if livestock leave their Conservancy but visit another Conservancy that is in the project area. Leakage was measured using information about numbers, classes, locations, and movement directions of livestock recorded in monthly grazing reports by the Rangeland Coordinators at each Conservancy. Review of the 624 submitted grazing reports revealed monthly measures of leakage for each Conservancy, as recorded in a Tab for each Conservancy in “Leakage and Encroachment Verification Calculations 2017 to 2020 v1 Sept 2021.xls”. These monthly totals were summed to determine the number of animal-days livestock of each class were either on their Conservancy, off their Conservancy but still on the project area, or off the project area entirely (Table 10).

Table 10. Summary of animal-days each year on Conservancy, off-Conservancy but on-project, and off the project area (Leakage) for the Northern Kenya Grasslands Carbon Project, with proportion of total animal-days spent in leakage.

Year	Off-Project Days	Total Livestock Days	Proportion Off-Project
2017	53,384,491	284,437,455	0.188
2018	17,478,612	291,838,079	0.060
2019	42,000,848	295,740,179	0.142
2020	60,972,561	325,626,616	0.187

Leakage, or movements of livestock off the project area, was relatively high (~18%) in 2017 and 2020, likely due to the drought and locust swarms mentioned previously. Because of the high cattle numbers in Namunyak, substantial leakage (though still below baseline levels of use) was caused by livestock use of the Mathews Range, which lies outside the project area.

Leakage can result from displacement of livestock off the project area (displacement leakage) and from replacement of livestock, reduced intentionally from project activities to reduce methane emissions, by producers outside the project area to meet market demand (market leakage).

Displacement leakage

Movement of livestock could result in losses of carbon from higher levels of overgrazing off the project area, or displacement leakage. Displacement leakage can be determined by using the tool VMD0040.

Leakage from Displacement of Grazing Activities for a measured approach or by using the penalty approach based on a reduction in net removals proportional to the total livestock-days spent off the project area. The project will use the penalty approach.

In this case, displacement leakage (LDt) must be calculated as a proportion of net removals from increased soil carbon in year t (PRSt), based on the proportion of total project livestock-days in project year t, 365 x PNc,t, that occurred outside the project area.

$$LDt = \frac{\sum \sum DN_{c,xkc}}{\sum \sum PN_{c,t}} = \frac{1365 \times \sum PN_{c,t}}{1365 \times \sum PN_{c,t}} = 1 \times PRSt \quad (41)$$

Where,

LDt = Leakage emissions from displaced livestock (tCO₂e)

DNC,x = Number of livestock of each category c that were off the project area on day x (head)

d = Total number of days livestock of class c were off the project area

k = Total number of livestock categories

PNC,t = Number of animals of each category c in year t (head)

PRSt = Project removals due to changes in SOC stocks in year t (tCO₂e)

Leakage was judged to be negligible, following VCS Standard 3.7 guidelines, if

$$\text{BEM} + \text{PEMt} + \text{PRSt} > 0.95 \cdot (\text{LDt} + \text{BEM} + \text{PEMt} + \text{PRSt}) \quad (42)$$

Uncertainty in displacement leakage, ULDt, if leakage is not negligible, is conservatively assumed to be the same as uncertainty in project net removals (equation (32))

$$\text{ULDt} = \text{UPRSt} \quad (43)$$

Market Leakage

Utilized the VCS AFOU Project Market Leakage tool to quantify market leakage. Market leakage was judged to be negligible since livestock reductions were not a project activity and livestock numbers are driven by range capacity and rainfall rather than markets. In the case of methane emissions by livestock, the Methodology VM0032 prohibits accounting for reduced methane emissions by livestock from reduced livestock numbers in order to avoid market leakage. In the case of the Northern Kenya Carbon Project. Kenya is not a beef exporter due to disease quarantines, so unlike Ethiopia, herders on the Conservancies do not have access to international markets. Consequently changes in market leakage are considered negligible.

Total Leakage

Consequently, with market leakage = 0, then total leakage,

$$\text{LEt} = \text{LDt} \quad (44)$$

And total leakage uncertainty

$$\text{ULEt} = \text{UPRSt} \quad (45)$$

3.2.4 Net GHG Emission Reductions and Removals

Net emissions and removals, after accounting for methane emissions, successful implementation of grazing management and movements of livestock off the project were about 4.1 million tCO₂e for the entire monitoring period. Subtracting the non-permanence buffer of 15% (626,928 tCO₂e) required by the Verra Standard yields approximately 3.5 million credits (tCO₂e) available for sale from 2017-2020.

Net GHG emission reductions and removals claimed during this monitoring period, along with their associated uncertainty, are summarized in Table 11. Total project uncertainty is < 30% in all years of the current monitoring period, so no deductions are taken as specified by section 8.4.4 of VM0032. Methane emissions, removals to soil carbon, and leakage were combined to determine net project removals (Table 11).

Table 11. Net emissions and removals of the Northern Kenya Carbon Project, 2017-2020

Year	Baseline Emissions, tCO ₂ e	Project Emissions, tCO ₂ e	Net Project Emissions, tCO ₂ e	Net Change in Carbon Stocks, tCO ₂ e	Total Change in Carbon Stocks and Emissions, tCO ₂ e	Leakage, tCO ₂ e	Total Net Emissions and Removals, tCO ₂ e	Total Annual Project Uncertainty %
2017	140,973	251,918	110,945	1,876,464	1,765,519	352,183	1,413,336	9.62
2018	140,973	251,918	110,945	1,419,546	1,308,601	85,019	1,223,582	10.75
2019	140,973	251,918	110,945	257,663	146,718	36,593	110,125	6.74
2020	140,973	251,918	110,945	1,899,008	1,788,063	355,583	1,432,480	9.69
Total	563,892	1,007,672	443,780	5,452,682	5,008,902	829,378	4,179,523	5.62

3.3 Optional Criterion: Climate Change Adaptation Benefits

Project activities provide climate change adaptation benefits to high conservation value ethnic groups as well as women in the poorest 25% of community members.

3.3.1 Activities and/or processes implemented for Adaptation (GL1.3)

The project zone includes several community High Conservation Values that are threatened by climate change.

1.) Sustaining livelihoods and cultural identity of indigenous groups. (General Benefits)

The project zone includes more than half the remaining people of several indigenous ethnic groups that have practiced pastoralism in northern Kenya since at least the mid-19th Century. The project helps sustain the livelihoods of endemic groups that include the Samburu (speakers of a Nilo-Saharan language) in Kalama, Oldonyiro, Meibae, Westgate, Sera, and Namunyak Conservancies, the Rendille (speakers of a Cushitic language) in Melako Conservancy, and Borana (a subgroup of Oromo (also Cushitic) speakers in Biliqo Bulesa Conservancy. These people account for more than 80% of the people in the project zone (Table 2).

2.) Support economic viability on traditional livestock grazing lands.

The ability of people in the project zone to continue pastoralism is threatened by development and land privatization, mineral exploration, and the potential for the government to designate lands as protected game reserves or other types of conservation areas without human occupancy.

3.) **Provision of ecosystem services.**

Livestock production, if sustainably managed, could meet an ever-growing demand for meat in Nairobi and elsewhere in Kenya, thus providing an important ecosystem service from land that has too poor of soils and too little rainfall to support intensive agriculture. The project itself will lead to removal of more than fifty million tonnes (CO₂ equivalent) of greenhouse gases, thus providing a valuable global ecosystem service.

4.) **Regional peace and security.**

Finally, successful, sustainable occupancy of the land by pastoralists provides the basis for peace among traditionally antagonistic groups such as Turkana or Borana and Samburu and serves as a bulwark against the import of terrorism from Somalia on Kenya's western border.

The project activity, planned rotational grazing both within Conservancies and across Conservancies as part of a regional grazing plan, will have positive effects on all four of these community HCVs (Figure 12).

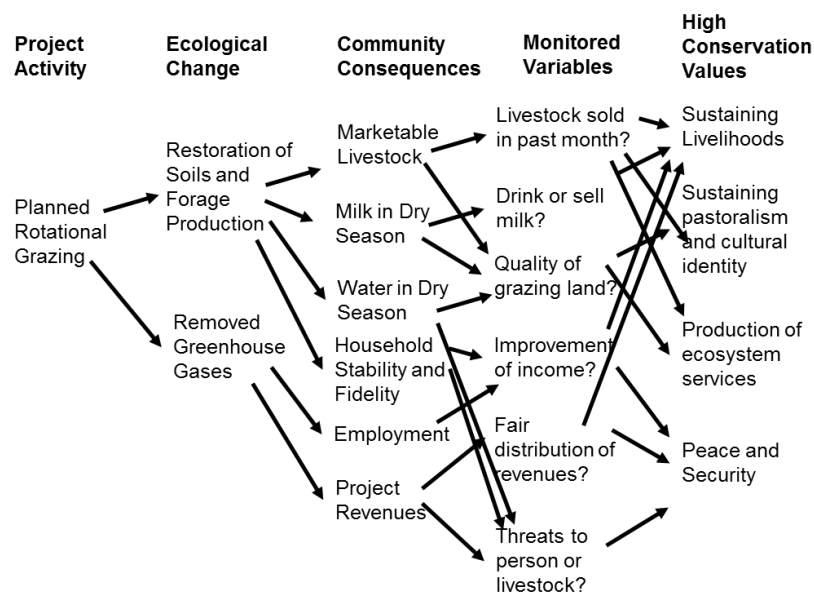


Figure 12. Causal model linking project activity to community High Conservation Values (HCVs) under the project scenario, which are threatened by climate change. The project activity should produce key ecological changes that lead to multiple community consequences. These consequences can be monitored and demonstrated with SocialComms (See Section 4) to improve all four HCVs for three principal endemic indigenous groups.

HCV 1: Sustaining livelihoods and cultural identity of indigenous groups

Samburu

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Samburu (similar to Maa Nilotic language, < 120,000 speakers)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Samburu and traditional enemy groups such as Borana and Pokot
Focal Area	Oldonyiro, Westgate, Kalama, Sera, and Namunyak Conservancies

Rendille

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Rendille (unique Cushitic languages < 100,000 speakers)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Rendille and traditional antagonist groups such as Borana and Pokot
Focal Area	Melako Conservancy

Borana

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Borana (Oromo language, Cushitic indigenous to Ethiopia, Muslim)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Borana and traditional enemy groups such as Samburu
Focal Area	Biliqo Bulesa and Nakuprat-Gotu Conservancies

HCV 2: Support economic viability on traditional livestock grazing lands. This support enables the unique cultures in the northern Kenya landscape to persist in the face of climate change rather than dissolve under pressure for land development and loss of livestock due to drought.

High Conservation Value	Support economic viability on traditional livestock grazing lands
Qualifying Attribute	The project, as executed from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies, will promote a shift in the economic valuation of forage and livestock production over previous wealth valuation of livestock numbers. This will allow communities to operate increasingly on a cash basis, maintained by remote cell phone banking mechanisms like M-Pesa ³⁴ and allow greater flexibility in managing herd sizes during drought. These adaptations will reduce the risk of catastrophic livestock loss during severe droughts and perhaps reverse the shift in livestock holdings to sheep and goats that pressure even very low forage production.
Focal Area	Entire project area, all indigenous groups

³⁴ <https://www.safaricom.co.ke/personal/m-pesa/do-more-with-m-pesa/m-pesa-global>

HCV 3: Provision of ecosystem services

The principal ecosystem services provided by the project are increased surface water availability, wildlife to support ecotourism. All these aid communities in adapting to climate change.

High Conservation Value	Ecosystem services provided by the project are increased surface water availability, wildlife to support ecotourism
Qualifying Attribute	The project will, as it develops, will build up soil organic matter, thereby increasing forage productivity and water-holding capacity of the soil. These outcomes result from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes should alter hydrological cycles, holding soil moisture longer into dry seasons and rejuvenating springs and groundwater sources, which will allow broader distribution of livestock on the landscape and provision of milk and meat through extended dry periods and drought. Wildlife populations are expected to increase, providing greater opportunities for and revenues from ecotourism. These changes will increase food security and reduce risk of ethnic conflicts over solitary sources of water and forage during dry seasons. The need for risky, long-distance migrations may decrease as well.
Focal Area	Entire project area, all indigenous groups

HCV 4: Peace and Security

High Conservation Value	Peace and security
Qualifying Attribute	Greater forage and water availability from improved rangeland conditions will result from project activities derived from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. Outcomes should allow herders from ethnic groups with histories of conflict to avoid convergent space use and temptation for historical cultural activities like stealing livestock or dominating resource locations at gunpoint. Encroachment of livestock outside regional grazing plans should also decrease. Security has been a major driver of sedentary settlement, and greater peace should allow for more peaceful dry season migrations. Greater game scout presence supported by project revenues will also help decrease deep

	encroachments of herders from outside the project area and quickly enforce justice on livestock thieves. Greater security and peace should also facilitate programs to develop infrastructure for finishing livestock for cash markets.
Focal Area	All Conservancies, but especially dry season ranges in Sera, Melako, and Biliqo Bulesa, as well as dry season refuges near the Ewaso Nyiro River (Kalama, Westgate, Oldonyiro, Nasuulu, Nakuprat-Gotu Conservancies)

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

The project provided substantial economic benefits to community households, as measured by the Social CoMMs surveys of over 2000 households in 2014 and 2015 and again in 2020 (project). Between baseline and project surveys, a higher proportion of households reported increased income over the previous year and ownership of cattle but lower incidence of illness. Climate change adaptation also increased, as the proportion of households with access to milk increased by 20% and time spent gathering water declined.

SocialCOMMs surveyed showed the project provided benefits to women, as women-led households surveyed also showed the increases in income, ownership of cattle, and lower incidence of illness.

Community Group	Women and Poorest 25%
Impact	% of households that claimed to benefit from implementation of grazing plans on home Conservancies and on other participating Conservancies
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	90% claimed grazing plan was “Helpful” or “Very Helpful” (Table 12).

Community Group	Women
Impact	% households whose income decreased in past year:
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Decreased from 55% in the baseline scenario to 26% in the project scenario (Table 12)

Community Group	Poorest 25%
Impact	Employment
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Employment among poorest 25% increased from 5% in the baseline scenario to 12% in the project scenario (Table 12)

Community Group	Women
Impact	% of households headed by women requiring less than 2 hours/day to acquire water
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Increased from 23.0% in the baseline to 68.9% in the project scenario

Community Group	Women
Impact	% of households headed by women that consumed milk in the previous week
Type of Benefit/Cost/Risk	Actual Benefit
Change in Well-being	Increased from 45.5% to 63.8%

Community Group	Women
Impact	% Households with livestock
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Increased from 68.4% in the baseline scenario to 80.8% in the project scenario (Table 12).

Community Group	Women
Impact	% Households headed by women that found grazing lands to be good or very good
Type of Benefit/Cost/Risk	Actual benefit

Change in Well-being

Increased from 38.3% in the baseline scenario to 68.5% in the project scenario (Table 12)

4.1.2 Negative Community Impact Mitigation (CM2.2)

1. More migration with livestock in order to increase rest may expose herders to encounters with antagonistic ethnic groups or livestock theft.

Mitigation: rapid response teams, anticipatory shared grazing agreements and peace meetings, regional grazing plans, and better communication between grazing coordinators and game scouts to facilitate conflict resolution.

2. Family members away from home for longer time periods.

Mitigation: Many herders are already away from home for up to 9 months of the year in response to drought and low forage availability. Proper management, with accompanying recovery of vegetation on home Conservancies, may actually increase time at home.

3. Carbon revenues may be converted into more livestock, creating little net positive benefit.

Mitigation Conservancy revenue distributions go mostly for public goods and are decided at open community meetings. There will likely be little personal return of cash, and thus capital to buy more livestock, to individual households.

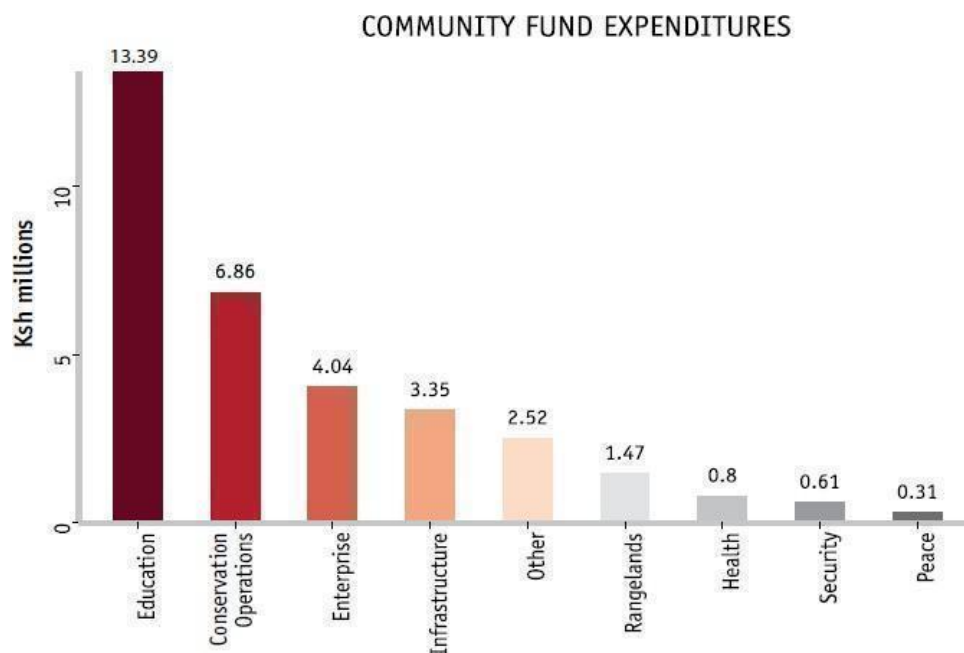


Figure 13. Expenditures for the NRT Pooled Conservancy Fund, showing the majority of allocation for education, including payments from bursaries for school fees for the poorest

community members. The majority of the rest is allocated to conservation operations, new enterprises, and infrastructure.

These benefits extend in many different positive ways to the poorest 25% (largely without livestock), women heads of household, young herders (morani) or other young men and women from poor households that may lack capital for livestock but whose education (funded by bursaries) may make them employable. The benefits extend to communities in Conservancies throughout the project area and thus help people from multiple indigenous ethnic groups.

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

The project introduces multiple positive community benefits. However, insufficient time has passed since the project start to the current crediting period for most of these benefits to be realized, but here several positive indicators for key population segments indicating exceptional community benefits are shown.

First, Livestock ownership provided a clear metric of the poorest 25% of people (Figure 13), and our household survey identified opinions of women, including households with female heads.

Benefits were classed as to whether they affected livelihood, health, or ability to adapt to climate change. Household survey questions generally are framed in terms of a choice of 2-4 alternatives from which respondents must choose. Consequently, data were analyzed according to the proportion of households choosing one or more responses to specific questions. Livelihood indicators were the proportion of households that (1) increased in income over the past year and (2) owned cattle. Health indicators were the proportion of households in which (3) an adult or (4) a child had been sick in the past month. Adaptation indicators were the proportion of households that (5) found the grazing program very helpful, (6) judged the quality of grazing lands to be “good” or “Very Good,” (7) drank milk the previous day, (8) found security to be “Improved” versus “Stayed the Same” or “Worsened,” and (9) spent less than two hours/day gathering water.

These assessments were based on household interviews at 2663 households in 8 different Conservancies during 2014 and 2015 (Baseline), and 2,637 households across five Conservancies in 2020 (Project). The results (Table 12) showed that overwhelmingly, across all households, the project activity was associated with dramatic improvements in all three classes of benefits.

Benefits to Women

Table 12 also clearly supports the idea that women benefit in multiple ways under project activities. Most strikingly, female-led households report increases in family income much more frequently, perceive grazing lands as “Good” or “Very Good” more often, and perceive improved security in 80% of cases relative to about a third of households under Baseline conditions. Interestingly they demonstrate larger increases in perceived security than men, but men-led households experience sickness less frequently and show larger increases in perceived quality of grazing lands and access to milk under project activities. Overall, the results are highly consistent with strong exceptional benefits to women.

Table 12. Proportions of households receiving benefits in the Northern Kenya Grasslands Carbon Project in 2014 and 2015 (Baseline conditions) and in 2020 (Project conditions) for all, female-led, and male-led households. Bold numbers indicate significant changes between baseline and project survey respondents.

	Head of Household	All		Female		Male	
	Number of Households	2663	2637	609	756	2054	1881
	Time	Baseline	Project	Baseline	Project	Baseline	Project
Livelihood	Greater Income	30.3%	52.4%	29.2%	51.9%	30.6%	52.6%
	Own cattle	75.1%	86.2%	68.4%	80.8%	77.1%	88.4%
Health	Adult Illness	49.3%	28.9%	28.9%	29.0%	55.3%	28.9%
	Child Illness	25.3%	19.3%	24.3%	19.3%	25.6%	19.3%
Adaptation	Grazing Program Helpful	62.0%	65.6%	64.0%	63.6%	61.4%	66.3%
	Grazing Lands Good or Very Good	33.0%	62.4%	38.3%	61.5%	24.3%	62.7%
	Access to Milk	36.8%	69.3%	45.5%	64.8%	34.2%	71.1%
	Security Improved	37.1%	73.2%	33.0%	80.7%	38.3%	70.1%
	Hours to Water < 2	23.0%	68.9%	21.9%	68.5%	23.3%	69.0%

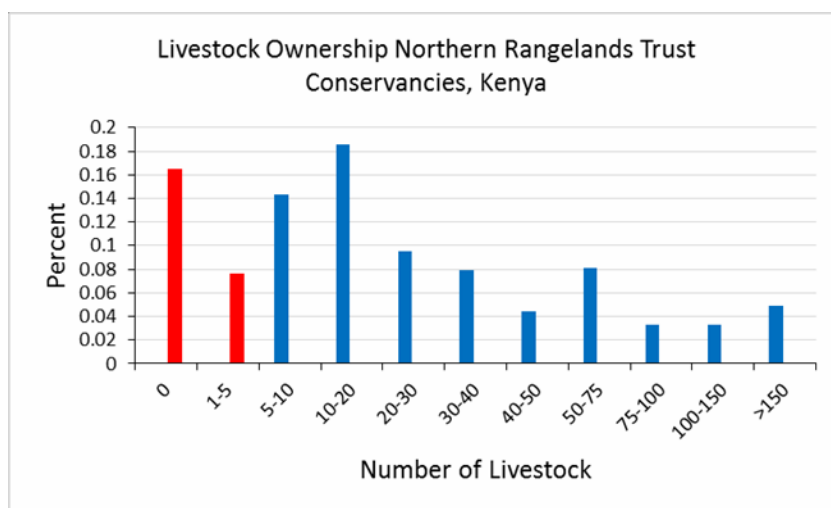


Figure 14. Livestock ownership distribution within four Conservancies in the project area. Red bars indicate the cumulative 25% of households with fewer than 5 livestock and no cattle that comprise the 25% poorest people in the project communities.

Second, the project scenario conveyed numerous financial benefits. Project Conservancies (> 2 years history with a grazing plan) employed 30% more people in total, employed twice the % of employees from vulnerable groups (Figure 16) than Baseline Conservancies (grazing plan in

effect < 1 year). There was essentially no difference in employment of women between Baseline and Project Conservancies. Project Conservancies had fewer people than Baseline Conservancies and experienced lower income than the previous year (Figure 16) and a greater proportion of households that experienced increased income (Table 12). Project Conservancies also featured a significantly lower proportion of households that thought security had improved or stayed the same than Baseline Conservancies (Figure 15). Households in communities under the project scenario drank milk at a higher frequency than in the baseline scenario (Table 12, Figure 16) and were less likely to be sick.

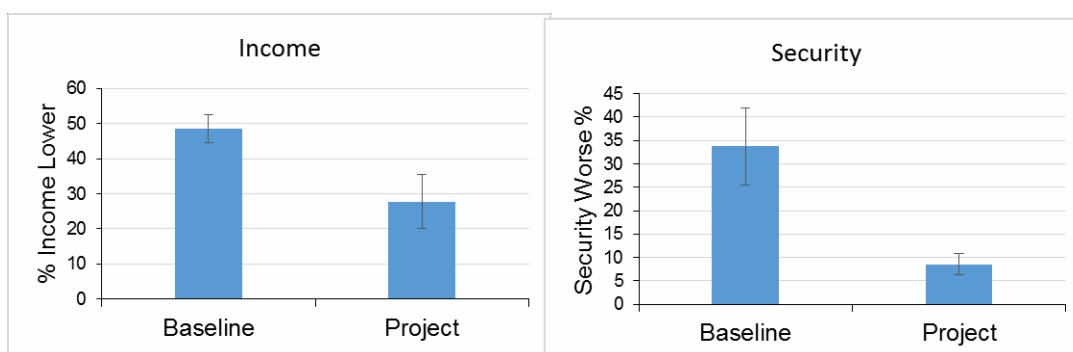
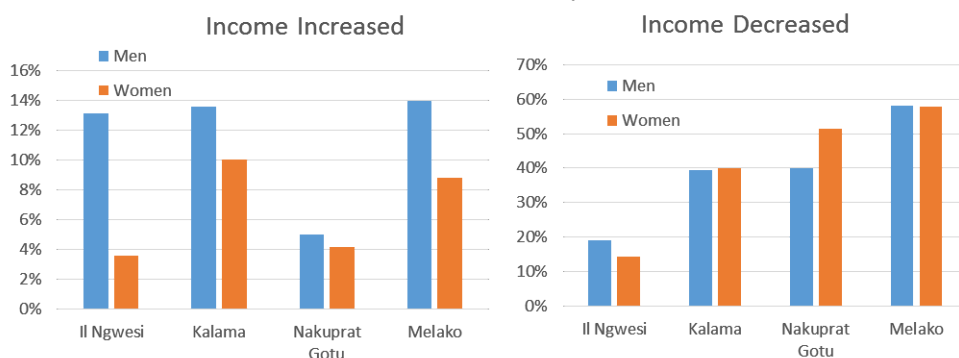


Figure 15. Comparison of benefits associated with Baseline (grazing plan in first year) and Project (Grazing plan > 2 years): A. percent that experienced lower income in the past 12 months, B. Percent that experienced reduced security in the past 12 months.

Financial Capital



Human Capital and Health

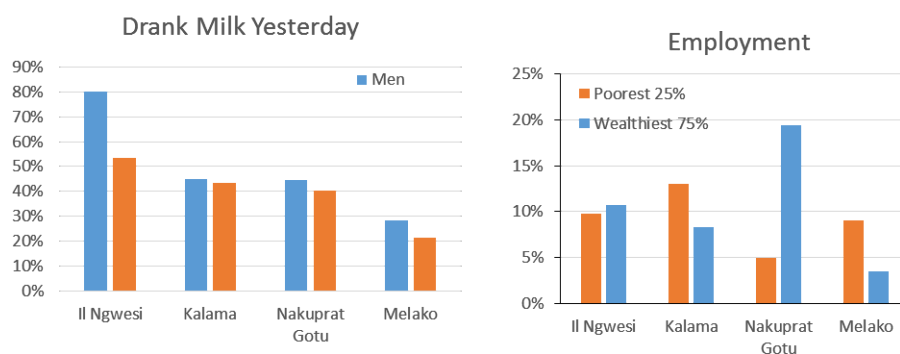


Figure 16. Economic and health indicators for **Project** Conservancies (Il Ngwesi, Kalama) with a longer history of planned grazing) versus **Baseline** (< 1 year with a grazing plan) (Nakuprat-Gotu, Melako) in the Northern Kenya Grasslands project area.

Finally, response of communities to planned grazing is overwhelmingly positive, as the vast majority (> 65%) of households find their Conservancy's grazing plan "very helpful" as compared to "not helpful," "somewhat helpful," or "don't know" (Figure 17). This positivity was expressed across households with men or women as heads, among the poorest 25% (largely not cattle owners) and owners of more than 5 livestock, and among people from Conservancies with a longer history of planned grazing versus not.

All these data indicate some positive trends that have accompanied the onset of project activities. Future data collected across additional Conservancies will allow a better assignment of cause and effect than is possible now, but the results suggest that the project activities are providing some social benefits, even as the resource benefits in livestock and grass await better weather conditions with higher rainfall.

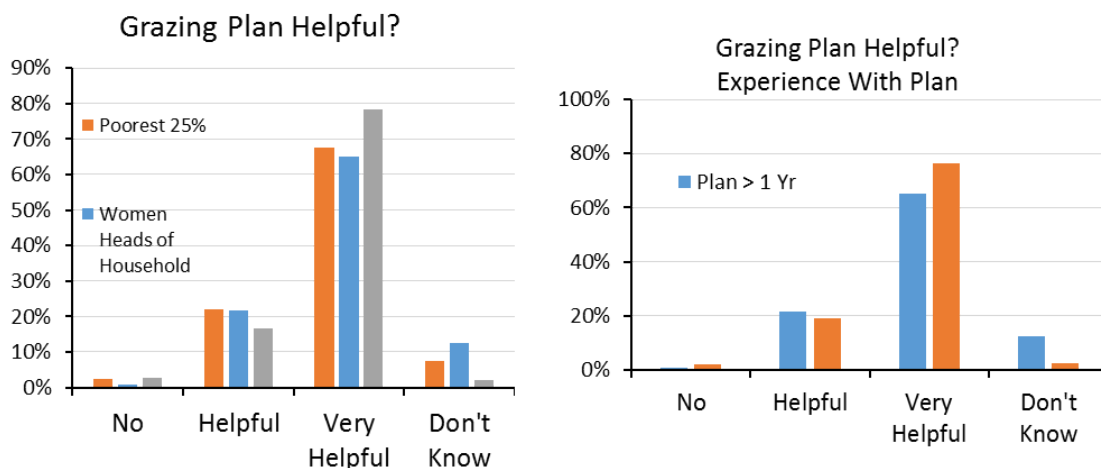


Figure 17. Responses to the question "Is the grazing plan helpful?" across different key groups of households in the project area. Data indicate overwhelming positive support for the project activities across the population at large and for vulnerable groups (women and poorest 25%). Enthusiasm for the grazing plan was not significantly different between **Baseline** (Plan < 1 Yr, blue bars) and **Project** (orange bars).

Anticipated climate change in the project zone is somewhat uncertain, as climate models differ in their predictions of whether mean annual rainfall will increase^[26] or decrease^[27]. However, climate variability is expected to increase regardless^[27, 28]. Increased variability and warmer conditions are expected to increase the frequency and intensity of droughts, shrink the length of the season soil moisture is available, reduce overall mean forage production, and potentially reduce livestock capacity by 10%^[26, 29, 30]. Assuming that temporal variability extends to spatial variability, climate change may increase the frequency of periods where pastoralists are in conflict over water or forage^[27, 31]

The rebuilding of soil organic carbon from the project activities of widespread PR grazing practices will likely induce dramatic improvements in the ability of people in the project area to adapt to climate change. Recent climate trends in the region^[32, 33] suggest a reduction in mean precipitation, with longer periods of consecutive years with either above- or below-average precipitation. The principal defense for pastoralist communities against such projected changes is to build heterogeneity in grazing use of the landscape so as to create opportunities for “chasing” spatially and temporally variable precipitation within a season and the establishment of grass banks as a hedge against drought. The PR grazing practices are fully expected to restore perennial grass cover and increase soil organic carbon. The occurrence of greater soil carbon across the project area would result in the following aids in climate adaptation

1. Increased infiltration and retention of limited precipitation in shallow soil layers^[34, 35].
2. Green, higher quality forage available for more months during dry seasons/years.
3. Reproduction by livestock will extend further into the dry season or occur during dryer years, which will supply milk to families year-round^[4, 36]. Baseline social surveys indicate that milk available for household use or to sell is available in communities.

Figure 18. Climate adaptation benefits A. Mean ($\pm$ SE) liters of milk sold per household under **Project** Conditions (grazing plans implemented for ≥ 2 years, $N = 5$ Conservancies) compared to **Baseline** conditions ($N = 4$ Conservancies, > 250 households interviewed/ Conservancy) (t-test, $df = 7$, $P < 0.003$). B. **Project** conditions produce significantly shorter mean ($\pm$ SE) travel distances to water (t-test, $df = 7$, $P=0.03$).

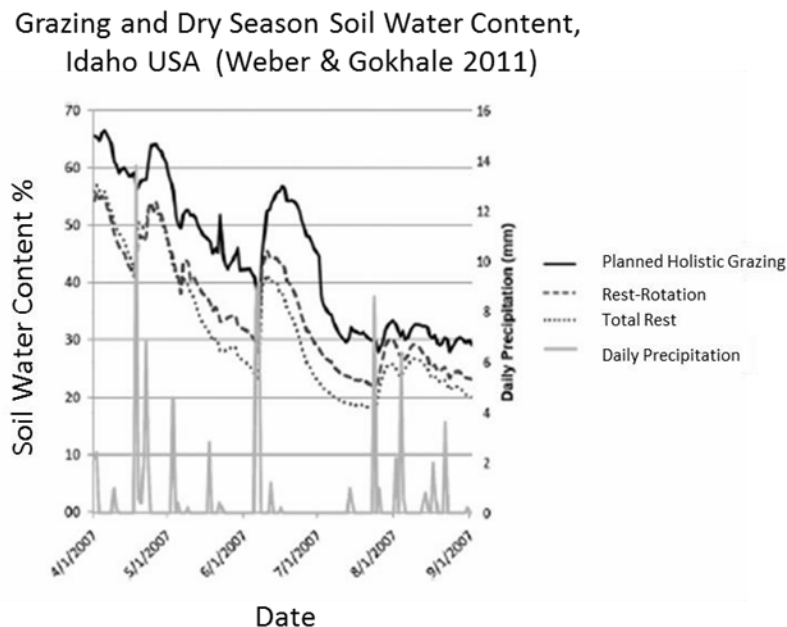


Figure 19. Soil water content to a depth of 1 m in areas receiving different livestock grazing management during the growing season in a temperate shrub steppe^[37]. Note the persistently

higher water content in planned “holistic” rotational grazing treatments as compared to total rest or limited rest-rotation management.

4. Greater retention of water in shallow layers due to higher soil carbon^[37-39] (Figure 19) supports the advent of standing water in a greater number of places across the landscape to support cattle, which have a greater dependence on water than smaller livestock, particularly goats. These effects are evident even in tropical savannas that form hardpan surfaces following degradation^[40, 41]

5. Greater standing biomass and water in the dry season allows greater allocation of livestock to cattle and away from sheep and goats.

Overall, the project activities should greatly enhance communities’ ability to cope with climate change relative to their baseline condition.

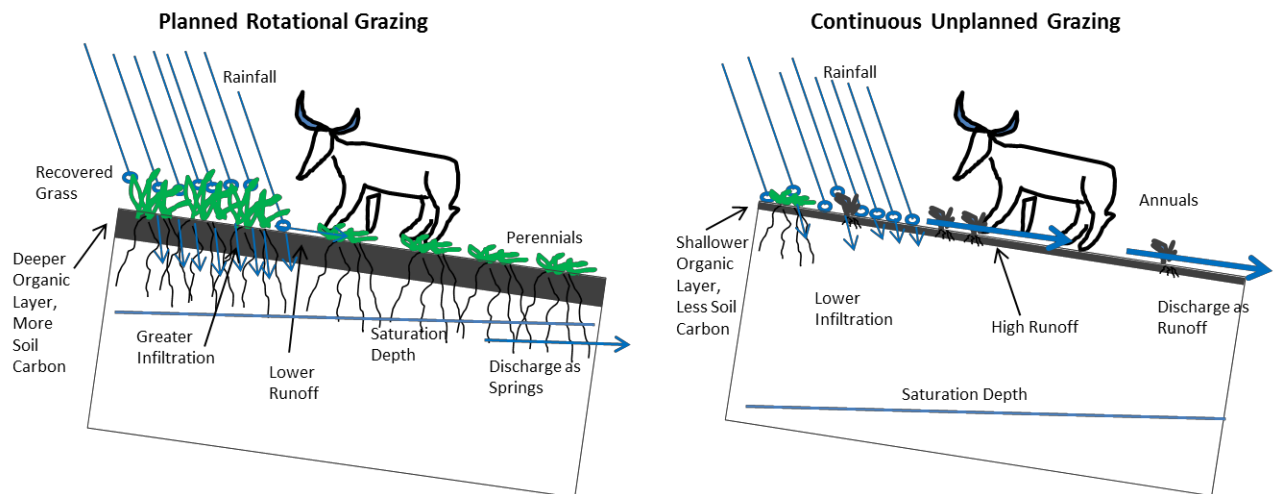


Figure 20. Diagrams to contrast the surface hydrology of rainfall in landscapes with planned rotational grazing (project activity) and continuous unplanned grazing (baseline activity).

4.1.4 Protection of High Conservation Values (CM2.4)

Implementation of planned grazing has not increased and in some cases has even reduced incursions of project livestock into HCV areas. For example, satellite-based assessments of livestock impacts using MODIS imagery and calculations of NDVI show that project activities reversed a trend of increasing use of the Mathews by livestock during the baseline period (Figure 21).

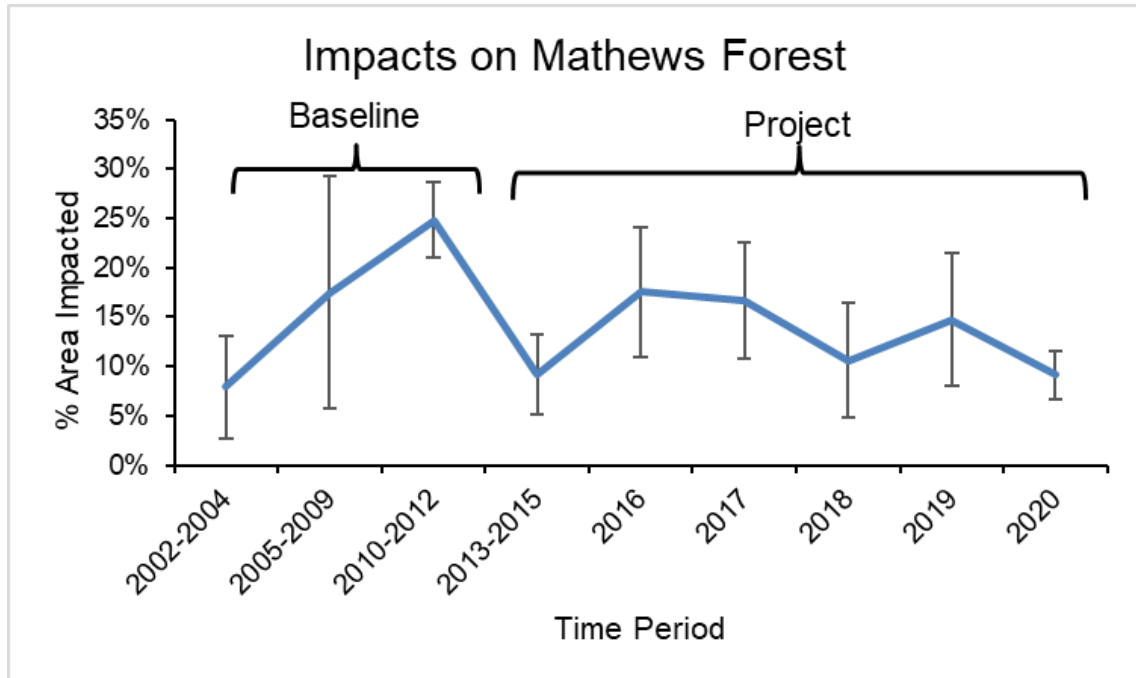


Figure 21. Mean ($\pm$ SE) extent of area experiencing net negative impacts ($SUM\Delta NDVI > 0.5$) of livestock in the Mathews Forest, averaged over different time periods. Note the trend of increasing impact on the Forest over time during the Baseline period followed by a dramatic reduction to near 2002-2004 levels under the project activities. These data likely reflect increased awareness of the need for herders to avoid the Mathews among grazing committees and managers in Namunyak Conservancy (which surrounds the Mathews) and more rapid movements through the Forest in grazing plans.

Livestock monitoring data from the Wildlife CoMMs data covering 2011-2012 (Baseline) and the monitoring period 2013 – 2016 (Project) indicate a clear reversal of trend in reducing encroachment into the Mathews Forest (Figure 22). Even in a drought year, when encroachment is usually at its worst, livestock produced negative impacts on only 20% of the Mathews Forest in 2014 (Figure 22). Impacts during the project (e.g., 2014) have mostly occurred on the edges where degradation has already occurred, compared to over 30% of the forest negatively impacted, and mostly in the high biodiversity rainforests at high elevation, that occurred in 2008, a comparatively wet year (Figure 11). As rotational practices continue and grass banks are established across a wide range of locations in the project area, encroachment should decrease further, thus addressing the number one threat to the Mathews Forest.

2008
Baseline Conditions

2014
Project Scenario

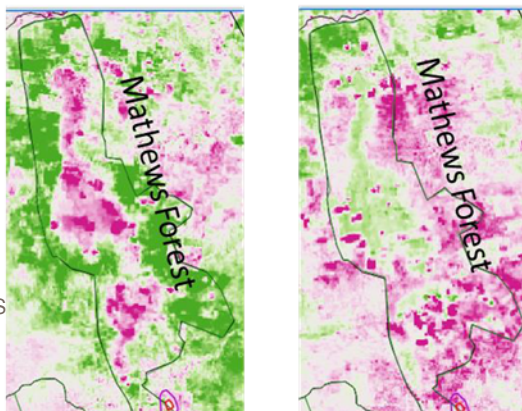


Figure 22. Satellite based monitoring of net annual grazing impacts, using the sum of monthly Δ NDVI (see section 5.4.3) on the Mathews Forest, comparing those in a wet year (2008), conservatively representing baseline conditions, and those in drought year under a preliminary regional grazing plan to mitigate effects of drought in western side of the project area. Note that impacts in 2014 were mostly along the edge of the forest, while impacts in 2008 were concentrated in the interior high elevation rainforest areas of the Mathews.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

The principal offsite stakeholders are herders from other Group Ranches or Trust lands, and these stakeholders may be negatively impacted by the enforcement of project boundaries and the perception that traditional use of other communities' grazing lands during droughts or dry seasons is no longer permitted. A key mission in designing planned rotational grazing will be to continue to have negotiated agreements among grazing committees of different Conservancies or communities both on and off the project area. These agreements will allow continuance of cultural traditions while maintaining the integrity of the project. They also may enhance social cohesion because acceptance of off-project livestock will be easier once sufficient grass banks are established.

To this end, NRT has established a **regional grazing plan** to manage movements and numbers of livestock across Conservancies during droughts or extended dry seasons. This plan uses near real-time information on rainfall in the previous 1-3 months from USDA-USAID Famine Early Warning System (FEWS) <http://earlywarning.usgs.gov/fews/> and from the University of Reading's (UK) Tropical Applications of Meteorology using SATellite data, or TAMSAT <http://www.met.reading.ac.uk/~tamsat/data/>, to identify areas experiencing drier and/or wetter than normal conditions. From this information system, a large-scale cross-Conservancy rotation of livestock can be designed that eases grazing pressure in drier areas while sustainably using forage available within the project area. This regional grazing plan can also accommodate movements of herders from outside the project area as a way to limit encroachment and better manage leakage.

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

Community feedback

The carbon project, whereby yet another benefit of planned rotational grazing is presented, has been communicated much more recently due to high uncertainty in the carbon market and a reluctance to raise hopes about potential revenues. However, the Conservancies began on 15 May 2015 and then again on 15 August 2016, a solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff in the form of pamphlets, and a posted announcement on the NRT website (<http://nrt-kenya.org>). Responses to the proposed carbon project were made by cell phone text to a standing repository, which was then downloaded.

These announcements were followed by community meetings in September and October 2016, open to all members in each Conservancy, in each of the 13 participating Conservancies. Meeting times and dates were posted at all Conservancy headquarters and were communicated verbally in villages in each Conservancy by NRT Social CoMMs supervisors.

Between 2017-2020 Annual Governance Meetings were held with representatives from each project conservancy. Starting in 2021, a Carbon Project Oversight Committee will be formed from conservancy representatives and meetings will be held to focus on developing additional internal governance procedures based on community feedback.

Information about the project's participation in the voluntary carbon market (VCM) inclusive of all aspects of the project (costs, risks, benefits and requirements) was communicated with stakeholders through signed agreements with conservancies including, consent and waiver forms and letters of intent. Both of these agreements include language on carbon credits and anticipated revenue including, "we intend to participate in these actions, with the expectation of revenues from the sale of carbon credits, for at least the next 30 years." Examples of these signed agreements have been provided to the VVB for review. As the project was not validated until 2020 and verified/issuing credits until late 2021/early 2022, there was uncertainty if the project would take off prior to 2020 and thus the delayed development of the benefit sharing mechanism.

Adaptation to community impacts

In meetings, community grazing committee chairmen communicated how grazing plans had been implemented or not in the past and reviewed past compliance with grazing plans, identified barriers to compliance, and communicated expected benefits of project activities as rotational or migratory grazing. The response of communities, especially after September 2016 meetings and workshops, were improved efforts to make, write, and implement grazing plans within Conservancies. This provides evidence that community members understood the benefits of project activities and approved intensified efforts to conduct grazing plans.

Evidence from social surveys indicate that a majority of members thought that they or the community were benefited by the implementation of grazing plans. These baseline data were, and future data will be, collected through NRT Social CoMMs surveys and focus groups. SocialCoMMs is an outgrowth of training of community members to conduct social surveys by professional social scientist Dr. Jessica Musengezi of The Nature Conservancy, supported by permanent NRT staff. Through the guidance of TNC and NRT personnel, focus groups are chosen by community chairmen to represent the public, rather than by the grazing committee. Household surveys are conducted at randomly selected households, and every effort is made to interview at least the matriarch of each household.

As an example of how this worked, our baseline SocialCoMMs survey in 2014 identified the risk of encroachment onto Conservancies as the number one concern. Implementation of grazing management training and education, which at that point was the main emphasized activity, was reduced in effort, and effort was increased to work with grazing coordinators, grazing chairmen, Conservancy managers and community chairmen to resolve disputes due to encroachment from off-project herders. More effort was also applied to developing a regional grazing plan to allow herders from within-project Conservancies to use grazing lands from other within-project Conservancies. This shift allowed NRT to have a rapid response to conflicts between herders from different ethnic groups and avoid fatalities that occurred under similar circumstances in the same period in 2014. The formation of a regional grazing plan also facilitated discussions of County-wide grazing plans in October of 2015, which delivered information to and coordinated activities. Since 2017, quarterly grazing committee meetings have been held to discuss, review and plan for rangeland matters. In addition, grazing committees have continuously been trained on rangeland management and their roles as rangeland committee members.

Between 2017-2020, all net impacts on other stakeholders continued to be tracked through annual Social CoMMs reports. Project Social CoMMs results are shared with the respective conservancy manager. Conservancy managers are tasked with discussing the Social CoMMs results with their conservancy board. The Social CoMMs results are also fed into 5-year Conservancy Management and Development Plans aimed at linking results to socio-economics and social development projects such as access to water, education centres and healthcare.

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The key variables determined from the three types of surveys reflect specific aspects of financial, human, natural, physical, and social capital that may be influenced by the carbon project. Some very important variables to consider for NRT and the NKCP are given below. Social data collection involves gathering information from approximately 5 groups of key informants from Conservancies with representation from women, elders, and morani and at least 10-15 focus groups [2 or 3 in each of five Conservancies] from within the entire project area.

Key informants were identified from existing NRT administrative conduits, including village chairmen, grazing chairmen, and community public health professionals (if available). In addition key informants included community elders from various community committees, such as grazing and settlement allocation committees. Typically at least five key informants were interviewed from each Conservancy. Interview questions were similar in theme to those in household surveys, except that informants were asked to reflect their opinion of community-wide issues and needs.

Focus groups were conducted by extending a general invitation within each of 3-4 villages (100-500 households in size) within each Conservancy to attend a meeting 5-10 days after the initial announcement in each village to engage in open discussions of issues related to well-being and grazing, the core project activity. In addition, village chairmen were asked to identify and extend invitations to individuals from vulnerable and marginalized groups, in particular women and the poorest 25% (nearly always single head-of-household households with few or now livestock)

Household Surveys

The goal was a sample size sufficient to demonstrate significant change over some period of time during the implementation of the project. Given typical standard errors in human data of 10-20% and frequent non-normal distributions, at least 25 households is necessary to ensure a sample large enough to successfully compare different subgroups^[42]. This sample size was selected based on reducing the coefficient of variation to less than 30%, based on standard deviations in responses from Louise Glew's thesis^[1] comparing socio-economic results from communities with and without Conservancies. This sample size should allow 15% improvement in conditions during the project crediting period to be found statistically significant ($P < 0.05$).

Sufficient effort will be made to include some of the same individuals upon re-sampling (samples taken after the start of the project) in order to reduce sampling error.

Surveys (N=1030) were conducted in November and December in 2015 in each of four Conservancies. Two Conservancies had been implementing grazing plans since 2013 (Kalama and Il Ngwesi) and represented the project scenario. The other two Conservancies (Melako, Nakuprat-Gotu) had just been organized and were still developing grazing plans and represented the baseline scenario. These Conservancies were selected because they encompassed most of the ethnic groups represented in the project area (Il Ngwesi, Maasai; Kalama, Samburu; Nakuprat-Gotu, Turkana, Pokot, and Borana; Melako, Rendille). Village chairmen and Conservancy managers were contacted and interviewed to identify key stakeholders, women, and underrepresented groups for focus groups and to plan the random selection of households for household surveys. Details of survey questions are provided below in subsection 4. Outcomes of surveys are summarized in section 4.1.3.

Focus Groups: Wealth or well-being ranking of local or community stakeholders

The tendency in most project types is for 'elite capture' and for the poorest to lose out, as has happened with the majority of community forestry experiences (McDermott & Schreckenberg 2009). A wealth ranking exercise should also lead to a better understanding of local perceptions of well-being and poverty, and can also generate a useful sampling frame for household surveys. As described by PROFOR (2008) a wealth or well-being ranking exercise involves the following stages:

- Undertake a community mapping exercise to identify and list all households
- Write down the names of the household heads on cards, one per card
- Consult with key informants and focus groups on poverty or well-being categories. In the case of the Nepal Swiss Community Forestry Project (PROFOR, 2008), this resulted in six wealth categories: capable, improving poor, coping poor, declining poor, extreme poor and incapable poor. In other studies, the number of months of household food reserve is a common wealth 'numeraire' (note: this stage could be undertaken first)
- Select 'representative', respected and knowledgeable key informants, including some women
- Get the key informants to classify the cards (households) into the wealth or well-being categories .

Use of Venn Diagrams for Stakeholder Analysis

Participants firstly cut three sizes of circles – at least two sets of circles using different colored cards. One color is for ‘insider stakeholders’ and another is for ‘outsider stakeholders.’ For each ‘outsider stakeholder’, the participants need to decide the importance of their involvement in the project, and select the corresponding size of circle:

- * Little importance = small circle
- * Some or significant importance = middle sized circle
- * Very important = large circle

The name of the ‘insider’ stakeholder or stakeholder groups is then written on the appropriate sized circles/cards. This is repeated for all ‘insider’ stakeholders using the other colored card. When all stakeholders have been represented, the circles are organized and stuck to a flipchart, grouping and placing the circles according to the relationships between the stakeholders: the closer the relationship between two stakeholders, the closer the circles should be on the flipchart.

The next stage is to cut three sizes of triangles from different colored cards. For each stakeholder (group), a small, medium or large triangle is chosen to represent the degree of influence that the stakeholder has on the project. The triangle is then stuck on the edge of the stakeholder circle. A stakeholder with a small ‘importance circle’ could have a large ‘influence triangle’ and vice versa. The overlap of the circles represents the extent of the relationship between stakeholders (see Figure 5). Once the diagram is complete, it is reviewed by the wider group, which should continue to discuss the relative importance and influence of each stakeholder or stakeholder group until a consensus is reached.

Conduct of Household Surveys

Surveyors use the following introductory script:

Hello, my name is and I am involved in a project sponsored by the Northern Rangelands Trust to investigate the livelihoods and well-being of people who live in this conservancy. The goal is to better understand the impact of the conservancy on your daily life.

<ASK TO SPEAK TO THE HOUSEHOLD HEAD, OR IF THE HOUSEHOLD HEAD IS NOT THERE OR BUSY, ASK TO SPEAK TO ANOTHER ADULT MEMBER OF THE HOUSEHOLD. THE PERSON INTERVIEWED MUST BE AT LEAST 18 YEARS OLD.>

I would like to ask a number of questions related to you and your household. The interview will last about 30 minutes. Please note that all the information you provide will be kept confidential and will not be shared with anyone else. The information will only be used to characterize the area in which you live. All individual information will be added together to determine the average for the whole community, so nobody will be able to identify individual participants. There are no risks from participating and there are no right or wrong answers. It is your opinion that is important to us. Do you have any questions?

Thank-you.

Can we find a quiet and comfortable place where I can ask you the questions?

Questions will be asked to both heads of household (mostly men) as well as head women of households so as to monitor project impacts on women as well as different groups of men.

Twelve indicators evaluated project impacts on women:

1. Have you been employed by the Conservancy in the past 12 months
2. Do you feel safe enough?
3. Does the Conservancy have a grazing plan?
4. Is the grazing plan helpful?
5. How has your household income changed in the last 12 months (1=decrease, 2 = stay the same, 3= increased)
6. How far to travel to water?
7. What were sales of livestock products in the past month?
8. Did anyone in the household sell milk yesterday?
9. Did anyone drink milk yesterday?
10. Did anyone in the household eat meat yesterday?
11. Were any adults sick?
12. Were any children sick?

These indicators were chosen to compare project scenario with no project scenario (prior to implementing project activities) on economics (questions 1, 5, 7), security (2), access to resources (6, 8-10), household health (11, 12), and recognition of benefits of the project activity (3, 4).

For assessing past livestock numbers, the methodology VM0032 requires surveys of at least 30 households, or 10% of households, whichever is greater, in a Conservancy. Surveys, conducted in October-November 2014, solicited the number and kind of livestock households owned, how these are grouped with others or otherwise managed to achieve rotational grazing, and their locations through the year. In addition, heads of household were asked for the number and kind of livestock owned in 2004, 2005, 2007, 2008, 2009, and 2011. Interviews were conducted by rangeland coordinators of livestock owners, typically while out herding livestock, and so did not include households without livestock. This bias was corrected by determining the fraction of households without livestock from the general household surveys, and appropriately discounting estimates of livestock numbers.

Livestock numbers for each Conservancy were obtained from monthly grazing reports by rangeland coordinators during 2014, 2015, and 2016 to provide project livestock numbers for the purpose of estimating leakage and methane emissions. For Conservancies where household surveys were conducted during any of these years, livestock numbers were estimated by multiplying livestock per household by number of households and corroborated against rangeland coordinator reports for that Conservancy.

In total, survey results indicated that exceptional community benefits have already occurred under the project scenario as income, security, and access to milk and water for multiple stakeholder groups has increased relative to the baseline scenario (Section 4.1.3). Further benefits are expected to be realized relative to the baseline scenario as the project continues and more

surveys are done.

4.3.2 Monitoring Plan Dissemination (CM4.3)

Each year in September or early October, we hold a Conservancy-wide Project Meeting attended by Conservancy chairs, community grazing chairpersons, grazing coordinators and interested public. This meeting is announced on the NRT website and notices will be posted in major community kiosks and markets. At the meeting a report of the previous 12 months' monitoring results will be presented. Data included are livestock counts, leakage and distribution within the project area, any soil or vegetation or biodiversity surveys, and results of the previous Social CoMMs surveys. Following this presentation, plans for the ensuing 12 months are discussed, with feedback from managers and the community welcomed. An Annual Report of the project, with a monitoring plan modified in response to community feedback from the Project Meeting and a summary written in Kiswahili, is posted on the NRT website. Links to the posted report are sent by text through the grazing management and grazing coordinator network and be communicated via email to other NGO and government stakeholders.

These meetings were held in September 2016 and each year subsequently.

The translation of the monitoring report to Kiswahili will be done after the monitoring report is approved and sent to Verra. This was done most recently for the 2013-2016 monitoring report that was translated to Kiswahili and thereafter shared as a monitoring report summary in 2021. Once the 2017-2020 monitoring report is approved and finalized, a monitoring report summary will be translated and sent to the conservancies.

4.4 Optional Criterion: Exceptional Community Benefits

The project activities will bring considerable community benefits, particularly to women and the poorest 25% of community members.

4.4.1 Short-term and Long-term Community Benefits (GL2.2)

As demonstrated in section 4.3, project activities have already helped increase employment and mitigate decreases in household revenue. Households have better food security in the form of higher and more consistent milk availability. These benefits have been demonstrated for marginalized groups (poorest 25% and women).

Longer-term benefits of improved forage, higher water tables and greater surface water availability have yet to be realized due to severe droughts in 2011, 2015 and 2017 that prevented substantial increases in grass cover during wetter years (2012, 2014) from persisting. However, the proportion of project area in which grazing of sites only once per year has remained above 80% (Figure 9), which suggests that recovery of forage and rangeland productivity is still expected.

4.4.2 Marginalized and/or Vulnerable Community Groups (GL2.4)

Community impacts of the project activities were monitored from surveys in 2014 and 2015 of more than 1000 households and were conducted by community members within their Conservancies, as part of NRT assisted Social CoMMs activities. Data were compared between Conservancies with longer histories of implemented grazing plans (Project Scenario) and

Conservancies that had just joined the project (Baseline Scenario). These household surveys contained multiple questions that pertained to the benefits discussed in section 4.1.1.

4.4.3 Net Impacts on Women (GL2.5)

Sections 4.3 and 4.1.3 demonstrate that women benefit from the project in multiple ways, including increased household food security (milk available to drink and sell).

4.4.4 Benefit Sharing Mechanisms (GL2.6)

No revenues have been generated from the project so there is no example of benefit sharing yet.

4.4.5 Governance and Implementation Structures (GL2.8)

The governance structure of the project that links NRT to community leaders to grazing committees and public hearings (Figure 2, section 2.3.10) allows households with few livestock, women, and underrepresented groups to participate in the formation of grazing plans and therefore in project activities. This participation is facilitated by community leaders who seek out and invite smallholders, women and minorities to committee meetings.

4.4.6 Smallholders/Community Members Capacity Development (GL2.9)

The project helped build the capacity of smallholders, women, and minorities to participate in the project. Each year of the monitoring period, families from these groups provided personnel for project-related employment, including game scouts, soil and vegetation surveyors, social surveyors and livestock herding. Community members were also employed each year of the monitoring period as processors of vegetation, wildlife, and social data. Education efforts by grazing coordinators and NRT and StfF personnel were provided to women and children in Kalama, Westgate, Meibae, Naibunga, and Melako Conservancies in 2014 and 2016.

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

Change in Biodiversity	Increased populations of trigger species Grevy's zebra, elephant, beisa oryx, black rhino, gerenuk, and reticulated giraffe
Monitored Change	Heat maps of sightings of five of these species show expansion of distributions and 3-5 x higher sightings per year of five species (see section 5.1.3 below). NRT, Sera Conservancy, and the Kenya Wildlife Service have established a black rhino sanctuary in central Sera Conservancy.
Justification of Change	The change has occurred through greater protection by project-employed game scouts, who also help manage leakage and livestock incursions from outside the project area.

	Geographic centers of population counts in the project area are associated with areas of more successful implementation of grazing plans, such as central Sera, Kalama, and southern Melako Conservancies
Change in Biodiversity	Increased species richness and species diversity indices of savanna biodiversity indicator taxa, including dung beetles, grasshoppers, herbaceous plants, and birds
Monitored Change	Diversity indices and species counts were sampled at biodiversity camps
Justification of Change	The change has occurred through greater protection by project-employed game scouts, who also help manage leakage and livestock incursions from outside the project area. Geographic centers of population counts in the project area are associated with areas of more successful implementation of grazing plans, such as central Sera, Kalama, and southern Melako Conservancies

5.1.2 Mitigation Actions (B2.3)

In the initial phases of the project, there is a strong need to create long periods of rest from grazing across large areas in order to allow some vegetation recovery. Given the widespread occurrence of degraded soil and vegetation conditions across the project area, herders, particularly from Namunyak Conservancy, may be tempted to move livestock into the Mathews Forest Reserve to cut limbs and small trees as forage for livestock. Chronic use of the forest for this purpose in the past has led to a reduction in forest cover and degradation, such as dominance of the forest by fast-growing early successional species²¹.

This potential negative impact should disappear as the project proceeds because successful planned rotational and migratory grazing should produce substantial grass banks that can be used to provide short-term rest and drought forage bases. These grazing options should greatly lessen the motivation for herders to enter forests and thus reduce potential impacts. Consequently, any potential negative effects should be temporary and produce little lasting effects on biodiversity. The impact of grazing management and movements of herders and avoidance of protected areas when possible is evident in the net impacts of grazing on the Mathews Forest, a major contributor to the High Conservation Value of the project area (see section 7.3).

The lack of protection of wildlife and continued rangeland degradation outside the project zone may further deplete biodiversity found there relative to that found inside the project zone, where rangeland restoration, funding for community conservation and monitoring activities, and community-based protection and law enforcement have already increased wildlife numbers and biodiversity. Such a result represents an *unmitigated* negative effect of the project activity. Despite ready access to firearms, improved socioeconomic conditions associated with the project do not

lead to increased wildlife poaching among the principal pastoralist groups, as their cultural traditions largely taboo the consumption of wild meat.

Unmitigated negative impacts on biodiversity outside Project Zone	Net Positive Biodiversity benefits with project
Degraded soil and vegetation from overgrazing and foraging of livestock	Implementing grazing plans using rotational grazing to reduce overgrazing and rebuild habitat for herbivores.
Wildlife poaching	Habitat improvements in the Sera Conservancy contributed to the establishment of a Rhino sanctuary by the Kenya Wildlife Service. Declines in elephant caresses found that resulted from illegal killing began with the start of the carbon project from 2012-2015.
Lack of wildlife protections	Game scouts reported sightings of Reticulated Giraffe, Grevy's Zebra, Beisa Oryx, wild dogs, and lions increased from 2012 to 2015 in Melako and Sera Conservancies as these conservancies began implementing planned grazing across full Conservancy in 2014. Game scouts reported sightings of Reticulated Giraffe, Grevy's Zebra, Beisa Oryx, wild dogs, and lions increased from 2012 to 2015 in Melako and Sera Conservancies as these conservancies began implementing planned grazing across full Conservancy in 2014. Kenya Wildlife Service will address rhino conservation and Grevy's Zebra Trust will address zebra conservation. GZT specifically focuses on restoring water and forage for zebras.
Violent conflict for limited forage creating greater insecurity and furthering degradation of soil, vegetation, and wildlife poaching	Project activity includes creating and enforcing agreements to allow herders from outside the project area to graze inside the Project Zone when instructed to support the planned rotational grazing.
Minimal data gathered over time to show trends in species biodiversity	Project includes data gathered for wildlife counts of major species as well as species diversity of key indicator groups plants, dung beetles, trees, grasshoppers, and trees.

Project activities follow the precautionary principle, i.e. the project does not introduce new products or processes whose ultimate effects are disputed or unknown. The project does not employ any new vegetation treatments or restoration procedures other than adjustment of animal husbandry.

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

Project activities that engage rangeland coordinators and game scouts on the landscape have led already to demonstrable biodiversity benefits. Black rhinos (*Dicornis bicerors*) have successfully been introduced into the Sera Conservancy in 2016 and remain dependent on sustained, diverse forages and safety from monitoring by engaged Conservancy members. Illegally killed elephants have sharply declined since the start of the project and three umbrella wildlife species that are endemic to northern Kenya and/or endangered are reticulated giraffe, Grevy's zebra, and beisa oryx, as discussed in section **5.4 Exceptional Biodiversity Benefits** below, have expanded their range and increased in density since the start of the project. These positive responses have occurred in part because Conservancy members are engaged across the wider Conservancy landscape in wildlife monitoring and shifting grazing activities. However, rangeland recovery associated with planned grazing appears to be important as well because the highest densities of wildlife occur in "Core" and Buffer" areas that previously experienced the most protection from grazing and typically were only grazed by livestock in the dry season. As rangeland vegetation recovers in settlement areas, wildlife populations should expand into those areas as well.

The project will not specifically undertake specific management actions to increase any specific component of biodiversity or endangered or vulnerable species. However, other stakeholders and partners with NRT, specifically the Kenya Wildlife Service in addressing rhino conservation and Grevy's Zebra Trust in addressing Grevy's zebra conservation will have their efforts augmented by project activities. Indeed GZT's priority management action for the past 8 years has been to improve livestock grazing practices to restore water and forage for zebras.

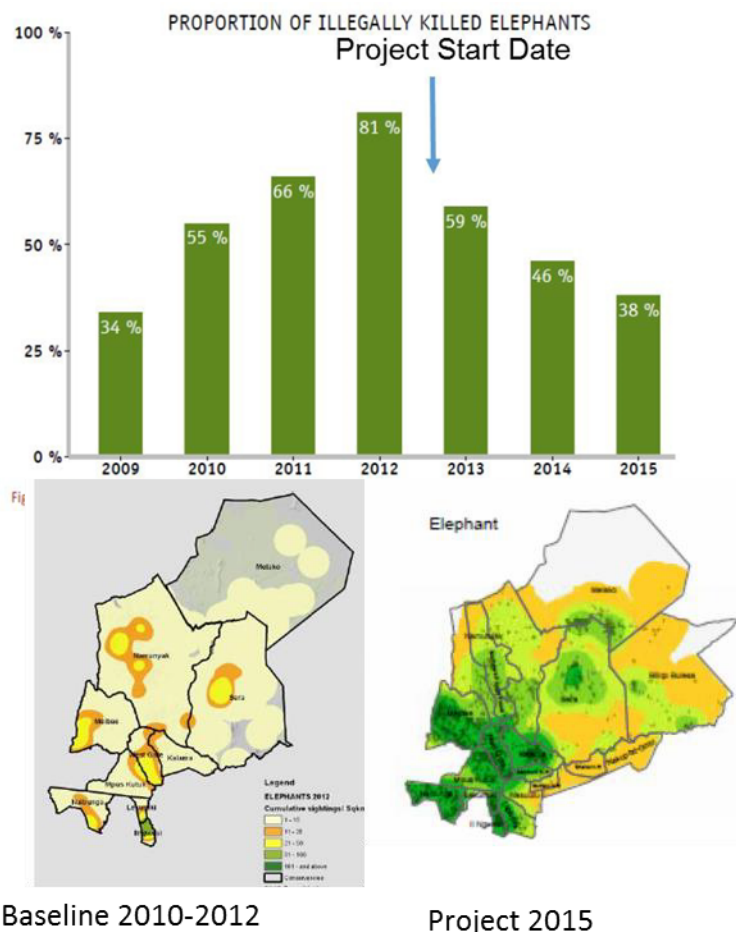


Figure 23. Upper: Proportion of elephant carcasses found that resulted from illegal killing (PIKE) from 2009 – 2015. Note that declines began following the start of the carbon project, again likely due to increased activity of game scouts and herders across the landscape in an effort to implement regional grazing plans in the dry season and to more fully utilize forage within the project area. Other background factors, such as a decline in ivory prices and better international enforcement likely also contribute to this trend. **Lower:** Distribution and abundance of elephant sightings on the NRT Conservancies, 2010-2013, under Baseline conditions compared to Project conditions (2015). Note the expansion of range (any elephants observed) and local density (shift from yellow to green) across the project zone. Note in particular the dramatic increases in numbers in the southern group of Conservancies and along the Melako-Sera Conservancy border.

The project benefits the High Conservation Values associated with the forest reserves and other protected areas adjacent to the project area. The anticipated increase in grazing options within the project area should reduce pressure to use protected areas for forage during dry seasons or droughts. Previous studies in the NRT or northern Kenya area^[1, 43-46] strongly suggest that planned rotational grazing will increase plant diversity^[47-50]. However, the impacts of rangeland restoration and soil carbon storage on other potential conservation values are poorly documented outside major, well-studied protected areas. Project activities that engage rangeland coordinators and game scouts on the landscape have led already to the demonstrable biodiversity benefits

discussed above. During the same period (2005-2015), reflecting baseline conditions, Ogotu et al^[51] report that virtually all species of Kenya wildlife even inside reserves have declined on average by 63%. This contrasts with the considerable increases in elephants (Figure 23) the three endangered species (Figure 24) observed in comparing baseline years 2009-2012 and with reports of wildlife from game scouts (Wildlife CoMMs) in 2015-2016 during the project period.

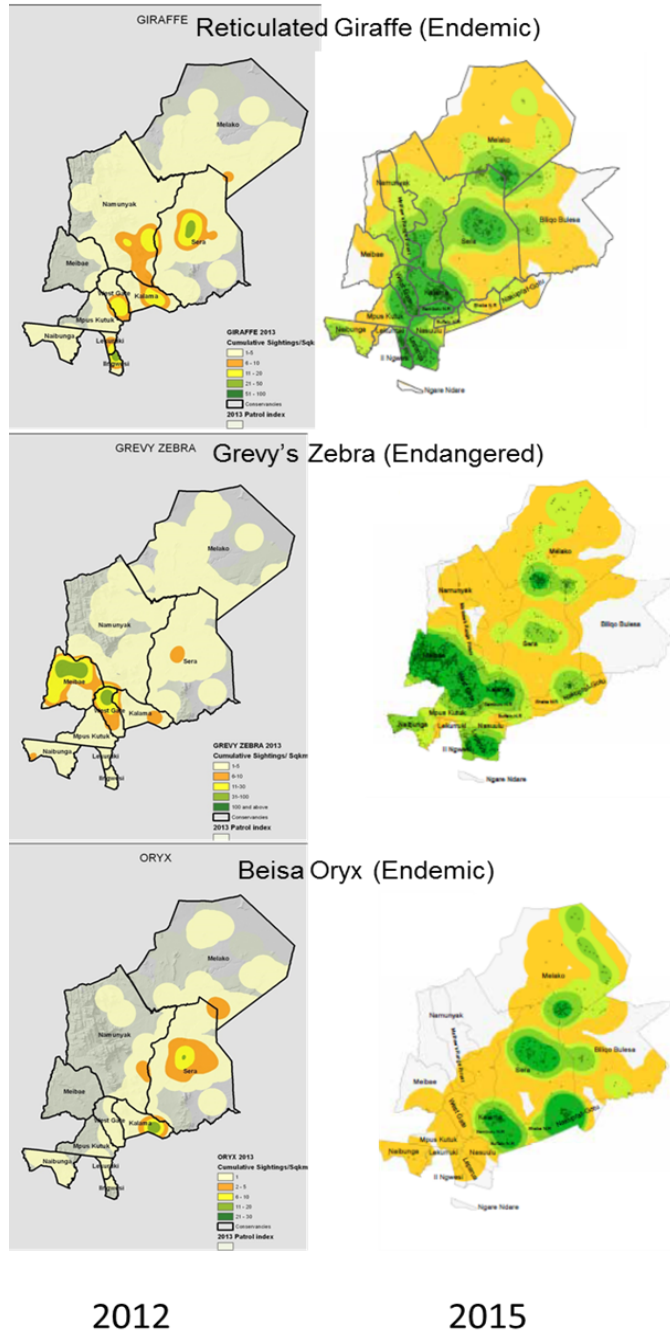
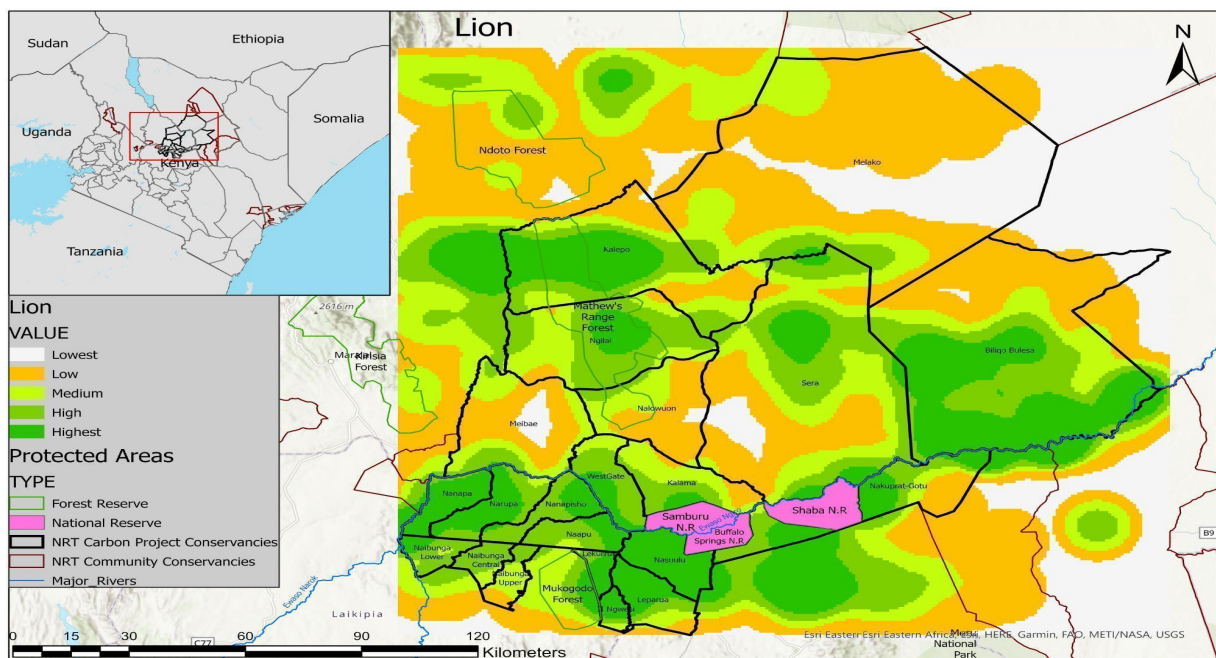
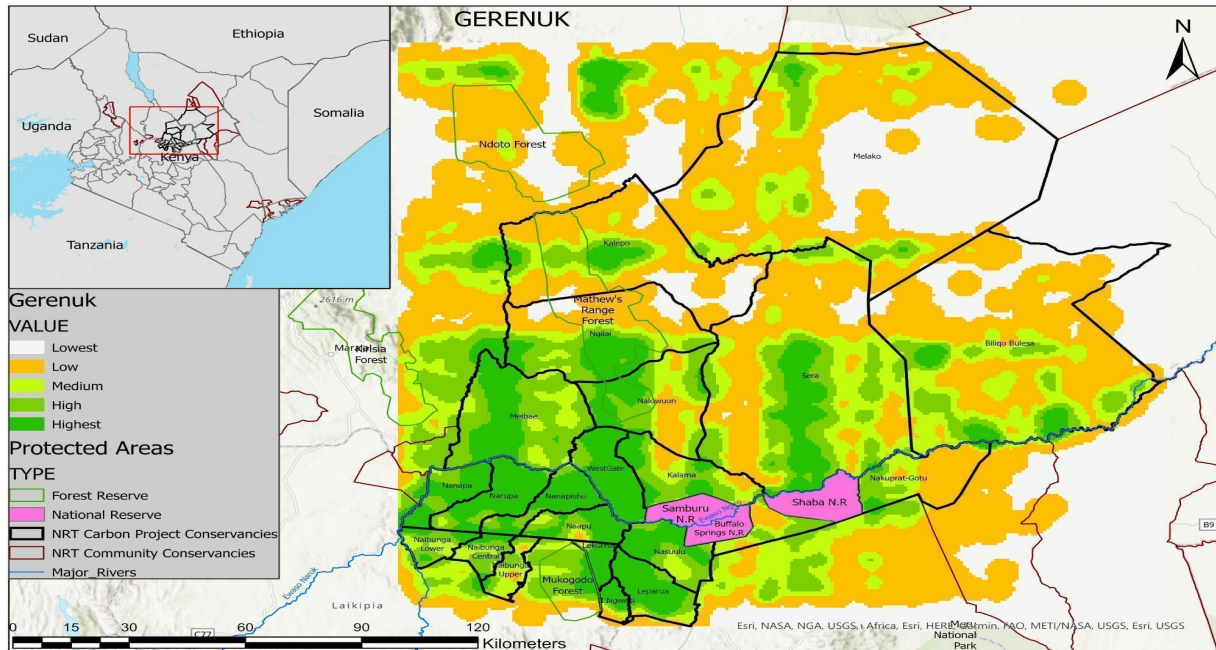
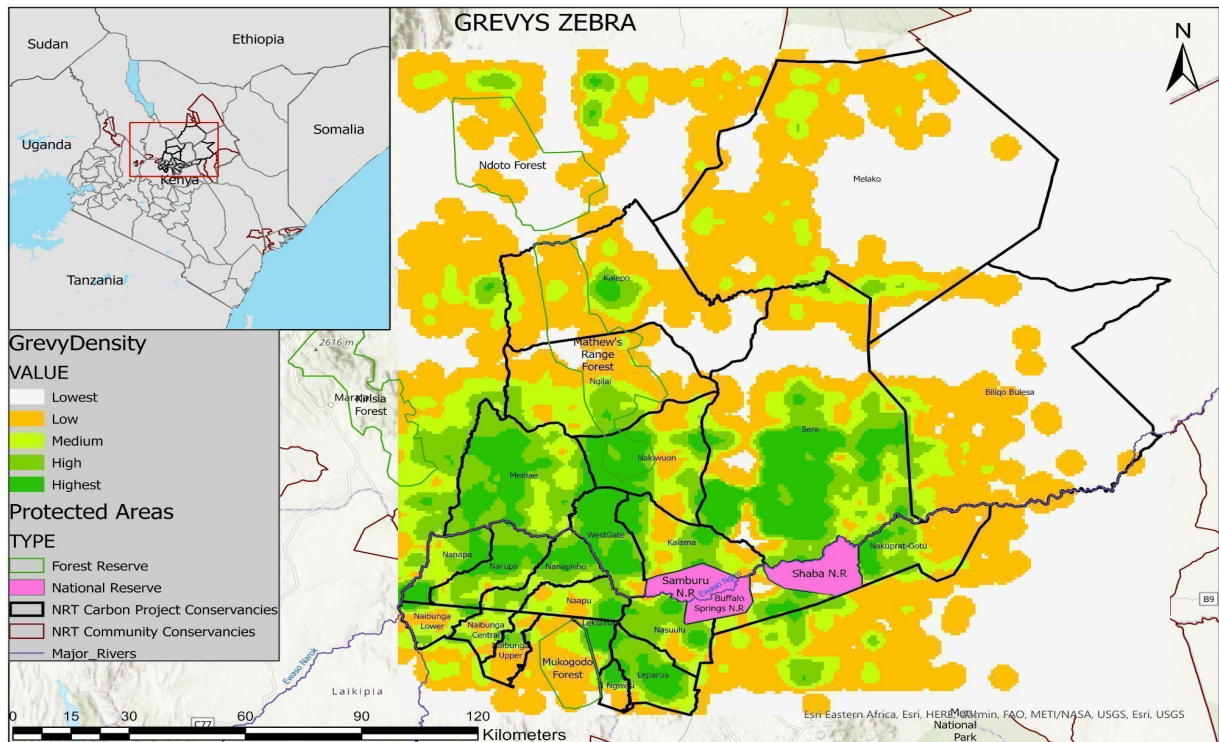
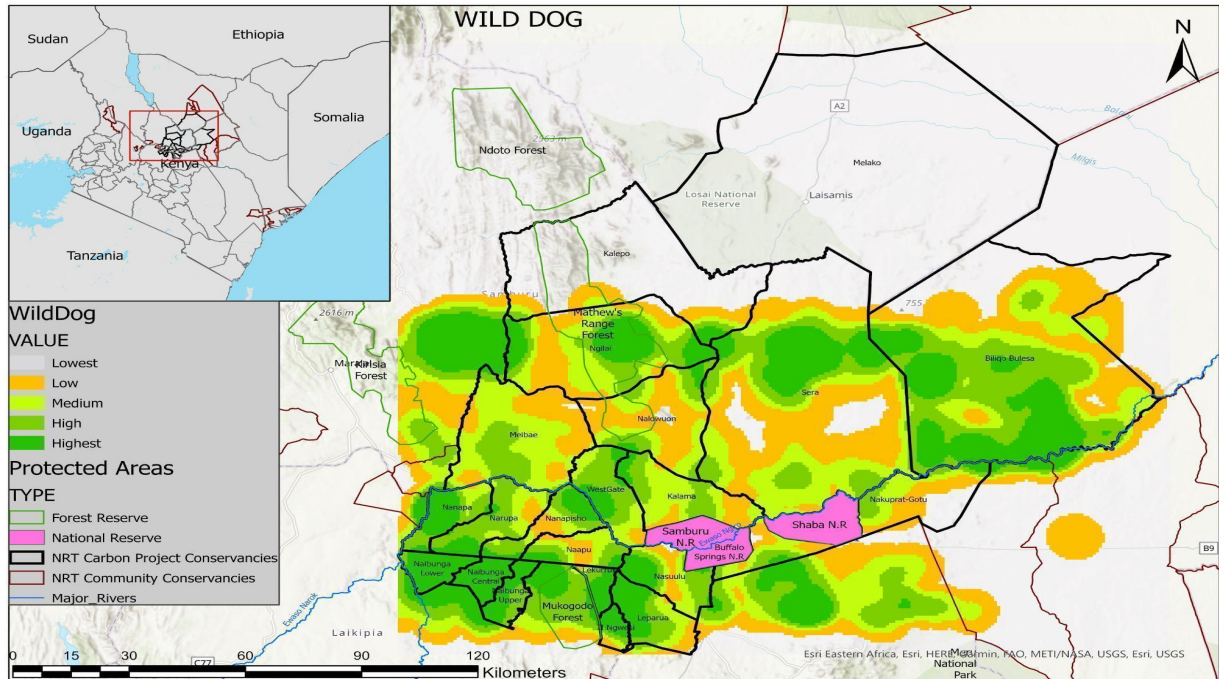


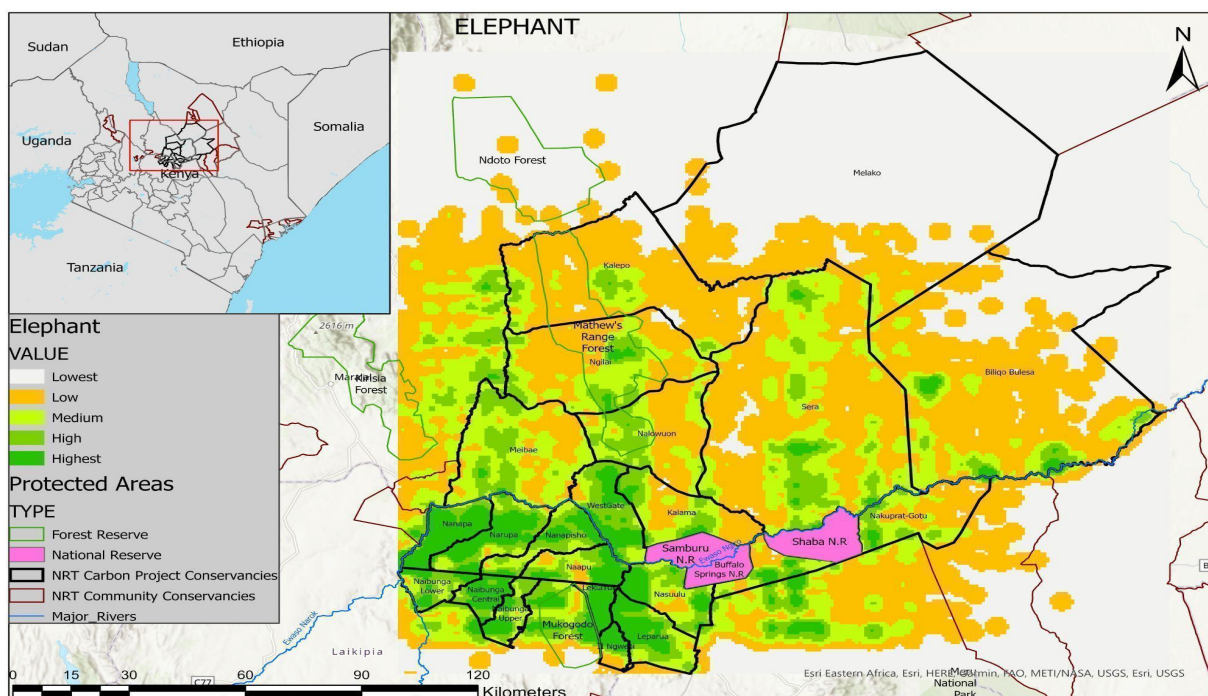
Figure 24. Heat maps of three umbrella, endemic and/or endangered species in the project zone, based on reported sightings by game scouts for 2012 (baseline conditions) and 2015 (project scenario). Green areas indicate the highest density of sightings, yellow and lighter tan areas the

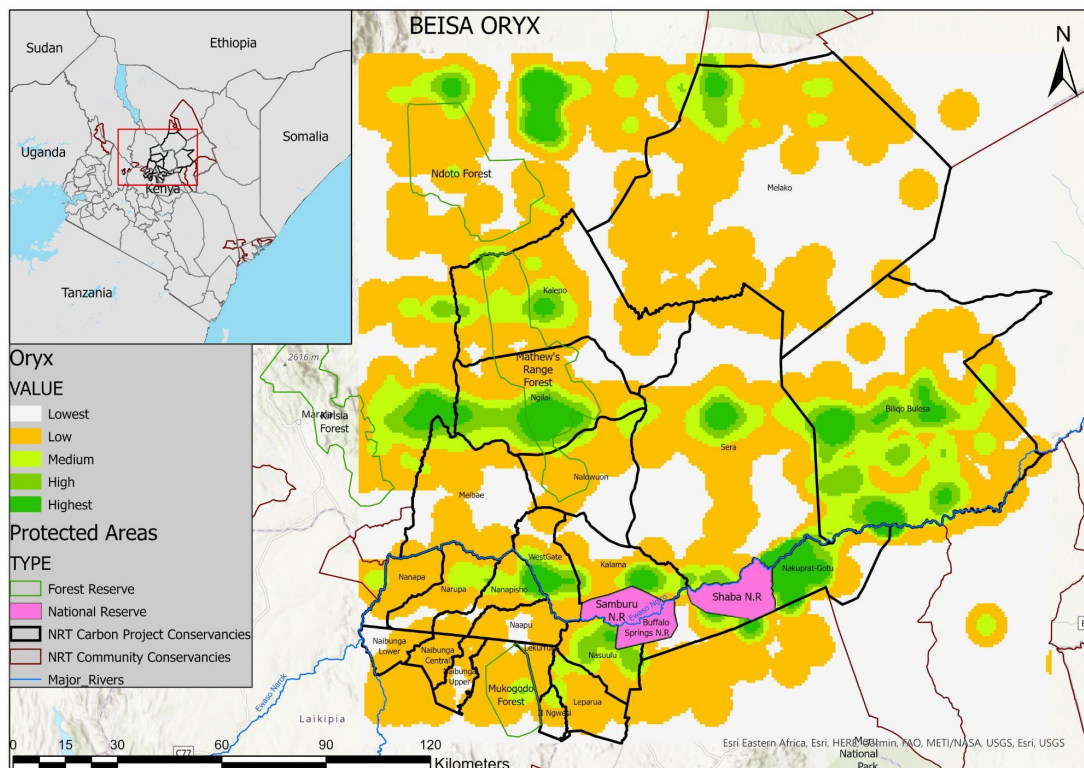
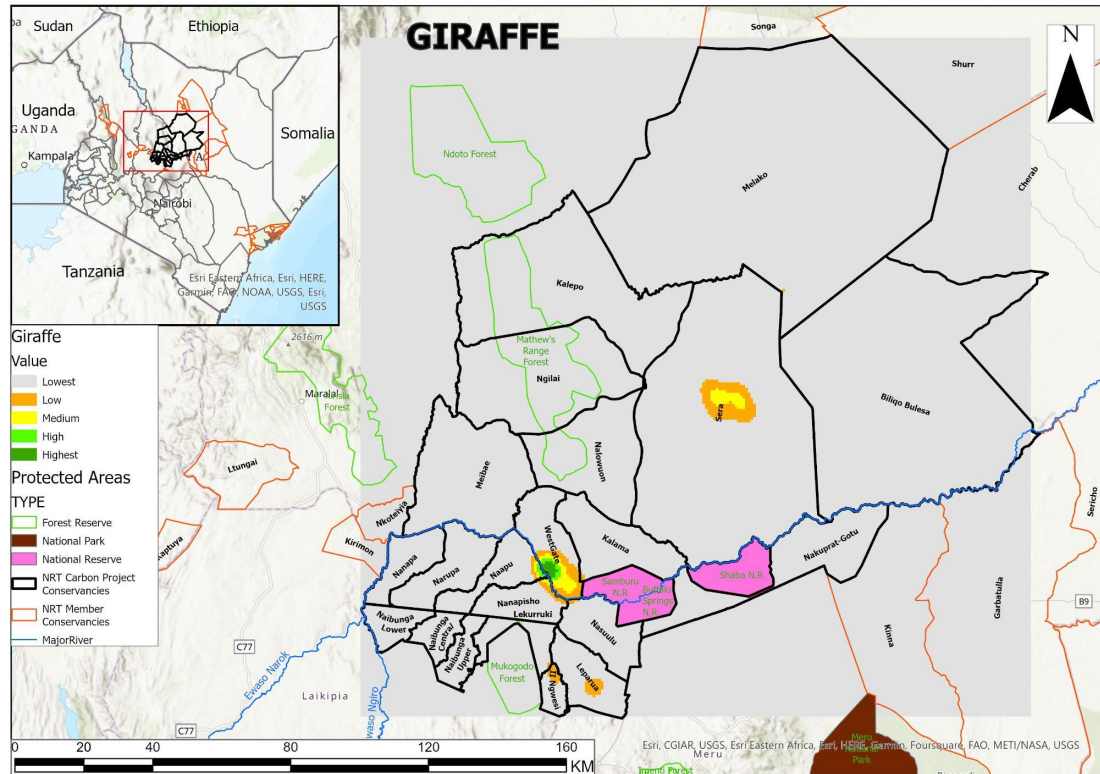
lowest density. Note the expansion of higher density areas in Melako and Sera Conservancies as these Conservancies began implementing planned grazing across the full Conservancy in 2014. These results, and similar results for wild dogs and lions (data not shown) show the potential for High Conservation Values in biodiversity as a result of project activities

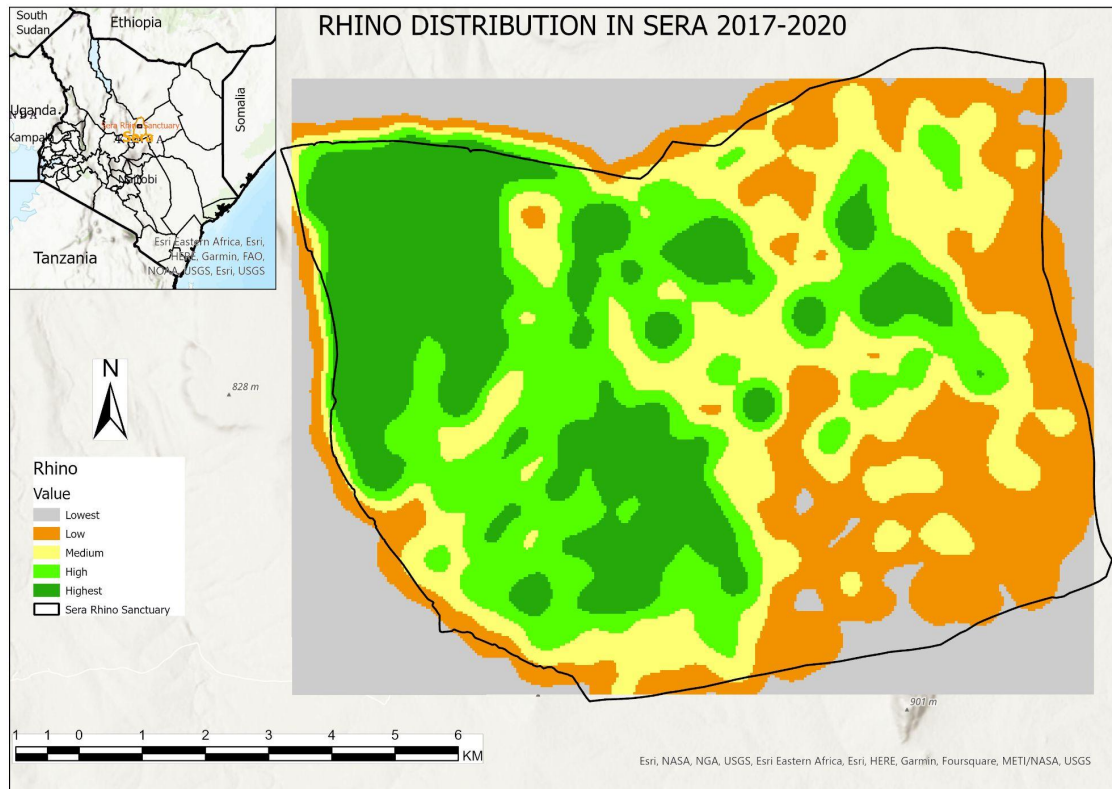
[1] <https://www.lewa.org/wp-content/uploads/2020/09/Kimiti-et-al-2020.-Evaluation-of-Restoration-success.pdf>











Figures 25-30d. Heat maps for the trigger species of Gerenuk, Lion, Wild Dog, Grevys Zebra, Cheetah, Elephant, Reticulated Giraffe (30b), Beisa Oryx (30c), and Black Rhino (30d) from data collected from 2017-2020. As an example, when Grevys Zebra is compared in heat maps that represent data from 2012 and 2015 (shown in Figure 24), there is a noticeable increase in the amount of density represented across the project area.

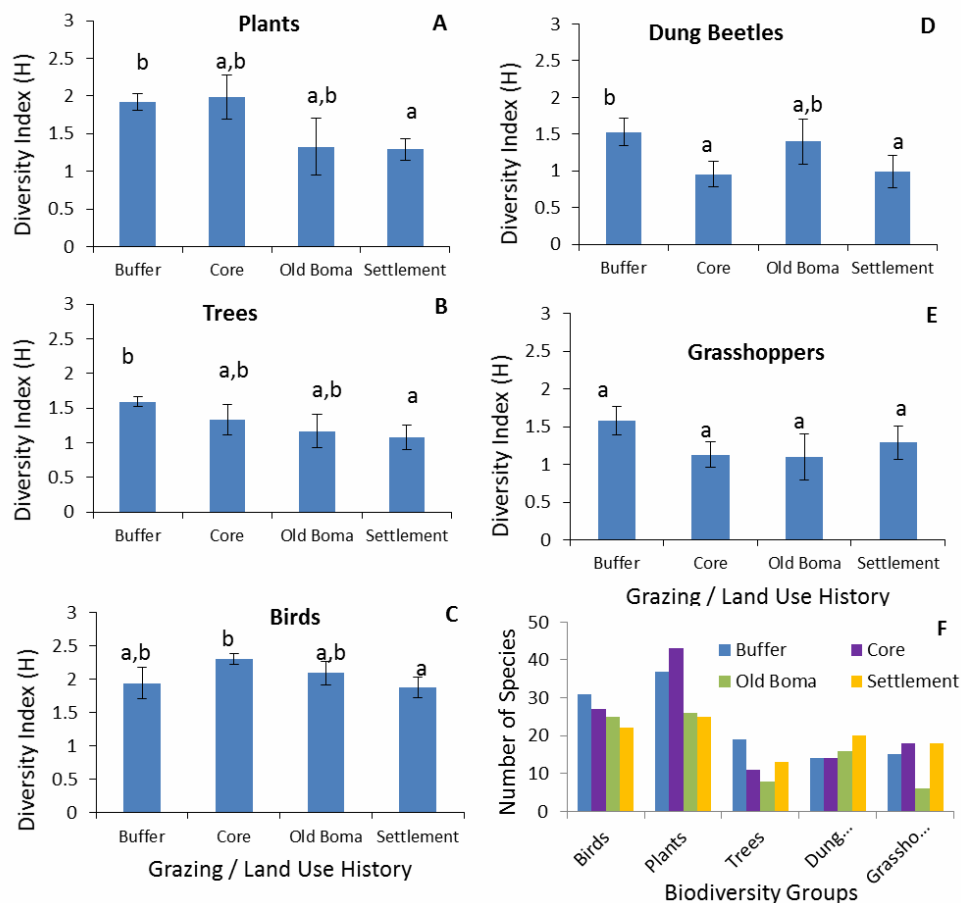


Figure 31. Mean ($\pm$ SE) Shannon-Wiener diversity indices for five different biodiversity indicator groups (A-E) sampled from each of four different land uses/management areas in three Conservancies (Kalama, Westgate, and Sera) in northern Kenya: Buffer areas with only dry season grazing, Core areas with rare dry season grazing only, Old bomas (livestock corrals) abandoned 10-20 years previously, and Settlement areas subject to continuous grazing. F) Cumulative number of species of the different indicator groups across all three Conservancies for each of the different land management/uses.

In addition the project is expected to increase species diversity of multiple taxa, including plants^{[46],[52, 53]}, birds^[48, 54] and arthropods^[47, 48, 50, 55]. Our baseline data for areas with different grazing history suggest that project activities will enhance diversity of multiple previously under-sampled groups.

Baseline measurements of the species diversity of key indicator groups, plants (trees and herbaceous), birds, grasshoppers and dung beetles were made at six sites at each of three different Conservancies: Kalama, Sera, and Westgate (Figure 31). This study demonstrates a proof of concept that relaxed grazing pressure from PRM grazing practices will increase diversity. The species diversity index of all herbaceous plants and trees was significantly higher in Core versus Settlement areas and for dung beetles in protected (Core or Buffer) areas versus Settlement areas. The data clearly show that PRM grazing practices are likely to improve a broad spectrum of biodiversity across the project area. The baseline diversity study sampled a total of 746 trees, 1481 birds, 1161 dung beetles and 351 grasshoppers. Number of herbaceous plants

was not available because cover was measured, rather than individuals. 67 herbaceous plant, 24 tree, 83 bird, 31 grasshopper, and 32 dung beetle species were observed or collected across the 18 sites.

Species richness and Shannon-Wiener diversity indices were calculated for the cumulative observed or captured species for each taxonomic group. The standard Shannon-Wiener Index^[56] was H , where

$$H = - \sum_{i=1}^n p_i \ln(p_i) \quad (44)$$

Where p_i is the relative (proportional) abundance (or in the case of herbaceous plants, canopy cover) of the i th species in the sample.

Mean species richness per site was not significantly associated with land use or past grazing management for any of the taxonomic groups (Figure 31F). Cumulative numbers of species across all three conservancies for each grazing history/land use showed different patterns : birds, herbaceous plants and trees exhibited a greater cumulative number of species across conservancies in Protected areas than in Settlements, while dung beetles exhibited fewer cumulative numbers of species in Buffer and Core areas compared to Settlement areas.

Buffer areas (seasonal grazing only) had higher species richness or diversity indices than Settlement areas in 10 out of 10 tests (sign test, $P = 0.001$, one-tailed test) and Buffer and core areas combined had higher species richness and diversity indices than Settlement areas in 16 of 20 tests ($P = 0.011$, one-tailed tests).

Ten species endemic to northern Kenya or the Horn of Africa (Ethiopia, Djibouti, Eritrea, Somalia, and northern Kenya) were found among birds, dung beetles, and grasshoppers combined. Of these endemic species, 5 occurred in Core areas, 4 in Buffer areas, and 1 in Settlement areas, and thus significantly more frequently than expected by chance ($P = 0.02$) in Protected areas than in Settlement.

Taken together, these baseline data show how project activities enhance biodiversity. Scheduled future biodiversity surveys will be expanded to include Melako Conservancy, thus expanding the range of sampling for indicator taxa of biodiversity.

In 2021, in Westgate, Kalama, and Sera biodiversity camps, a total of 1,362 birds of 73 species from 329 records were observed. Likewise, 992 dung beetles of 51 species were captured. We found almost no grasshoppers, and then only newly hatched nymphs, in 2021. By comparison, the 2014 (baseline) study counted 746 trees, 1,481 birds, 1,161 dung beetles and 351 grasshoppers. Number of herbaceous plants was not available because cover was measured, rather than individuals. The 2014 survey found 67 herbaceous plant, 24 tree, 83 bird, 31 grasshopper, and 32 dung beetle species

These somewhat lower overall animal numbers in 2021 compared to 2014 (baseline) reflect the fact that sampling in 2021 occurred during a time period, April – June, that would normally feature 200-300 mm rainfall. Instead, the project area experienced no rains in the previous short rainy season (November- December) or during the early wet season just prior to sampling^[1]. The dry conditions that persisted on the project area from June 2020 to May 2021 may have contributed to essentially a failed early wet season hatch of grasshoppers. In addition, locust swarms were widely sprayed throughout the project area during 2019 and 2020^[2]. These insecticide sprays

may have dramatically depleted grasshopper populations and egg-laying across virtually all Orthoptera taxa. Thus, a possible change in grasshopper diversity and species richness cannot be determined from 2021 sampling and will await future biodiversity surveys.

The weather pattern, and possibly locust control insecticide spraying, may also explain shifts in bird species composition in our survey. The year 2014 featured 8 more insectivorous flycatcher and sunbird species than did 2021, and large gallinaceous birds like vulturine guineafowl, yellow-necked spurfowl, and black-faced sandgrouse were observed in far fewer transects (4 of 19) in 2021 than in 2014 (13 of 18) and only 151 individuals were observed in 2021 as compared to the 694 observed in 2014. In addition, the frequency of encounters with large colonies of weavers declined from 12 of 19 surveys in 2014 to 4 of 19 in 2021. Other patterns in the bird data suggest that these declines were not related to the project activity. Six species not observed in 2014 and commonly found in tallgrass habitats in East African protected areas were observed in the project area: Plain-backed pipit, Rattling cisticola, Fischer's sparrow lark, Side-spotted barbet, and Steel-blue and Straw-tailed whydahs. This suggests that improvements in vegetative cover observed during the monitoring periods 2013-2016 and 2017-2021 may have attracted tall grass-associated bird species in some areas. Moreover, 8 species endemic to northern Kenya or the Horn of Africa (eastern Ethiopia, Somalia, and northern and eastern Kenya) were found in 2021 but not in 2014.

These observations for the entire project area are consistent with a decline in bird species richness per site from 2014 to 2021 (Fig. 32A). However, bird diversity index H' per site was not significantly different between years (Fig. 33A). This result suggests that the disappearance in our surveys of large, locally abundant gallinaceous birds and large colonies of weavers may have made the bird species we did encounter more similar in abundance, which would yield a higher diversity index for the same number of bird species

^[1] <https://www.nature.com/articles/d41586-020-02698-3>

^[2] <https://www.theafricareport.com/23745/kenya-opts-for-pesticides-to-contain-locust-infestations/>

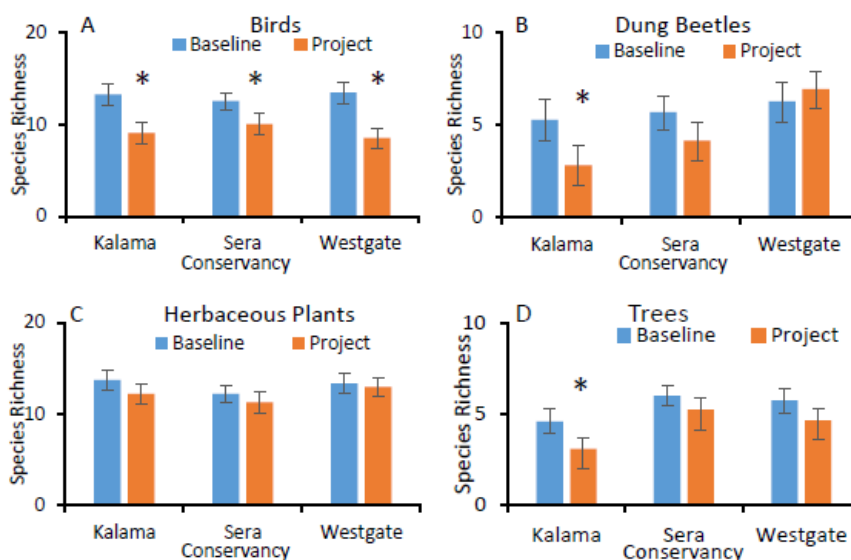


Figure 32. Mean $\pm$ SE species richness of four indicator taxa in 2014 (Baseline) and 2021 (Project) at three different Conservancies in the project area. Asterisks indicate difference significant at $P < 0.05$ in a Generalized Linear Model

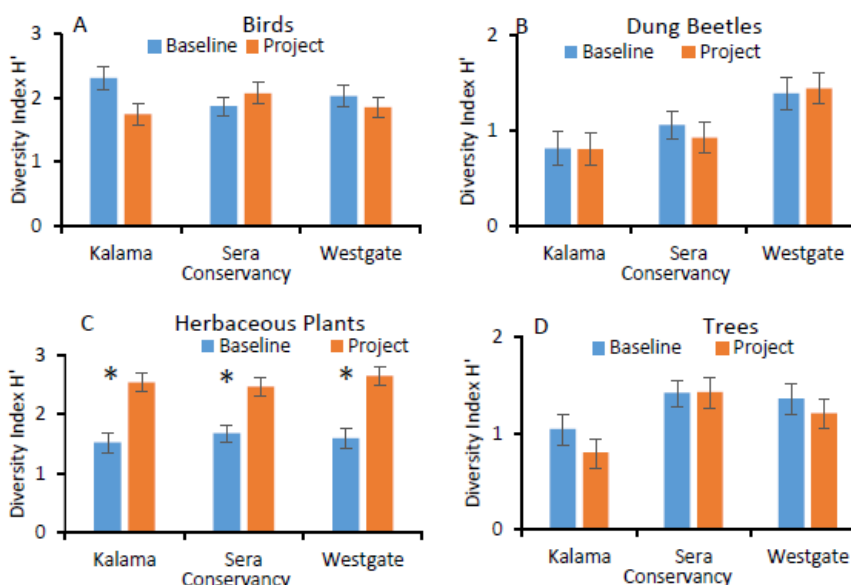


Figure 33. Mean $\pm$ SE Shannon-Wiener diversity Indices for four indicator taxa in 2014 (Baseline) and 2021 (Project). Asterisks indicate differences significant at $P < 0.05$ in a Generalized Linear Model

In contrast with birds, overall species richness of dung beetles was much higher in 2021 (51) than in 2014 (32), a trend that would be expected from the observed improved vegetation cover and expanding wildlife populations associated with project activity. Despite this increase in overall project area species richness, mean dung beetle species richness and diversity index H' per site showed no difference between 2014 and 2021 (Fig. 32B, 33B). These two patterns combined suggest that in comparison to 2014 dung beetle species communities in 2021 differed in composition more strongly between different locations. This is a pattern consistent with higher beta diversity associated with project activity.

Herbaceous plant species richness across all sites increased over time across the entirety of the project area, as we found 81 species in 2021 as compared with 62 species in 2014. This change

was associated with an increase in mean perennial grass species richness per site (Figure 32C). In 2014, we detected only 8 perennial grass species across the project area, but found more than 16 species in 2021. Moreover, % cover of perennial species per site increased from an average of 1.2% (± 2.4 SE) in 2014 to 5.2% (± 2.1 SE) in 2021. These changes produced large increases in the diversity index for herbaceous plants at all three Conservancies sampled between 2014 and 2021 (Fig. 33C).

Previously, we compared the diversity of these taxa among areas receiving different livestock use as a proof of concept that relaxed grazing pressure from PRM grazing practices will increase diversity (Fig. 31). In 2014, the species diversity index of all herbaceous plants and trees was significantly higher in Core versus Settlement areas and for dung beetles in protected (Core or Buffer) areas versus Settlement areas. In contrast, in 2021, bird and dung beetle species richness and diversity index showed no difference between protected (previous buffer and core) and settlement areas (Fig. 34). Such a lack of difference may reflect an increase over time in these metrics in settlement areas.

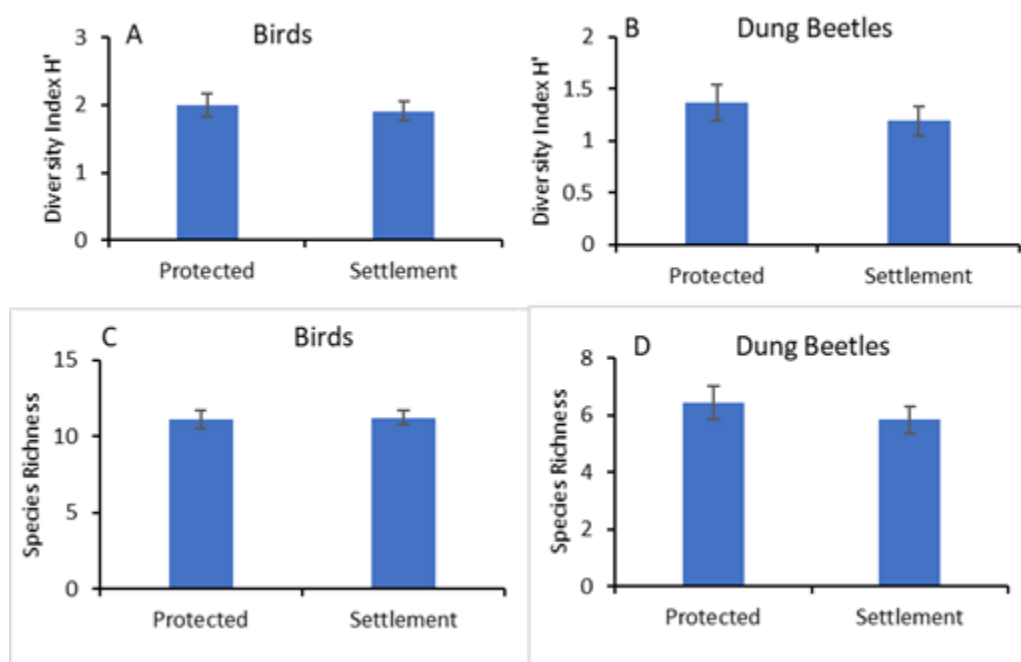


Figure 34. Mean ($\pm$ SE) diversity index (A,B) and species richness (C,D) for birds and dung beetles, two indicator taxa, respectively, in areas with grazing restricted to dry season or short periods in the wet season (Protected) vs areas subject to planned rotational grazing in the wet season (Settlement) under the project activity. Means were not different ($P < 0.05$) in Generalized Linear Models

In contrast, mean herbaceous plant and tree diversity indices and herbaceous plant species richness continued to be higher in “protected” areas with restricted grazing in 2021, as they were in 2014 (Fig. 35).

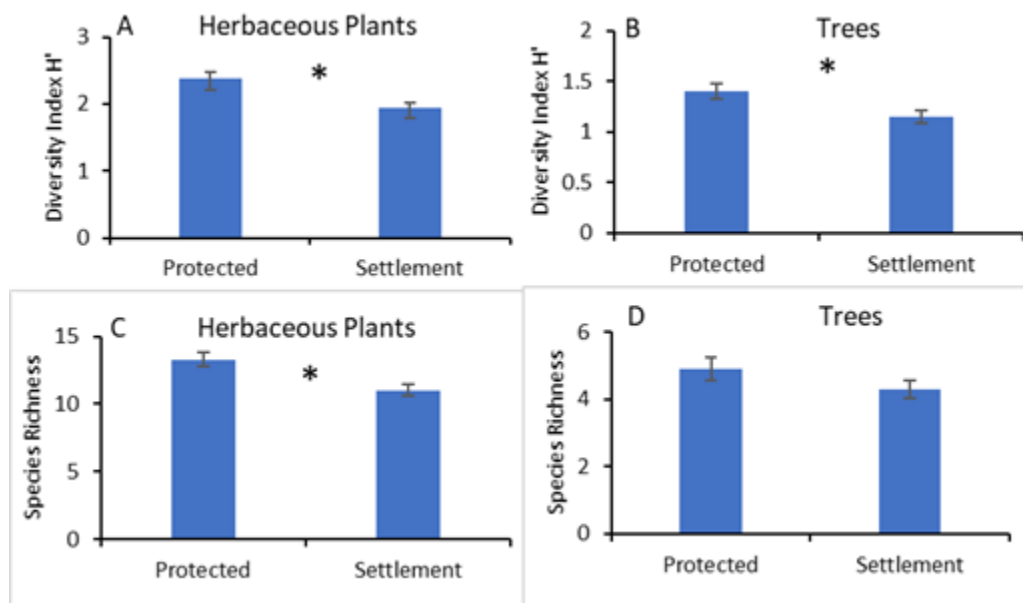


Figure 35. Mean ($\pm$ SE) diversity index (A,B) and species richness (C,D) for birds and dung beetles, two indicator taxa, respectively, in areas with grazing restricted to dry season or short periods in the wet season (Protected) vs areas subject to planned rotational grazing under the project activity grazing in the wet season (Settlement) under the project activity. Asterisks indicate significantly different ($P < 0.05$) differences in Generalized Linear Models.

These persistent differences due to grazing management that occurred prior to the onset of the project activity suggest that vegetation is recovering in both types of areas from poor grazing practices that occurred decades before grazing restrictions were imposed. Fig. 33C shows that herbaceous plant species diversity indices are increasing in in the project area and the Generalized Linear Model suggests that diversity indices increased similarly under the project activity in protected and settlement areas (interaction between management and year is not significant, $X^2_{2,29} = 0.672$, $P = 0.715$).

Tree numbers and species richness would be expected to change slowly with time, and with the exception of tree species richness in Kalama Conservancy, we observed no difference in tree species richness between 2014 and 2021 or between protected and settlement grazing management histories (Fig. 32D, 33D, 34D, 35D). The decline in tree species richness in Kalama Conservancy may be due to attempted control of the invasive tree *Acacia reficiens* at several of our sampling sites^[1].

Overall, our results suggest that, given the prolonged low rainfall conditions and the prevalence of insecticide spraying to control locusts prior to sampling, the 2021 biodiversity survey may not provide a worthy evaluation of change in biodiversity of animal indicator taxa. Nevertheless, other data suggest the project activity is increasing species richness of dung beetles and herbaceous plants over the entire project area. Herbaceous plant diversity indices were substantially higher in 2021 than in 2014 and this change was driven by an increase in the number of perennial grass species detected. Results for total species numbers across the three Conservancies sampled in both 2014 and 2021 suggest that the project activity is associated with higher beta diversity (greater variation in species composition across sites) of plants and dung beetles. Even for birds, we found increased numbers of species of endemics and species associated with tall grass habitats that were absent from the 2014 survey. Thus we conclude that the project activity has improved several important aspects of biodiversity of these indicator taxa relative to baseline conditions.

5.1.4 High Conservation Values Protected (B2.4)

Use of Biodiversity HCV sites by livestock has been decreased by project activities (Figure 21). Livestock monitoring data and analysis of satellite imagery for relative change in NDVI for 2011-2012 (Baseline) and the monitoring period 2013 – 2020 (Project) indicate a clear reversal of trend in reducing encroachment into the Mathews Forest (Figure 21, 22). Even in a drought year, when encroachment is usually at its worst, livestock produced negative impacts on only 20% of the Mathews Forest in 2014 and approximately 15% in 2017 (Figure 22). Impacts during the project (e.g., 2014) have mostly occurred on the edges where degradation has already occurred, compared to over 30% of the forest negatively impacted, and mostly in the high biodiversity rainforests at high elevation, that occurred in 2008, a comparatively wet year (Figure 22). As rotational practices continue and grass banks are established across a wide range of locations in the project area, encroachment should decrease further, thus addressing the number one threat to the Mathews Forest.

5.1.5 Invasive Species (B2.5)

- i. No invasive species have been introduced into the project zone, nor have established invasive species spread through the project zone, due to project activities.

5.1.6 Impacts of Non-native Species (B2.6)

No non-native species have been used as part of project activities. Small-scale grass restoration efforts have used a native grass *Cenchrus ciliaris* in these restoration efforts.

5.1.7 GMO Exclusion (B2.7)

No GMOs have been used to generate GHG emissions reductions or removals, as re-seeding or plantings are not a project activity. Project activities merely adjust the way in which livestock are herded and moved across the landscape.

5.1.8 Inputs Justification (B2.8)

No fertilizers are allowed according to the applicability conditions of methodology VM0032. Project activities do not include addition of pesticides or other potentially harmful chemicals into the environment.

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

As far as we are aware, there have been no negative offsite biodiversity impacts related to project activities. Our efforts to reduce herder excursions off the project area has likely reduced impacts on off-project forest areas (see section 5.1).

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

No explicit evaluation of biodiversity off the project area is available. However, given the rapid expansion of areas used by the six trigger species (Figures. 25-30) and possible dispersal of animals off the project area, it is likely that project activities increased biodiversity off the project area.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

- Biodiversity of indicator taxa birds, dung beetles, grasshoppers, herbaceous plants and trees was measured in April and May 2021 at 6-7 sites at each of five biodiversity camps in the project area (Fig. 36). Measurements of the species diversity of key indicator groups, plants (trees and herbaceous), birds, grasshoppers and dung beetles were made at 6-7 sites at each of five different Conservancies: Kalama, Sera, Il Ngwesi, Melako, and Westgate, for a total of 31 sites sampled. We sampled at the same sites (approximately) as were sampled in Sera, Kalama, and Westgate in 2014, which allowed for a time comparison over the course of the monitoring period for those three Conservancies.
- Monitoring biodiversity occurs primarily through 1) wildlife counts by game scouts during patrols on their home Conservancies as part of the Conservancy's community monitored and management activities (Wildlife CoMMs), and 2) through periodic sampling of indicator taxa (insects, plants, birds) in the vicinity of five biodiversity camps distributed across the rainfall gradient in the project area.
- The results of these monitoring efforts for the monitoring period were all presented in section 5.1.3 (Figures 25-35).

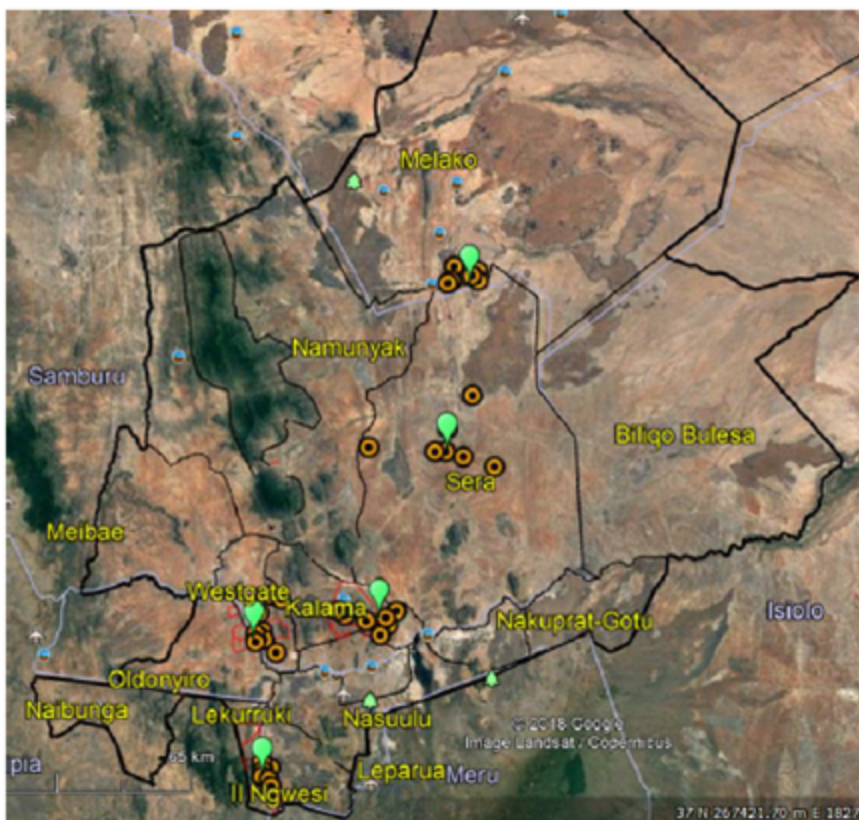


Figure 36. Map of locations of biodiversity sampling across the project area in 2021. Each point is clustered around a biodiversity “camp” from which measurements are made at each of 6-7 sites that include both sandy and black soils and areas with restricted (protected) and unrestricted grazing (settlement). Five biodiversity camps were sampled that encompassed the project-wide mean annual rainfall gradient from 300 mm in the north (Melako) to 800 mm (Il Ngwesi) in the south.

Species richness (number of species per site) and Shannon-Wiener diversity indices were calculated for the cumulative observed or captured species for each taxonomic group. The standard Shannon-Wiener Index^[56] was H , where

$$H = \sum_{i=1}^n p_i (-\ln(p_i)) \quad (44)$$

where p_i is the relative (proportional) abundance (or in the case of herbaceous plants, canopy cover) of the i th species in the sample.

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

In September 2016, we held a Conservancy-wide Project Meeting attended by Conservancy chairs, community grazing chairpersons, grazing coordinators and interested public. This meeting was announced on the NRT website and notices were posted in major community kiosks and markets. At the meeting, a report of the previous 12 months’ monitoring results for both implementation of the grazing plans and of wildlife and indicator taxa monitoring was presented. Results of the previous Social CoMMs surveys were also presented. Following this presentation,

plans for managing grazing during the ensuing 12 months were discussed, with feedback from Conservancy managers and the community attendees.

Since 2016, and as mentioned in Section 3.14 'Dissemination of Monitoring Plan and Results,' NRT continues to share communications notifying conservancies of upcoming biodiversity surveys. The results of these surveys are shared with the conservancies who then communicate the results to communities. All conservancies have been briefed on the project and its ongoing monitoring requirements in terms of soil sampling, biodiversity surveys, remote sensing, livestock movements and numbers, etc. At the end of every year, each conservancy receives a summary of their Wildlife CoMMS data that highlights their progress.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

The project activities will provide exceptional biodiversity benefits. Project activities protect and promote populations of trigger species, protect sky island habitat for forest endemic species and reduce livestock pressure on permanent wildlife water sources protected by national game reserves.

5.4.1 Trigger Species Population Trends (GL3.3)

In the *project scenario*, a specific rhino restoration project begun in 2015 in Sera Conservancy has now yielded 20 rhinos. Active conservation methods and range improvements resulting from grazing management education by stakeholder Grevy's Zebra Trust³⁵ are associated with expansion of area used and number of sightings of Grevy's zebras (Fig. 24 in section 5.1.3) The Conservancies represent the only growing population of Grevy's zebra in Kenya (see section 5.1.2 and 5.1.3). Trends for gerenuk and elephants are positive in the project zone (Figs. 25, 30) in contrast to negative trends even in protected Reserves found in the Ogutu et al. (2018) study. Active conservation measures in the project zone include persistent patrols of community game scouts, which have reduced elephant mortality in the project zone by more than 60% over the past five years. Rangeland recovery spurred by planned rotational and migratory grazing should further aid conservation efforts through habitat restoration, as shown by the dramatic increase in wildlife species in livestock "refuges" (core and buffer areas) since the project start.

Trigger Species	Black rhino (<i>Diceros bicornis</i>)
Population Trend at Start of Project	No animals on project area
Without-project Scenario	Continued lack of presence on the project area and further declines in nearby National Reserves and Lewa Conservancy
With-project Scenario	A specific rhino restoration project begun in 2015 in Sera Conservancy established the Sera has now yielded 20 rhinos and reproduction had been documented for this project as heat maps until 2017. Community-based game scouts have successfully protected the sanctuary for more than two years. See 2013-2016 Monitoring Report for further data on the Black rhino.

³⁵ Grevy's Zebra Trust is a local stakeholder in the project, <http://www.grevyszebratrust.org>

Trigger Species	Elephant (<i>Loxodonta africana</i>)
Population Trend at Start of Project	Small herds near Samburu National Reserve and Mathews Forest, declining.
Without-project Scenario	Habitat degradation between preferred elephant habitats further isolates maternal groups, reducing migration and gene flow and increasing risk of mortality during drought and poaching in more restricted migration routes.
With-project Scenario	Improvement in wet season grass forage biomass and project-sponsored increases in game scouts creates safer habitats for elephants. Percent of elephant carcasses poached has declined from 81% to 30% since project start. Elephant sightings now occur throughout the project area with > 450 mm annual rainfall.

Trigger Species	Oryx (<i>Oryx beisa</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and near Samburu National Reserve, declining.
Without-project Scenario	Continued lack of presence on the project area and further declines in nearby National Reserves and Lewa Conservancy.
With-project Scenario	Protection from game scouts and improved forage increased sightings 20-fold. Found predominantly in project area < 400 mm annual rainfall in Sera, Melako, eastern Namunyak, and Biliqo Bulesa.

Trigger Species	Reticulated giraffe (<i>Giraffa camelopardus reticulata</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and Namunyak Conservancy, declining.
Without-project Scenario	Continued range degradation and poaching without game scouts.
With-project Scenario	Protection from game scouts and improved forage increased sightings 50-fold. Found in all 13 Conservancies but predominantly in project area > 450 mm annual rainfall in Namunyak, Kalama, Meibae, Oldonyiro.

Trigger Species	Gerenuk (<i>Litocranius walleri</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and eastern Namunyak Conservancy, declining.
Without-project Scenario	Continued range degradation and poaching without game scouts.

With-project Scenario	Protection from game scouts and improved forage increased sightings 20-fold. Found throughout the project area < 450 mm annual rainfall in Nakuprat-Gotu, Sera, Melako and Biliqo Bulesa Conservancies.
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Trigger Species	Carnivores, lion (<i>Panthera leo</i>), cheetah (<i>Acinonyx jubatus</i>) and wild dog (<i>Lycaon pictus</i>)
Population Trend at Start of Project	Occasional sightings in Sera Conservancy and Kalama Conservancy near Samburu National Reserve, declining.
Without-project Scenario	Continued prey declines and poaching without game scouts. Continued persecution by pastoralists due to loss of wild prey, forcing predation on livestock.
With-project Scenario	Protection from game scouts and improved prey abundance increased sightings 10-fold. Found in remote areas of Melako, Sera, and Namunyak Conservancies and near Ewaso Nyiro River (Oldonyiro, Westgate, Nasuulu, Kalama).

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

Narrative Addressing Adherence to VM0032 Applicability Condition #3

Any pastoralist system of livestock management cannot be defined by hard boundaries, and pastoralist culture reflects norms that ensure reciprocity: a community will share grazing with another provided that no livestock are stolen under the idea that the hosting community has grass this month, the visiting community might have grass next month. Ethnic groups and family relationships are woven into this mutualistic fabric as well; when rains fail, herders often travel long distances to share the range with relatives (by descent or marriage) or with fellow tribe members. Consequently, pastoralist communities establish traditional migration destinations under unexpected conditions, as well as certain seasonal movements into unsettled lands.

With these traditions in mind, it is worth discussing how movements of livestock outside the project boundaries might or might not indicate a lack of “control” of project boundaries.

Despite traditional exchanges that often drive livestock movements off the project area, which will be discussed below, the project was designed to encompass the majority of traditional movements by pastoralists of different ethnic groups under a wide range of weather conditions. Samburu herders in particular have a long history of moving from wet season ranges in the southwestern portion of the project area and the west side of the Mathews Range to dry season ranges in eastern Samburu County (Sera and the east side of the Mathews Range in Namunyak) and northern Isiolo County (Biliqo Bulesa). Likewise, Melako and Sera herders have access to vast dry season ranges on the tops of lava plateaus within their Conservancies that are unavailable during the wet season due to their heavy clay “black cotton” soils, but that often offer abundant forage during the dry season. Consequently, herders from these two Conservancies rarely move out of their own Conservancy, much less the project area. Finally, the Borana herders in Biliqo Bulesa traditionally move from sandier soils in the east during the wet season to heavier

clay soils and lava plateaus in the western portion of Biliqo during the dry season. These annual movements are possible because of the lack of permanent water that discourages settlements and a mutually accepted ethnic buffer zone between Samburu and Borana in the eastern part of Sera and western portion of Biliqo. Collectively, these attributes allow 90% of the livestock movements of the communities in the project area to remain on the project area.

The project zone and surrounding areas do feature geographies and social dynamics that produce traditional movements of livestock off the project area. In our opinion, this should not be viewed as a lack of control of the project area but rather the consequence that no specific, finite boundary can be formed that would encompass 100% of herder movements. First, multiple boundaries of the project area are set by political borders, such as between Counties, Isiolo versus Laikipia Counties along the southern borders of Nasuulu, Kalama, Westgate, and Oldonyiro (formerly Mpus Kutuk), and Laikipia versus Isiolo County in the case of Naibunga.

The Ewaso Nyiro River sets several Conservancy boundaries that are straddled by the same ethnic group, e.g., Samburu on both sides of the river between Westgate and Oldonyiro or Borana on both sides where it flows between Biliqo Bulesa and Nakuprat-Gotu. Similar boundaries overlap across project borders, such as between Biliqo Bulesa and communities east of Nakuprat-Gotu. However, Naibunga, Lekurruki and Il Ngwesi Conservancies are dominated by Laikipiak Maasai and represent a northernmost extent of this ethnic group. Communities with related family groups are scattered throughout northern Laikipia County (lying to the south of the project area). Under extreme drought conditions herders in these Conservancies would take herds southward and share range with relatives, in other words, under informal agreement. Other instances of frequent exchange of livestock in response to patchy rainfall occur in Meibae, where related Samburu live to the west in and around the Kirisia Mountains and the town of Maralal and for Rendille herders in Melako to move north into Marsabit during severe droughts.

Even though these movements can be characterized as “traditional” the project still counts them as leakage. However, because these movements represent the outcome of multi-year reciprocal exchange, they should not be viewed as evidence that the project cannot control its boundaries. As the project shows for the current monitoring period, average annual methane emissions from movements of off-project livestock onto project conservancies represent <5% of annual average project methane emissions.

7 ADDITIONAL PROJECT IMPACT INFORMATION

No additional information is needed.

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9 APPENDIX A

Methods for Interpreting Grazing Reports to Determine Leakage Emissions and Encroachment Methane Emissions

Decision Tree for Interpreting Grazing Reports to Determine Leakage Emissions

Enter all data in the tab for the appropriate conservancy in the '[Leakage and Encroachment Verification Calculations 2017 – 2020 July2021.xlsx](#)' workbook.

1. Does the Report claim all livestock are on the project area? If Yes, Go to 2. If No, is there a claim that some livestock are in locations off of the project area (this could be an explicit statement or a list of off-project locations)? If Yes, Go to 5. If No, Go to 3.
2. **Leakage is 0. Assign all livestock numbers to either On Conservancy or On Project cells.**
3. Does the map indicate Conservancy livestock locations/movement with symbols and/or movement arrows? If Yes, Go to 4. Otherwise, assume that livestock are either on the conservancy or off the project area (no livestock in adjacent on-project conservancies) and Go to 7.
4. The map may show either symbols, arrows, or symbols and arrows. Are all symbols/arrows contained within the project area? If Yes, Go to 2. If No, Go to 5.
5. Are there symbols or arrows that show conservancy livestock off the project area? If Yes, Go to 6. If No, Go to 7.
6. Symbols and/or arrows indicate presence or movement of livestock off the project area.
 - a. **Option A. Only symbols are used to indicate locations of livestock.** Count the number of symbols attributed to Conservancy livestock (some symbols may indicate encroaching livestock) that are on the map. From this pool of Conservancy livestock symbols, count the number of symbols that are located off the project area. ***Enter as leakage the estimated livestock numbers for the Conservancy in that year weighted by the proportion of all symbols on the map representing Conservancy livestock that are off the project area.***
 - b. **Option B. Symbols and arrows indicate livestock movements and positions.** Consider only symbols and arrows that are attributed to Conservancy livestock (some symbols may indicate encroaching livestock). From this pool, are there arrows that are not attached to groups of symbols? If Yes, Go to Option C. If No, count the number of symbols in the indicated group(s) shown by arrows to be off the project area (i.e. the arrow's tip is off the project area). Additionally, count the number of symbols without arrows that are off the project area. ***Enter as leakage the estimated livestock numbers for the Conservancy in that year weighted by the proportion of all symbols on the map representing Conservancy livestock that are grouped symbols with arrows leading off the project area or symbols without arrows that are off the project area.***
 - c. **Option C. There are ONLY arrows indicating livestock movements.** Count the total number of arrows on the map referring to Conservancy livestock (some arrows may be encroaching livestock). From this pool, count the number of arrows with tips that are off the project area. ***Enter as leakage the estimated livestock numbers for the Conservancy in that year weighted by the proportion of all arrows representing Conservancy livestock with tips that are off the project area.***
7. Either there is a claim that Conservancy livestock are off-project but with no corresponding symbols/arrows (Q5 Pathway), or there are no claims about Conservancy livestock and no symbols/arrows (Q3 Pathway). Are there Conservancy livestock claimed/listed in adjacent on-project conservancies? (If coming from the Q3 Pathway answer must be No). If Yes, Go to 8. If No, are there livestock numbers reported for the Conservancy? If No, Go to 8. If Yes,

- then **enter as leakage the difference between estimated livestock numbers for the Conservancy and livestock numbers reported on the report.** These should be positive numbers (or zero for some livestock classes) that are less than the estimated livestock numbers for the Conservancy.
8. There is a statement listing livestock at multiple locations. Determine the number of locations (L) and which locations off the project area (n). **Multiply the estimated Conservancy livestock numbers for the livestock class(es) listed as in off-project locations by the ratio n/L and enter as leakage.**

Decision Tree for Interpreting Grazing Reports to Determine Livestock Encroachment Emissions

Enter all data in the 'Encroachment Records' tab of the '[Leakage and Encroachment Verification Calculations 2017 – 2020 July2021.xlsx](#)' workbook.

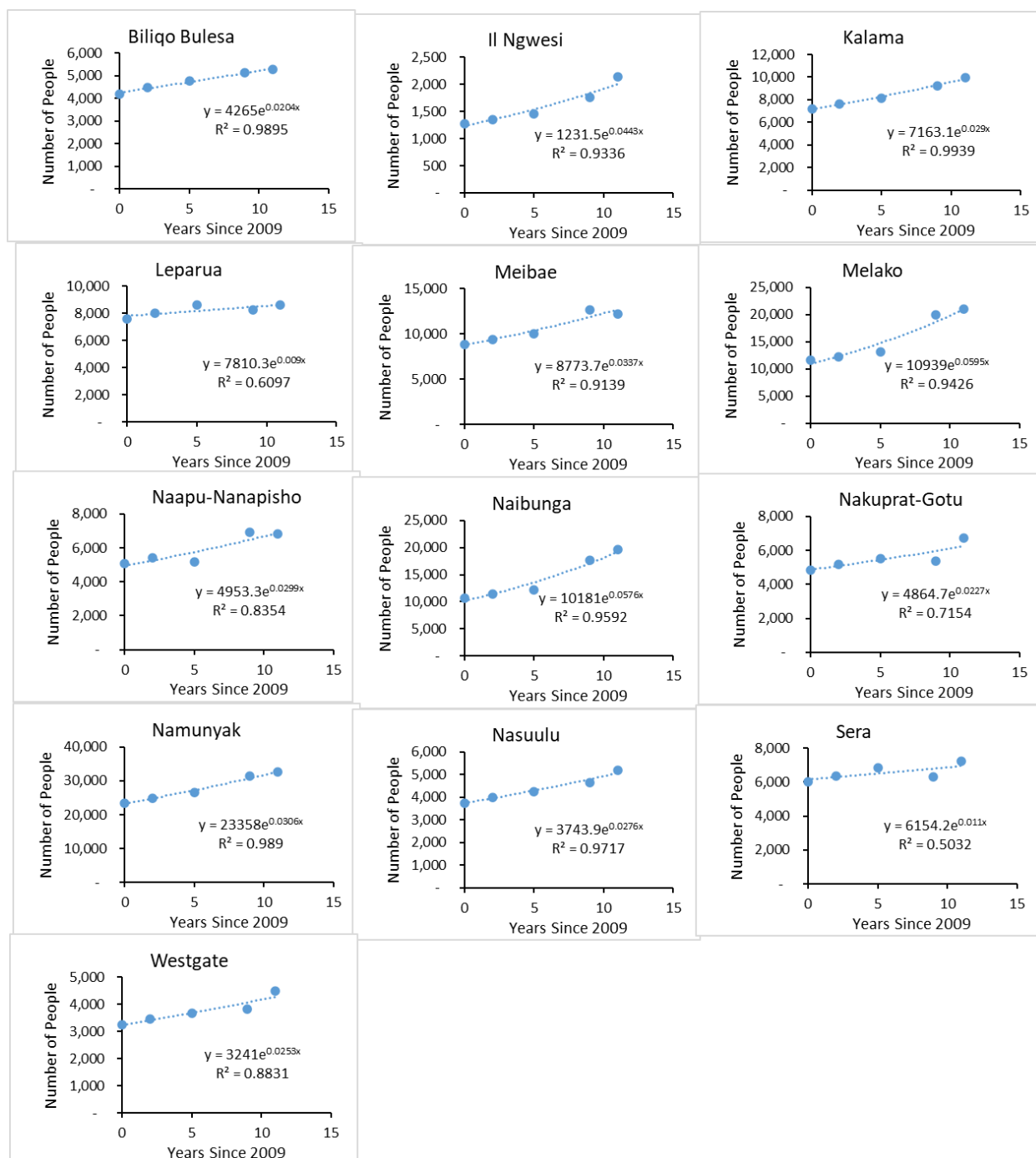
1. Does the grazing report mention an "encroachment" or "incursion" of livestock from some off-project source? Note that this does not include "encroachment" into Buffer or Core areas that is tracked by some Conservancies (Namunyak, Il Ngwesi, Sera, Westgate). Encroachment must be attributed to off-project sources.
If Yes, Go to 2, if No, Go to 12; If report missing go to 11
2. Is the number and class of encroaching livestock explicitly mentioned on the report?
If Yes, Go to 3; otherwise, Go to 4.
3. **Enter the reported encroachment numbers.**
4. Does the grazing report table list numbers of livestock in different conservancy blocks *and* the source conservancies of those livestock (there may be more than one conservancy listed)?
If Yes, Go to 5, Otherwise Go to 6.
5. **Divide the number of livestock reported in a grazing block by the number of sources (conservancies) with livestock reported in said grazing block. Multiply this quotient by the number of off-project sources contributing to encroachment. Add this product for each grazing block and livestock class and enter the resulting sum.**
6. Are there arrows or symbols on the map that are attributed to encroaching livestock? Are conservancy livestock also represented by arrows or symbols?
If Yes to both, Go to 7, if No to either or both, Go to 8.
7. **Divide estimated number of encroaching off-project livestock symbols or arrows by the number of symbols or arrows representing Conservancy livestock. Multiply this ratio by the estimated Conservancy livestock for the year. Enter this product.**
8. Does the grazing report table list numbers of livestock for any class that are greater than the livestock census estimate of that livestock class for the conservancy in question?
If Yes, Go to 9, if No, Go to 10.
9. For each livestock class, sum the total number of animals of each class reported in the grazing report table and then subtract the estimated Conservancy livestock numbers for that livestock class. **Enter the result.**
10. If Conservancy livestock are reported to be off the Conservancy, but livestock numbers are listed on the report and encroachment is mentioned, conservatively assume all numbers reported in the grazing report table are encroaching livestock. **Enter the sum total for each livestock class in the grazing report table.** If sources of livestock are mentioned, determine the total number of sources and the number of sources that are from off-project and **multiply the sum total for each livestock class by the ratio of off-project sources to total sources.** If encroachment is mentioned, Conservancy livestock are noted as present, but there is no evidence of elevated numbers or indications of symbols or arrows distinct from Conservancy livestock, assume

“encroachment” mentioned is of Conservancy livestock into the Core or Buffer management zones and thus is not off-project encroachment is negligible and **Go to 12.**

- 11. Missing report.** Do both neighboring months report encroachment in a way that can be quantified by the above methods? **Enter encroachment numbers from the neighboring month with most conservative (highest) encroachment estimate.**
- 12. No evidence for encroachment. Do not enter anything.**

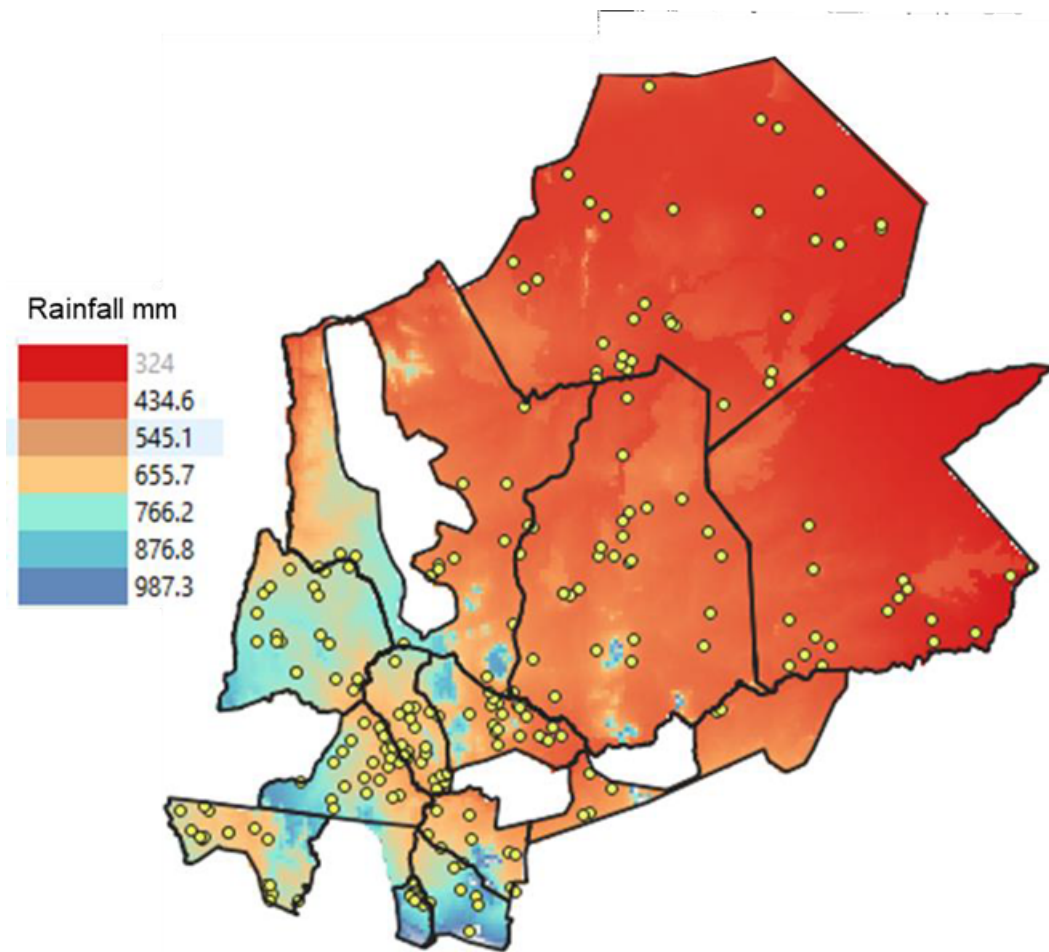
10 APPENDIX B

Population trends in the 13 Conservancies of the Northern Kenya Grasslands Carbon Project. Fitted curves assume a constant per capita population growth rate over the period 2009-2020.



11 APPENDIX C

Digital map of mean annual rainfall for the project zone, showing the boundaries of all the Conservancies. Data are averaged over summed monthly estimates of rainfall from WorldClim, World Resources Institute, an integrated weather station and satellite estimate of rainfall resolved at 0.5 x 0.5 km resolution. Consequently, rainfall amounts for the project area were assigned to individual sampling points (shown as yellow points).



12 APPENDIX D

Summary

For the Northern Kenya Rangeland Carbon Project, a remote sensing method is employed to verify the degree of successful implementation of the project activity and subsequently adjust the modeled output of the SNAP soil organic carbon model. The remote sensing method was calibrated during a pre-study following baseline soil sampling and during baseline biodiversity sampling in April - June 2014. The analysis evaluates the annual changes in green biomass at a site relative to that expected if there was no grazing. This annual relative change is calibrated relative to known temporal patterns of grazing impacts, and a threshold of annual relative change is determined that corresponds to more than one major grazing event per year, and thus unsuccessful implementation of project activities. The area that exhibits annual relative change in excess of the criterion is subtracted from the project area in determining total modeled estimates of soil carbon sequestration, this making claimed removals more conservative.

For 2017–2020 monitoring period, there were two deviations to the remote sensing method as compared to the 2013–2016 Monitoring Report. First, an adjusted set of ungrazed reference polygons is used for calculating annual relative change in biomass. This is justified due to conservancies with lower mean annual rainfall (relative to wetter reference polygons) exhibiting high relative changes in NDVI and thus being judged in error as unsuccessful. Secondly, the threshold value of annual relative change in NDVI is adjusted from 0.05 to 0.1. This is justified because a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry season grazing areas.

Assessment of successful pastoralist grazing plans - full methodological details

Remote sensing of grazing and implementation of grazing plans

In NRT conservancies, data to verify livestock activity were obtained from 72 MODIS satellite images and 468 monthly grazing reports for each conservancy during each month and year (2014-2016). These data include satellite images showing bi-monthly 16-day composite NDVI calculations and used to calculate mean NDVI scores for every 16-day period each year within designated grazing blocks within each Conservancy over the monitoring period.

NDVI was calibrated to actual biomass measures during the pre-study in July and August 2014 following May-June rains by clipping annually produced biomass (not gray litter) from thirty 0.125 m² quadrats within each of thirty separate MODIS pixels during a specific 16-day window that corresponded to a single 16-day composite NDVI image. Measured biomass was strongly correlated with NDVI for the designated pixel both before and after cattle and sheep herds moved through the pixel (Figure 1). Importantly, NDVI measurements estimated biomass across a relatively low range of biomass encountered across the degraded, high-percentage bare ground landscape in the project area.

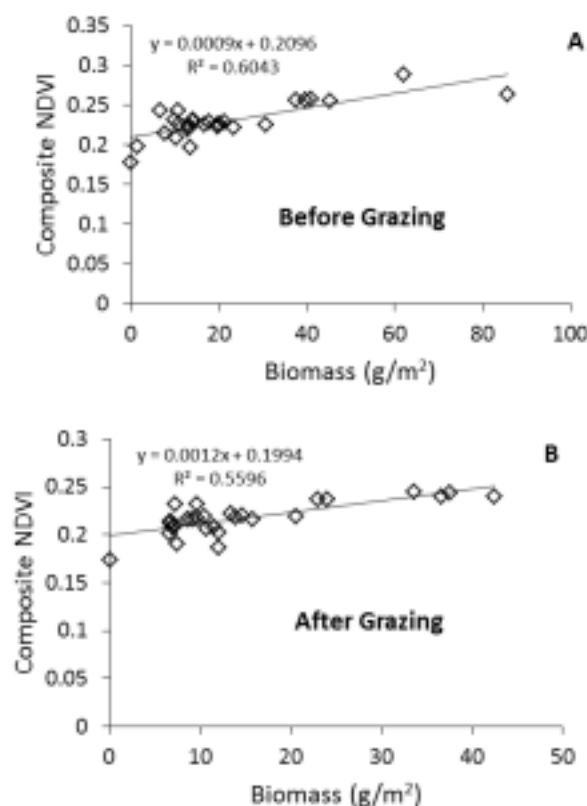


Figure 1. Relationships between NDVI measurements from individual pixels (250 x 250 m) of 15-day composite MODIS NDVI and measured mean forage biomass (shrub leaves and herbaceous plant biomass) before and after grazing. Forage biomass is from thirty 0.125 m² quadrats placed randomly inside each pixel in August 2014 in Kalama and Westgate Conservancies. Relationships are shown for A. before grazing by a herd of 30 or more cattle and/or additional sheep and goats, and B. immediately after grazing. These relationships demonstrate the strong connection between satellite based measures and actual forage biomass and support the use of satellite imagery in monitoring range condition and grazing impact. The relationships imply that a reduction in forage biomass from grazing results in a detectable reduction in NDVI as predicted by the regression lines in each graph, which are very similar. The y intercepts in each relationship reflect the NDVI signal of trees and shrubs, since forage biomass = 0.

To measure livestock grazing impacts, a monthly change in NDVI ($\Delta\text{NDVI}_{\text{Im}}$) was calculated by subtracting the later NDVI image from the earlier NDVI image, with more positive values indicating stronger reduction in forage biomass (Fig. 2; see below); this serves as a measure of livestock impact on forage during the 15 days between composite images. Further, a relative monthly change in NDVI ($\Delta\text{NDVI}_{\text{Im,rel}}$) (Eq. 1) was calculated by subtracting the $\Delta\text{NDVI}_{\text{Im}}$ for the same period from the mean $\Delta\text{NDVI}_{\text{Im}}$ ($\text{mean}\Delta\text{NDVI}_{\text{Im,REF}}$) across one or more ungrazed reference areas (inaccessible plateaus, fenced rhino sanctuaries, ecotourism areas from which livestock were excluded) to correct for expected changes due to recent rainfall or drying.

$$\Delta\text{NDVI}_{\text{Im,rel}} = \Delta\text{NDVI}_{\text{Im}} - \text{mean}\Delta\text{NDVI}_{\text{Im,REF}} \quad (1)$$

$\Delta\text{NDVI}_{\text{Im,rel}}$ for a MODIS pixel, was consistently correlated with change in forage biomass for that pixel (Figure 2), confirming the ability of the remote sensing method to estimate forage availability on the ground.

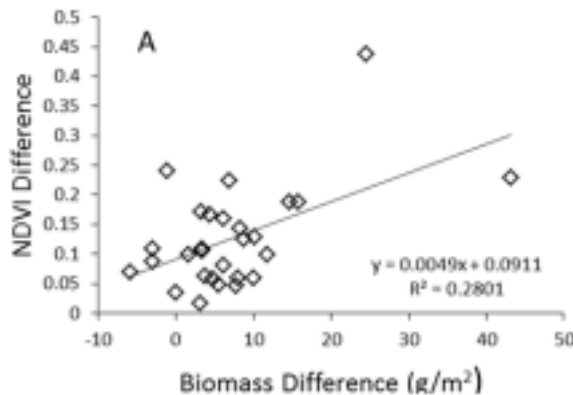


Figure 2. Relationship between measured forage biomass difference before and after grazing by 50 or more cattle and relative change in NDVI, $\Delta NDVI_{m,rel}$ (NDVI difference), between 28 July and 12 August, 2014 for thirty MODIS pixels within Kalama and Westgate Conservancies.

Monthly relative $\Delta NDVI_{m,rel}$ was corroborated with observed distributions of livestock by Conservancy Grazing Coordinators (CGCs) during the May-July 2014 pre-study. Random points were selected from within monthly $\Delta NDVI_{m,rel}$ maps of each conservancy to obtain estimates of $\Delta NDVI_{m,rel}$ for individual pixels, which were then compared to distribution maps of livestock occurrence submitted by grazing coordinators. Livestock occurrence values of 0 for livestock absent or 1 for livestock present were assigned to each pixel and thus to each value of $\Delta NDVI_{m,rel}$. This analysis produced a logistic model in which $\Delta NDVI_{m,rel}$ predicted absence/presence of livestock (Figure 3C).

Given the many potential sources of error in reporting by CGCs, such as mismatch in the period of days assessed by CGCs relative to satellite analysis, imprecision in the drawing of polygons relative to MODIS pixel sizes (Figure 4B), lack of knowledge of livestock distributions in relatively inaccessible areas, e.g. northern Namunyak Conservancy (Figure 4A), and the possibility of encroachment by herders from communities outside the project area, the strength of the logistic regression indicates a strong signal of livestock grazing impacts detected by the $\Delta NDVI_{m,rel}$ measurement.

In the initial ground truthing study in July-August 2014 in Kalama and Westgate Conservancies, $\Delta NDVI_{m,rel}$ was reasonably well correlated with biomass removals by a single livestock grazing event (Figure 2). Consequently, it was concluded that $\Delta NDVI_{m,rel}$ sufficiently represented a measure of rainfall-corrected monthly grazing impacts by livestock to determine success or lack thereof in implementing conservancy grazing plans.

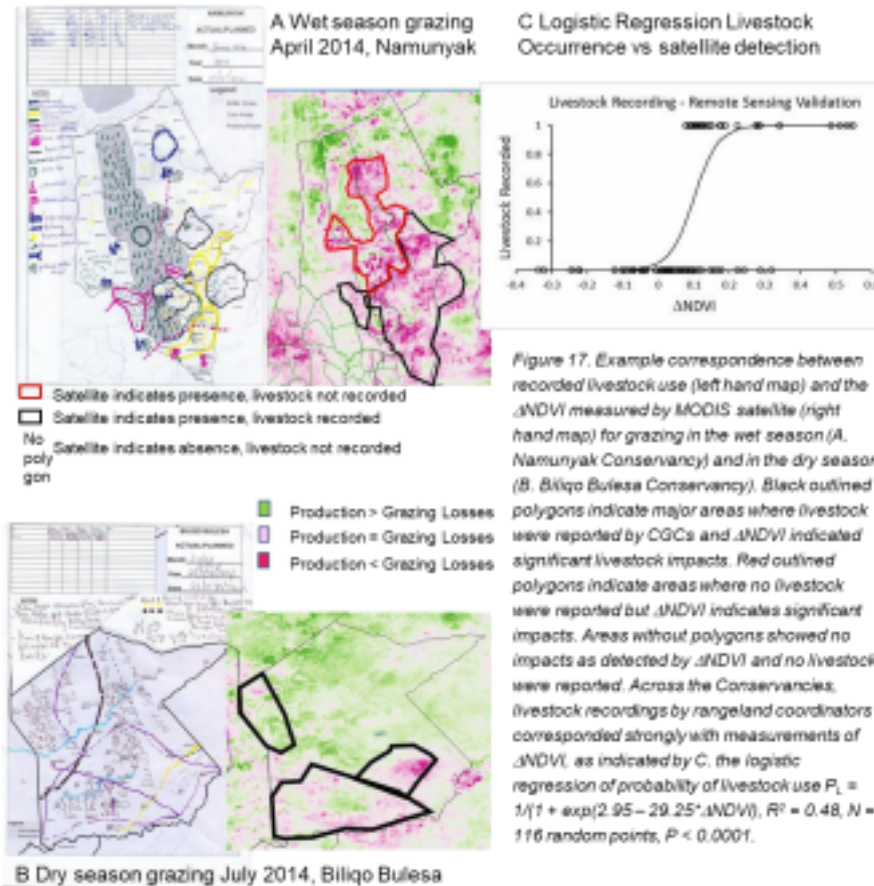


Figure 3. Example correspondence between recorded livestock use (left hand map) and the $\Delta NDVI$ measured by MODIS satellite (right hand map) for grazing in the wet season (A, Namunyak Conservancy) and in the dry season (B, Biliqo Bulesa Conservancy). Black outlined polygons indicate major areas where livestock were reported by CGCs and monthly relative $\Delta NDVI$ indicated significant livestock impacts. Red outlined polygons indicate areas where no livestock were reported but monthly relative $\Delta NDVI$ indicates significant impacts. Areas without polygons showed

no impacts as detected by relative monthly $\Delta NDVI_{m,rel}$ and no livestock were reported. Across the conservancies, livestock recordings by rangeland coordinators corresponded strongly with measurements of $\Delta NDVI_{m,rel}$ as indicated by C., the logistic regression of probability of livestock use $PL = 1/(1 + \exp(2.95 - 29.25 \cdot \Delta NDVI_{m,rel}))$, $R^2 = 0.48$, $N = 116$ random points, $P < 0.0001$.

Method for assessing successful implementation of project activities.

For Meibae, Westgate, Kalama and southern Namunyak Conservancies, monthly grazing impact (not relative) $\Delta NDVI_m$ was calculated for each 17 grazing blocks **for each month of 2014** (see examples in Figure 5) and summed to yield $SUM\Delta NDVI_m$. A similar annual reference sum of NDVI differences, $meanSUM\Delta NDVI_{m,REF}$, was calculated as the sum of mean monthly $\Delta NDVI_m$ across the reference polygons. An annual estimate of grazing impacts, $SUM\Delta NDVI_{rel}$, for each MODIS pixel was calculated as:

$$SUM\Delta NDVI_{rel} = SUM\Delta NDVI_m - meanSUM\Delta NDVI_{m,REF} \quad (2)$$

$SUM\Delta NDVI_{rel}$, the **annual relative change in NDVI** for a MODIS pixel, was strongly correlated with whether a MODIS pixel experienced successful implementation of project activities, i.e., a single major grazing event each year, as reflected by a single month (two consecutive 16-day composite images) in which $\Delta NDVI_{m,rel} > 0.1$ (as suggested by the logistic regression in Fig. 3C) This correspondence was measured by logistic regression of $SUM\Delta NDVI_{rel}$ versus whether one or no grazing events occurred (score 0) or (Figure 5B). The regression suggested a threshold value of $SUM\Delta NDVI_{rel} < 0.08$ was associated with a probability of successful grazing implementation > 0.5 , so, conservatively, we used

$SUM\Delta NDVI_{rel} < 0.05$ as the threshold for successful implementation of rotational grazing. Note that this criterion applies to situations where rotations went as planned, as in the grazing block examples (Figure 4) or if livestock were grouped and moved around a larger regional landscape, as occurred during each dry season.

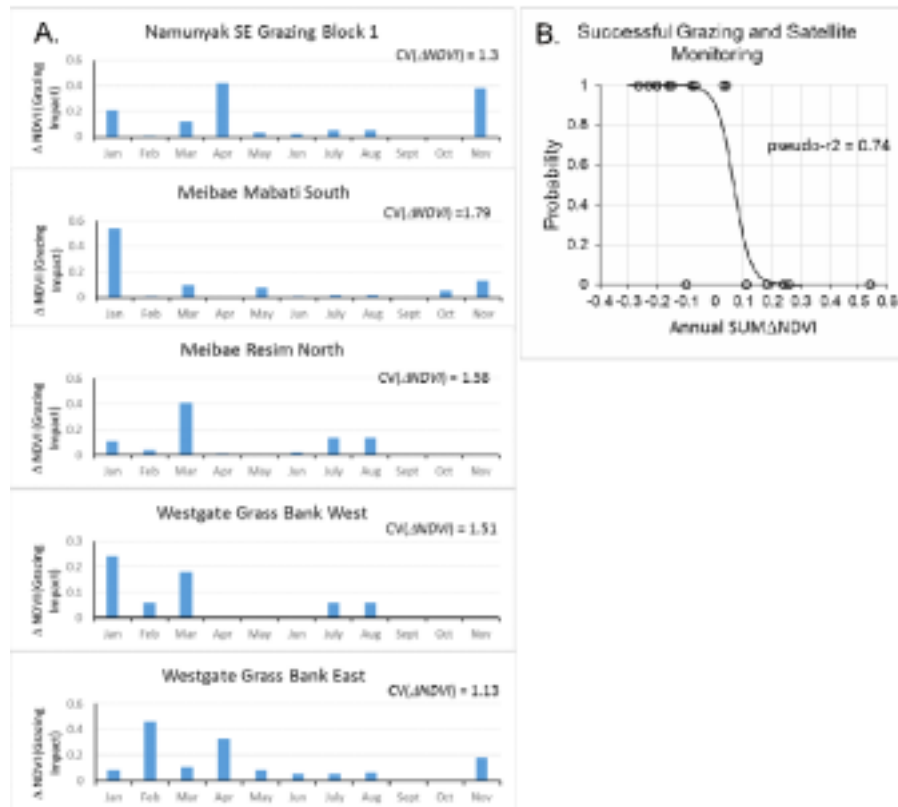


Figure 4. A. Example temporal patterns of monthly grazing impacts, measured as relative monthly change in NDVI ($\Delta NDVI_{m,rel}$), on selected grazing blocks with high quality grazing reports in southern Samburu County, Kenya from Meibae, Westgate, and Namunyak Conservancies for 2014. These examples clearly show the difference in temporal pattern between unsuccessful and successful rotational grazing practices. Unsuccessful patterns include Namunyak SE Grazing Block 1 and Westgate Grass Bank East, which were

grazed ($\Delta NDVI > 0.1$) three times during the year, and especially twice during the wet season. Successful patterns included Meibae Mabati South and Meibae Resim North, which were grazed only once. B. Logistic regression to determine probability of successful grazing implementation over the whole year, p , as a function of annual $SUM\Delta NDVI_{rel}$: $p/(1-p) = 2.3 - 35.2(SUM\Delta NDVI_{rel})$, approximately 74% of the variation explained.

Estimating area of successful implementation

To estimate the area of successful implementation of grazing plans **in the first monitoring period 2013-2016**, we used the criterion of $SUM\Delta NDVI_{rel} < 0.05$ to identify MODIS pixels (250 x 250 m) ((from NASA LP DAAC <https://lpdaac.usgs.gov/>) note that the Google Earth Engine database did not exist at that time) where rotational grazing practices were successful in each year using the Raster->Processes->Geoalgorithms->Zonal Statistics feature in QGIS 3.4 (2013-2016) and are now calculated using Google Earth Engine (2017-present). The proportion of area that was unsuccessful is the proportion of pixels of $SUM\Delta NDVI_{rel} > 0.05$ to total number of pixels for each Conservancy; this proportion discounts the total area used in the SNAP soil carbon model. Areas of unsuccessful implementation of rotational grazing practices for the 2013-2016 monitoring period are shown in Figure 5.

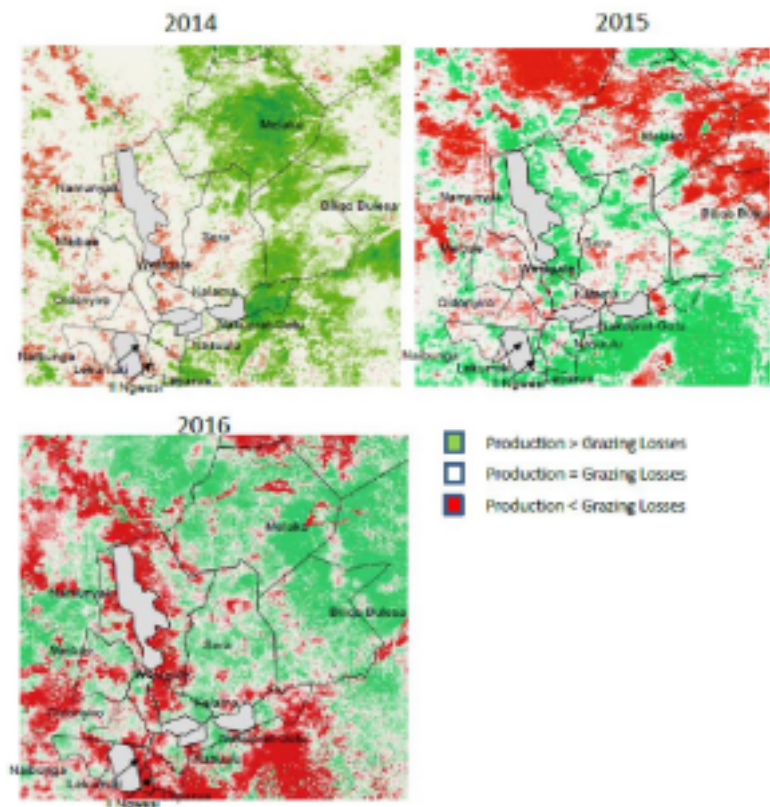


Figure 5. Evidence of successful project implementation (green and white areas) for 2014-2016 in the project zone

Justification for Adjusted sets of reference polygons for calculating successful implementation of grazing plans during 2017-2020.

The method of assessing annual grazing plan implementation depends on comparing annual relative changes in NDVI, $SUM\Delta NDVI_{rel}$, determined by summing monthly $\Delta NDVI_m$ across the twenty-four 16-day intervals for each year, $SUM\Delta NDVI_m$ (Eq. 2 above) at each MODIS pixel, as compared with mean annual sums of $\Delta NDVI_m$ across the same time intervals ($meanSUM\Delta NDVI_{REF}$) from reference polygons that are not grazed or only slightly grazed by livestock (see Eq. 2 above). These reference locations are embodied in multiple polygons that span the project area, including “Core” areas of Il Ngwesi and Kalama Conservancies, the tops of plateaus in Kalama Conservancy that are relatively inaccessible to livestock, and a large ~1,000 ha fenced rhino sanctuary in Sera Conservancy. In most years, the mean annual relative change in NDVI ($meanSUM\Delta NDVI_{REF}$) in these reference areas is relatively similar despite differences in mean annual rainfall. Consequently, the Google Earth Engine (GEE) calculations of annual relative change in NDVI ($SUM\Delta NDVI_{rel}$) for individual pixels represent a good estimate of livestock impacts in drought years (2017, 2018). However, in wet years following droughts, such as occurred in 2019 relative to 2018, the $meanSUM\Delta NDVI_{REF}$ in Il Ngwesi Core area and Kalama Plateaus are much bigger than the values for Kalama Core and Sera Rhino Sanctuary reference polygons. Averaging in these higher $meanSUM\Delta NDVI_{REF}$ from those reference polygons virtually guarantees that conservancies with lower mean annual rainfall will exhibit frequent pixels with high $SUM\Delta NDVI_{rel}$ and thus be judged in error as unsuccessful. Therefore, reference polygons appropriate to the mean annual rainfall of the different Conservancies should be used for each year (2017-2020). Reference polygon sets are broken into higher rainfall (wetter) and lower rainfall (drier) conservancies:

Reference set 1 is to be used with the higher rainfall conservancies – Il Ngwesi, Naibunga, Meibae, Nanapishu (Oldonyiro), Leparua, Nasuulu, Namunyak

Reference set 2 is to be used with the lower rainfall conservancies – Kalama, Westgate, Sera, Biliqo Bulesa, Melako, Nakuprat-Gotu

Thus, we calculated **relative** change in NDVI, $SUM\Delta NDVI_{rel}$ (Eq. 1) for the drier Conservancies Melako, Sera, Namunyak, Biliqo Bulesa, Nakuprat-Gotu, Nasuulu, Kalama, Westgate, and Leparua Conservancies using only the reference polygons from Kalama Core and the Sera Rhino Conservancy. Calculations for wetter Conservancies Il Ngwesi, Naibunga, Meibae, and Oldonyiro include all the reference polygons, as is coded in the current GEE model.

Justification for a Shift in the Threshold Value of Annual Relative $\Delta NDVI$

The threshold for unsuccessful implementation was obtained from two logistic regressions, which indicate the estimated $\Delta NDVI$ values that correspond to whether substantial livestock were present in a grazing block (Fig. 3), and whether substantial livestock were present for more than one two-week period during the year (unsuccessful implementation).

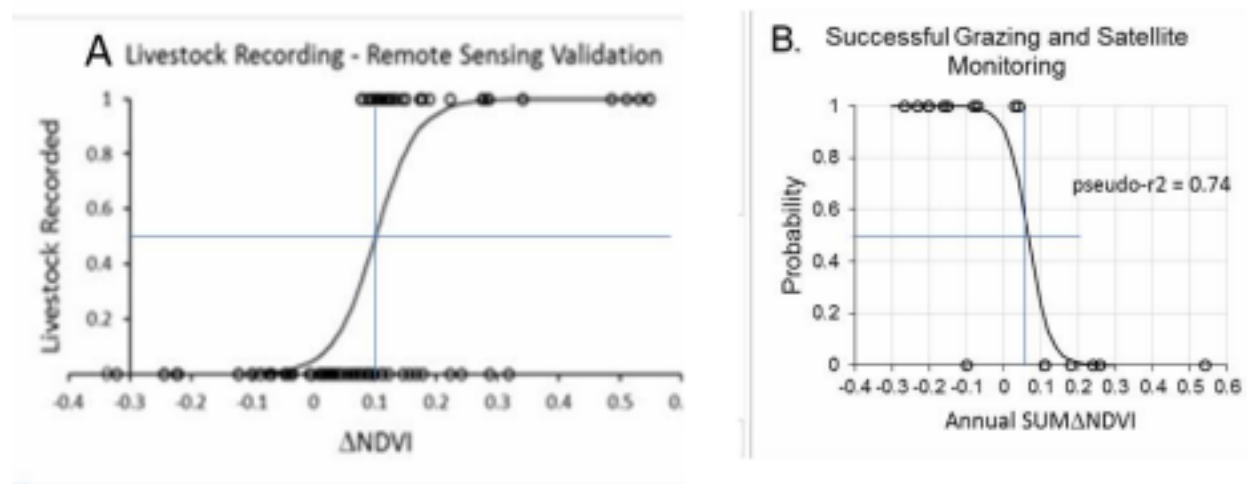


Fig. 1. Logistic regressions to assess successful implementation of grazing plans in the Northern Kenya Grasslands Carbon Project. Fig. 1.A. The logistic regression of livestock presence (livestock recorded = 1) and absence (livestock recorded = 0) versus calculated **monthly relative $\Delta NDVI$** ($\Delta NDVI_{m,rel}$, Eq. 1), the value of monthly relative $\Delta NDVI$ at which there was greater than a 50% probability (livestock recorded > 0.5) of recording substantial livestock present is 0.1 (intersection of two perpendicular lines). This logistic regression was previously shown in Fig. 3C. Also, livestock were only rarely encountered in map locations where $\Delta NDVI_{m,rel} < 0.1$. Fig. 1.B. At 17 grazing blocks where presence/absence of livestock was monitored over an entire year, a logistic regression of the occurrence of only a single grazing event (probability = 1) or more than one grazing event (probability = 0) versus the relative annual sum of $\Delta NDVI$ for the 2014 calendar year (**annual relative $\Delta NDVI$** , $SUM\Delta NDVI_{rel}$ from Eq. 2). This logistic regression was previously shown in Fig. 4B.

The threshold of 50% probability of unsuccessful implementation occurs at approximately annual relative $\Delta NDVI = 0.08$, but, due to the small sample size of grazing blocks, there is a large confidence interval around that estimate. Since **monthly relative $\Delta NDVI$** ($\Delta NDVI_{m,rel}$) must be > 0.1 for livestock to be detected at a location then it is reasonable that the annual sum, or **annual relative**

$\Delta NDVI$ ($SUM\Delta NDVI_{rel}$) should be > 0.1 . This is especially true for dry season grazing areas because they do not regrow immediately after grazing. Since it requires a relative change in NDVI of 0.1 to detect a grazing event (Fig. 1.A.), it is difficult for dry season grazing areas to show a relative

$\Delta NDVI < 0.1$ even if they are grazed only once. Thus using a threshold of 0.1 more accurately reflects the occurrence of single grazing events in **both** wet and dry season grazing areas.

This justifies a threshold of **annual relative $\Delta NDVI$** ($SUM\Delta NDVI_{rel} = 0.1$) for assessing whether grazing plans were successfully implemented at a location. Thus, where pixels have $SUM\Delta NDVI_{rel} \leq 0.1$, these areas are deemed to be successfully grazed in each year 2017-2020. Pixels with $SUM\Delta NDVI_{rel} > 0.1$, are deemed to be unsuccessfully grazed in each year 2017-2020.